rainier3d: a reproducible digital model of the subsurface and surface of Mount Rainier, Washington

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Abstract. Mount Rainier is an active volcano of the Cascade Volcano Arc that presents diverse geohazards relies on one-dimensional velocity models, and the regional three-dimensional models of Cascadia do not resolve the edifice, its altered core, its glaciers or the near surface. rainier3d is an open digital model of Mount Rainier and the West Rainier Seismic Zone (WRSZ) that places subsurface, near-surface, surface, geodetic and hydrological layers on one set of grids, over a 70 × 75 km box from the ground to 20 km below sea level. Its inputs are fetched by script from public archives, cached with checksums and rebuilt by numbered scripts in one environment; four model inputs and four viewer files are read from local or delivered copies. Every parameter names its source or is marked as an author choice.

The subsurface model gives P- and S-wave speed, density, attenuation, porosity and permeability on three grids, from 250 m × 50 m to 1 km. Mapped geology is extended to depth by rules, near-surface alteration comes from a helicopter electromagnetic survey, and a critical-zone model resolves soil, Quaternary cover and weathered rock in the top 150 m with the same water table and pore pressure as the rock below. The result is merged with the USGS Cascadia velocity model v1.7 and CRESCENT Gen0 in the wavenumber domain and calibrated on 1823 P and 1280 S analyst picks from 88 Pacific Northwest Seismic Network (PNSN) earthquakes, with every event relocated in every trial model. On 44 events withheld from the fit, the root-mean-square (RMS) residual falls from 0.120 to 0.093 s for P and from 0.325 to 0.188 s for S; the PNSN one-dimensional model gives 0.131 and 0.263 s. A local-earthquake tomography of 3770 earthquakes below the critical zone, given in the supplement, lowers the held-out residuals further, to 0.073 s for P and 0.121 s for S. Relocated with NonLinLoc, the 371 PNSN earthquakes of magnitude 1 or larger from 2023 to 2025 fit better in rainier3d than in the one-dimensional model (median RMS 0.109 against 0.119 s for 340 well-located events).

The surface layers describe elevation, geology, glacier thickness, soil properties, depth to rock, water table, Vs30, land cover, vegetation structure, terrain geometry and satellite imagery. GNSS strain rates and the strain of the edifice load are given on a 500 m grid in the model volume, and a catalogue holds 1,649 landslide, avalanche and debris-flow events, 19 of them located seismically, with the outlines of 463 debris flows and three lahar deposits.

The derived products are distributed under CC-BY 4.0 [TODO: Zenodo version DOI of the products and of the software]. A command-line and Python client writes them in the formats of eikonal solvers, NonLinLoc, SPECFEM3D, PyLith and the EarthScope Earth Model Collaboration, and a web viewer shows them in three dimensions.

1 Introduction

Mount Rainier is the most hazardous volcano in the Cascade Range, for two reasons. Its hydrothermally weakened edifice is covered by glaciers, and its lahars reach populated valleys (Hoblitt et al., 1998; Scott et al., 1995). Seismic monitoring at Rainier relies on one-dimensional velocity models, with station corrections absorbing the effects of topography and near-surface structure. The regional three-dimensional models of Cascadia resolve the crust at wavelengths of several kilometres or more (Wirth et al., 2025; He et al., 2026). They do not resolve the volcanic edifice, its altered core, its glaciers or the shallow units exposed around it.

Integrated models of a region exist elsewhere, but rarely for a volcano. Southern California has community velocity, fault and rheology models maintained by the Statewide California Earthquake Center (SCEC) and served through one software framework (Lee et al., 2014; Shaw et al., 2015; Plesch et al., 2007; Hearn et al., 2026; Small et al., 2017), and Cascadia has the USGS velocity model and the CRESCENT community model (Stephenson et al., 2017; Wirth et al., 2025; He et al., 2026). Instrumented volcanoes have three-dimensional velocity models from local tomography, at Yellowstone, Kīlauea and Mauna Loa, Mount St. Helens, Etna and Campi Flegrei (Huang et al., 2015; Lin et al., 2014; Kiser et al., 2016; Ulberg et al., 2020; Totaro et al., 2024; Calò and Tramelli, 2018). Rainier itself has been imaged by local-earthquake tomography, receiver functions with surface waves, and magnetotellurics (Moran et al., 1999; Obrebski et al., 2015; Flinders and Shen, 2017; McGary et al., 2014). Earth System Science Data has published three-dimensional geological and crustal compilations (Ahlers and Henk, 2026; Michailos et al., 2023; Livani et al., 2023). Each of these describes one kind of property, or compiles models without a scripted path from the archives to the grid. rainier3d puts subsurface, near-surface, surface, geodetic and hydrological layers on one grid, and rebuilds and checks them from the archives by script, in the spirit of the FAIR principles (Wilkinson et al., 2016). Digital twins of the Earth system assimilate observations as they arrive (Bauer et al., 2021); rainier3d is the static, reproducible state on which such a scheme for a volcano can start.

rainier3d connects these scales in one model with three properties:

- it is tied to the mapped geology at the surface: the 149 map symbols of the Washington 1:100,000 geologic map in the box (Washington Geological Survey, 2026a) are assigned to 14 surface model units by the rules of `configs/units.yaml` (S1) and extended to depth by explicit geometry rules (S3, Sect. 5), and an invariant test requires the top rock cell of the finest level (L1) to carry the mapped unit in at least 95% of bedrock columns;
- it is consistent with regional tomography at depth: S5 merges the geology model in the wavenumber domain with the USGS Cascadia velocity model v1.7 down to 9.9 km below the ground (Wirth et al., 2025) and CRESCENT Gen0 below it (He et al., 2026), keeping the regional model at wavelengths longer than a cutoff of 6 km at the surface, 10 km at 2 km depth and 20 km at 10 km depth (placeholder values in `configs/domain.yaml`), the geology model alone above 300 m depth, and a linear taper between 300 and 1000 m; the root-mean-square (RMS) difference of ln V between the low-passed fused and regional models is at most 0.011, against a tolerance of 0.03;
- it reproduces the travel times of the Pacific Northwest Seismic Network (PNSN): calibrated in S13 on 1823 P and 1280 S analyst picks from 88 PNSN earthquakes (M 2.0–3.4, 1 January 2015 to 1 September 2026) (U.S. Geological Survey,

Table 1. Outline of the paper.

Section	Content
Sect. 2	Domain and workflow
Sect. 3	How the data are compiled deterministically, and how to rebuild them
Sect. 4	Surface layers
Sect. 5	Subsurface model: petrophysical models, the critical zone, and the crust with its calibration and validation
Sect. 6	Strain and stress
Sect. 7	Geohydrology
Sect. 8	Catalogue of mass movements
Sect. 9	Access to the products
Sect. 10	Limitations

2026a), each relocated in 3D in every trial model, it lowers the RMS residual of held-out events after relocation from 0.120 to 0.093 s for P and from 0.325 to 0.188 s for S (Sect. 5).

The same grids carry the surface layers that describe soil, water, vegetation and ice, the geodetic strain field, and the stress of the edifice load. The model is a digital model rather than a digital twin: it assimilates no time-dependent data. Only the GNSS product is refreshed on a schedule.

This paper describes, in order:

2 Domain, grids and workflow

The model uses Universal Transverse Mercator zone 10N (EPSG:32610) coordinates, with elevation in metres above NAVD88 (EPSG:5703), positive up.

- **Model box.** Easting 553–623 km and northing 5149–5224 km, 70 × 75 km (`bounds_utm` in `configs/domain.yaml`): the geographic box 122.30–121.40° W, 46.50–47.15° N (the box of the companion mt-rainier-smart-sensing project), whose projected corners span 553.07–622.77 km E and 5149.84–5223.08 km N, rounded outward to whole kilometres. It covers Mount Rainier National Park and the West Rainier Seismic Zone (WRSZ).
- **Summit.** Columbia Crest (4392 m; 121.7603° W, 46.8523° N, `summit_lonlat` in `configs/domain.yaml`) lies at easting 594.5 km, northing 5189.5 km. The 4392 m is the maximum of the 3DEP mosaic (about 8.3 m) of the mt-rainier-smart-sensing project; the highest cell of the 100 m surface grid is 4380 m, centred at 594.45 km E, 5189.45 km N.
- **Single source of the geometry.** All grids are derived from one configuration file (`configs/domain.yaml`, read by `rainier3d.config.domain.load_domain()`), which sets the horizontal coordinate system (EPSG:32610), the vertical datum (EPSG:5703), the bounds and two resolution profiles: `m1`, used for every product in this paper (100

Table 2. Model grids. The master product is one xarray DataTree in Zarr v3 format (`model.zarr`) with one node per level and a surface node.

Level	Elevation range (m)	Horizontal spacing (m)	Vertical spacing (m)	Cells (z × y × x)
Surface grid	–	100	–	750 × 700
L1	4400 to 0	250	50	88 × 300 × 280
L2	0 to −6000	500	250	24 × 150 × 140
L3	−6000 to −20000	1000	1000	14 × 75 × 70

m surface grid and the levels of Table 2), and `full` (10 m surface grid; horizontal × vertical spacing 50 × 20 m on L1, 200 × 100 m on L2, 500 × 500 m on L3), which is defined but not built. The code never hard-codes bounds, spacing or coordinate system.

Properties are stored on three stacked regular grids whose spacing coarsens with depth (Table 2). Cells above the ground carry no properties. The depth of each cell below the local ground surface is stored with it.

The workflow is a sequence of numbered scripts, S0 to S37, each run as a task of one locked software environment (pixi). Fig. 1 shows how they connect:

- the model chain (S1–S5): S1 builds the 100 m surface stack of elevation, surface unit and ice thickness (`surface.zarr`), S2 the environmental surface layers (`surface_layers.zarr`), S3 the rule-based 3D geology on L1–L3 (`geomodel.zarr`), S4 converts the units to Vp, Vs, density, Qp and Qs (`properties_geology.zarr`), and S5 fuses them with the regional models into the master product `model.zarr` (`pixi run all` runs S1–S8);
- the calibration loop, in which travel-time residuals update the rock-physics parameters and the regional correction (S13, S14): S13 relocates the 88 PNSN calibration events in 3D in each trial model and fits, by Gauss–Newton iterations on P, S and S−P residuals, three multipliers on all rock units (zero-pressure Vp V0, crack-closure pressure P* and Vs) and a static depth-dependent bias of the regional models below 1 km depth, written to `configs/velocity_calibration.yaml`; S14 relocates the same events in any model and scores it by the residuals left;
- the checks and products that read the finished model: the travel-time check against the PNSN one-dimensional model (S6), VTK and HTML views (S7), the sensor atlas (S8), uniform grids for ray tracing and location (S9), the paper figures (S10), the 3D viewer layers (S11), the download packages (S20) the comparison with the iMUSH local-earthquake tomography (S21) and the terrain-geometry layers (S30, Sect. 4.5);
- the local-earthquake tomography of the crustal model (S37, Sect. 5.3.7 and Supplement S1), written to a separate store (`model_let.zarr`) beside the master product.

The domain and its sensors are mapped in Fig. 2.

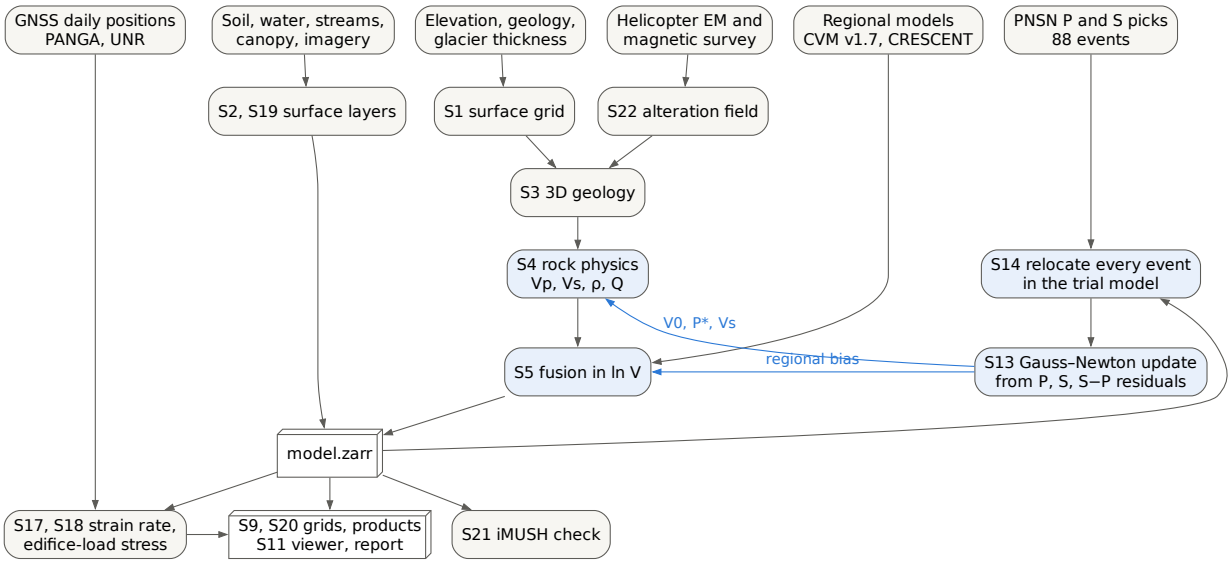


Figure 1. Workflow. Inputs (left) are fetched from their original archives. The model chain (middle) builds the surface grid, the 3D geology, the rock physics and the fusion with the regional models. The calibration (blue) relocates every PNSN event in each trial model and updates the rock-physics multipliers (S4) and a static correction of the regional models (S5) from the P, S and S−P residuals. Checks and products read the finished model (right).

3 A deterministic data compilation

3.1 Principles

The database is compiled by code, not by hand, following five rules.

1. **One registry.** `configs/sources.yaml` lists every data set and publication the model uses (99 entries). Each entry gives its DOI or service address, licence, the date and method of verification, and its role in the model. Every number in the configuration files names a registry key. A value chosen by the authors carries the key `m1_placeholder`, so the unsourced values can be listed and replaced. The bibliography of this paper is generated from the same registry (`pixi run bib`), with BibTeX keys equal to the registry keys.
2. **Original archives, cached.** Each stage downloads what it needs from the original archive, clipped to the model box where the service allows it. Examples are a window of a cloud-optimised GeoTIFF, an OPeNDAP subset, a feature-service query or a staged file. Each download is cached under `data/raw/<source>/`, and reruns read the cache. The GNSS stage also writes a manifest with the URL, retrieval time, size and SHA-256 of each file.

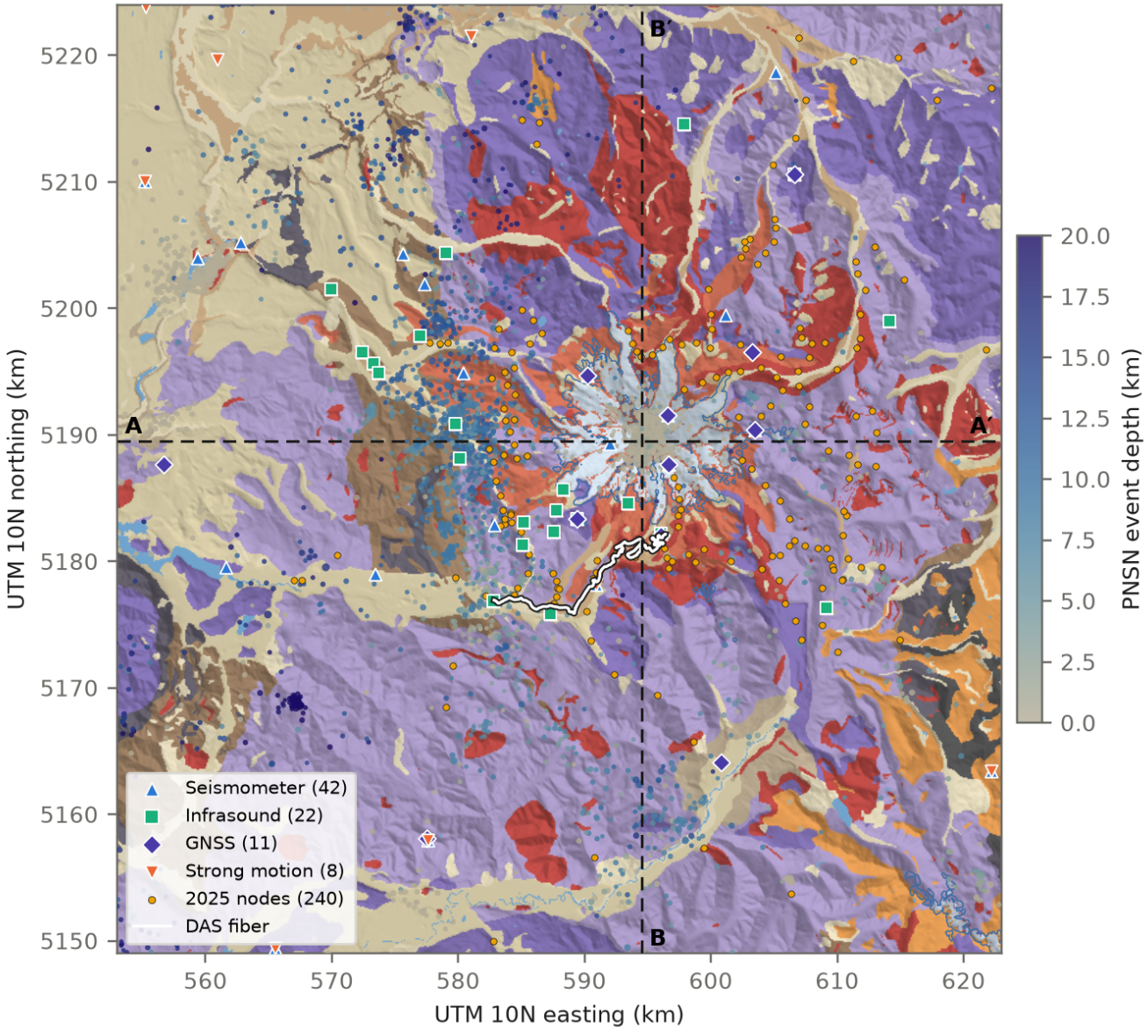


Figure 2. Model domain. The background shows the surface model units derived from the Washington 1:100,000 geologic map over a hillshade, with glacier outlines in blue. Symbols are operating sensors (FDSN and GNSS site metadata), and small circles the 2025 node array, which recorded in July and August 2025 (FDSN network Z5 (Schmandt, 2025); the legend counts all 240 nodes, of which 191 lie inside the box). The seismometer count of the legend is of operating broadband and short-period sites drawn on the map; the 46 permanent stations of Sect. 5.2 also include strong-motion sites. The white line is the Paradise–Nisqually Entrance distributed acoustic sensing (DAS) fibre, and dots are PNSN earthquakes since 2015 ($M \geq 0.5$), coloured by depth. The dashed lines mark sections A–A′ and B–B′.

3. **A checksum for every cached file.** `scripts/00_data_manifest.py` records the size and SHA-256 of all cached files in the committed table `docs/data_manifest.csv` (2,768 files, 5.16 GB; Table 3). `pixi run manifest -- --check` compares a rebuilt cache with it. Some services can return different bytes on a later request: ComCat can revise picks, and the USGS can re-stage NHDPlus. The check lists those files, so any difference between a rebuilt model and the published one can be traced to its input.
4. **One environment.** The software is pinned in `pixi.lock` for Linux and macOS (arm64). Two slim environments serve automation: `gnss` for the weekly strain refresh and `paper` for this report.
5. **Invariants.** The test suite checks the built model against rules that must hold whatever the data:
   - no properties above the ground: on L1, L2 and L3, Vp, Vs and density are NaN in every air cell (unit 0) and finite in every other cell;
   - Vp/Vs and density within physical bounds: Vp/Vs between 1.5 and 3.0 in rock cells (air, ice and magma mush excluded), fused Vp/Vs between the geology and regional ratios of the same cell (relative tolerance 10^{-4}), and density between 900 and 3100 kg m^{-3} in every non-air cell;
   - the long wavelengths of the fused model equal those of the regional model within a tolerance: per level and for Vp and Vs, the RMS of ln V between the low-passed fused and low-passed regional models is at most `lowpass_rms_tol` = 0.03 (`configs/domain.yaml`), checked on the report that S5 writes to `outputs/fusion_report.csv` (current maximum 0.011, L1 Vs);
   - the surface unit matches the unit of the top model cell: where bedrock (plutonic, supracrustal or cap units) crops out, the top non-air L1 cell carries the mapped surface unit in at least 95% of columns.

Tolerances are fixed in the configuration and are not relaxed to make a run pass.

Table 3: The raw input cache, from `docs/data_manifest.csv`. SOLUS100 and POLARIS are read directly from their cloud-optimised GeoTIFFs and virtual rasters and are not cached.

Source folder	Stage	Content	Files	Size (MB)
`dem`	S1	USGS 3DEP 1 arc-second (about 30 m) DEM (U.S. Geological Survey, 2026c), fetched with py3dep over the box plus 0.02° (`3dep_30m_rainier-v0.tif`)	1	59

Source folder	Stage	Content	Files	Size (MB)
`geology`	S1, S24	Washington 1:100,000 geology (GeMS) map units (feature-service layer 11) and faults (layer 7) (Washington Geological Survey, 2026a); Quaternary faults (Earthquakes and Faults map service, layer 12) (Washington Geological Survey, 2026b); one GeoPackage each; the Description of Map Units of the compiled quadrangle (layer 13, JSON), read for unit names and deposit thicknesses	4	7
`glaciers`	S1	IceBoost v2 per-glacier thickness and error GeoTIFFs for 219 glaciers (Maffezzoli, 2025), RGI 6.0 glacier list and statistics (two CSV files) (RGI Consortium, 2017), GlaThiDa survey table (World Glacier Monitoring Service (WGMS), 2020)	222	44
`ecology`	S2	ETH canopy height 2020, 10 m tile N45W123 clipped to the box plus 2 km (Lang et al., 2023); NLCD 2021 from the MRLC web coverage service on its native 30 m EPSG:5070 grid, box plus 1 km (U.S. Geological Survey, 2023)	2	27
`hydrology`	S2	NHDPlus HR geodatabases (HU4 1703, 1708, 1711) and the flowlines extracted from them (U.S. Geological Survey, 2026b), NHDPlus v2.1 flowlines (nhd), clips of the two water-table grids (Ma et al., 2026b; Fan et al., 2017)	7	924
`imagery`	S2	Sentinel-2 L2A median composite, 1 August – 30 September 2025, bands B02, B03, B04, B08 and B11 at 20 m (European Space Agency, 2025)	1	96
`soilgrids`	S2	Window over the box plus 2 km of the SoilGrids250m 2017 absolute depth to bedrock (`BDTICM_M_250m_box.tif`) (Shangguan et al., 2017)	1	0.3

Source folder	Stage	Content	Files	Size (MB)
`vs30`	S2	Window over the box plus 0.05° of the USGS global hybrid Vs30 grid (`global_vs30_box.tif`) (Heath et al., 2020)	1	0.01
`wells`	S31	Washington Department of Ecology well reports (hydrogeological fields only) (Washington State Department of Ecology, 2026); USGS groundwater sites and water levels (U.S. Geological Survey, 2026d); Washington Geological Survey subsurface database (Washington Geological Survey, 2026d); records up to 23 September 2026	4	71
`regional`	S5	Cascadia velocity model v1.7, levels L01 and L2 (0–9.9 km below the ground), clipped to the box plus 2 km (`cvm17_domain.nc`) (Wirth Moriarty et al., 2025)	1	60
`pnsn`	S6, S8, S13, S26	ComCat phase-data QuakeML for 432 events (the 91 events of `configs/validation_events.csv`, of which 88 pass the pick criteria of the calibration, and the 2023–2025 events relocated by S26), the P and S picks extracted from them and station tables (U.S. Geological Survey, 2026a); PNSN events of magnitude 0.5 or more from 2015 to the data freeze (S8)	437	65
`sensors`	S8	FDSN station and channel metadata of all networks in the box (EarthScope Consortium, 2026) and of the whole 2025 node array, network Z5 (Schmandt, 2025); GNSS site metadata (ear); the box query is frozen at 23 September 2026 and named by domain and date, and two earlier unfrozen copies remain	5	2
`das`	S8, S10	Paradise–Nisqually Entrance DAS channel table, staged from a project machine	1	0.4

Source folder	Stage	Content	Files	Size (MB)
`overlays`	S8	KMZ overlays of the USGS I-432 geologic map and the park soil-map poster, staged from a project machine	2	26
`smart_sensing`	S8	Synoptic weather, SNOTEL and streamflow station catalogues of the mt-rainier-smart-sensing project, staged	66	12
`viewer_cache`	viewer	3DEP terrain and 1 m summit lidar, National Map imagery, EarthScope stations and ComCat events for the 3D viewer, frozen at 23 September 2026	51	443
`mrms`	S29	MRMS hourly precipitation, the 216 hours of the December 2025 storm (Zhang et al., 2016)	216	120
`usgs_nwis`	S29	USGS instantaneous discharge at the gauges of the box during the storm (U.S. Geological Survey, 2026e)	1	1
`seis_hydro_2_sed`	S29	Seismic discharge estimates and atmospheric-river windows (sei)	5	3
`gnss`	S17	76 UNR daily position series (`.tenv3`) (Blewitt et al., 2018), the PANGA raw archive and its horizontal and vertical NAM20 velocity fields (Pacific Northwest Geodetic Array, 2026), download manifest	80	218
`emc`	S21	iMUSH local-earthquake tomography of Ulberg et al. (2020), in the netCDF format of the EarthScope Earth Model Collaboration (EMC)	1	5
`rainier_aerogeophysics`	S22	1996 USGS helicopter survey (Rystrom et al., 2000): apparent-resistivity grids at 33 kHz, 4737 Hz, 4341 Hz and 837 Hz (50 m), reduced-to-pole magnetic grid (62 m), electromagnetic (EM) and magnetic flight lines	8	19
`wgs_landslides`	S24	Washington landslide inventory (Washington Geological Survey, 2026c): lidar-protocol deposits, recent landslides, compilation (one GeoPackage each)	3	3

Source folder	Stage	Content	Files	Size (MB)
`allstadt2017`	S24	Seismically recorded mass movements, western United States (`Events.csv`) (Allstadt et al., 2017)	1	0.03
`usgs_rainier_hazards`	S24	Lahar hazard zones of 1998 (`rainier_98shapefiles.zip`) (Hoblitt et al., 1998)	1	0.2
`dem_3dep_1m`	S24	1,638 3DEP 1 m windows around landslide polygons (U.S. Geological Survey, 2026c); one GeoPackage of 3DEP source footprints	1,639	1,818
`canopy_storage`	S28	GEDI L3 global 1 km grids of mean and standard deviation of relative height 100 and of shot counts, 18 April 2019 – 9 July 2025 (Dubayah et al., 2021a), and four crops to the box (adding the standard error), written by the vendored canopy-storage code	7	1,135

Four inputs are not fetched by script:

- **Cascadia velocity model v1.7** (Wirth Moriarty et al., 2025). ScienceBase blocks scripted downloads with a captcha, so the files are downloaded by hand once: levels L01 (0–1200 m below the ground, 200 m spacing) and L2 (1500–9900 m, 300 m spacing) are read from a local path, clipped to the box plus 2 km and cached as `data/raw/regional/cvm17_domain.nc` (S5); level L3 (10.8–59.4 km) is not downloaded.
- **CRESCENT Gen0.** The model file is read from a local copy of its figshare record (He, 2026). It supplies Vs and its uncertainty below the 9.9 km base of CVM v1.7, with Vp from the Vs relation of Brocher (2005), and its depth axis (-4 to 100 km) is read as kilometres below sea level.
- **PNSN one-dimensional model.** The table is read from a local file (`vel_pnsn_wa.csv` of the cascadia_obs_ensemble repository, registry key `pnsn_1d_wa`): seven layers with linear gradients, from Vp 5.40 km s^{-1} at the surface to 7.80 km s^{-1} at 41 km depth, with Vp/Vs 1.73. It is the reference of the S6 check. Which PNSN model it is (P3 Puget Sound or C3 Cascades) is not documented [TODO: confirm the model name with PNSN].
- **Canopy-storage products.** The lidar canopy and soil-map products are delivered files (Sect. 4.4), read from the paths listed in `configs/canopy_products.yaml` (delivered 24 September 2026); in the current cache only the GEDI L3 canopy height is regenerated by the vendored pipeline (S28).

The code names each of these paths, and their registry entries say how to obtain them.

3.2 Rebuilding the database and the model

From a clean clone, the following commands download every input, rebuild the model and check the result. Stages that need credentials or delivered files skip those layers and say so.

```
git clone https://github.com/Denolle-Lab/mt-rainier-digital-model && cd mt-rainier-digital-model
pixi install                        # the locked environment
pixi run all                        # S1–S8: surface, layers, geology, rock physics, fusion, checks,
↪  figures
pixi run s2 -- --ma                 # optional: the Ma et al. (2026) water table (~1 GB download)
pixi run s22 && pixi run s3 && pixi run s4 && pixi run s5   # alteration from the EM survey, then
↪  rebuild
pixi run s17 && pixi run s18        # GNSS positions, velocities, strain, edifice load
pixi run s24                        # mass movements and faults: catalogue, figure, viewer layers (~1.8
↪  GB of 1 m windows)
pixi run -e canopy canopy           # canopy-storage pipeline: GEDI (Earthdata login), Sentinel-2 LAI
↪  (CDSE client)
pixi run s9                         # uniform grids for ray tracing and location
pixi run s31                        # wells and boreholes: water levels, bedrock picks, comparison with
↪  the grids
pixi run platform                   # the 3D viewer bundle, deleted and rebuilt from data/raw and the
↪  model products
pixi run manifest -- --check        # compare the rebuilt cache with docs/data_manifest.csv
pixi run test                       # unit tests and the invariants of the built model
```

The calibration (S13, 29 min on a 10-core laptop) and the relocation comparisons (S14) are rerun as described in `docs/joint_calibration.md`.

The platform is rebuilt, never patched. `pixi run platform` (S32, `configs/platform.yaml`) deletes the viewer bundle and the atlas data, copies the four inputs that exist only on a project machine (the DAS channel table, two KMZ overlays and the Synoptic station catalogue) into `data/raw/`, and runs the viewer, sensor, layer, catalogue and event steps in order. Every live query in these steps stops at one data-freeze date (23 September 2026), so a rebuild from the same cache gives the same files; S32 writes their SHA-256 to `SHA256SUMS` in the bundle and packs it into a tarball with fixed entry order, owner and time. The model products (the subsurface model and the other stores of S2–S30) are read, not recomputed, and a missing one stops the rebuild.

3.3 Redistribution

The pipeline is published in full. Derived products are published unless a source forbids it.

- **Excluded products.** A variable whose provenance (its `gaia:source_keys` attribute) includes a source flagged `redistribute_derived: false` in `configs/sources.yaml` is left out of the downloadable archives by

S20 (`scripts/20_publish_products.py`), and is listed in the catalogue's `excluded` field and in the archive's `excluded_variables` attribute.

- **Flagged sources.** Four sources carry the flag (registry keys `soilgrids_bdticm`, `ma2026_wtd`, `canopy_lidar_chm` and `soil_map_image`):
  - the SoilGrids 2017 depth to bedrock (Shangguan et al., 2017), whose record states no licence;
  - the Ma et al. (2026) water table (Ma et al., 2026b), whose licence (CC-BY-NC-ND 4.0) forbids derivatives; the 3D viewer shows it as a visualisation only, and `pixi run s2 -- --ma` rebuilds it from the open Zenodo record;
  - the lidar canopy height and vegetation cover (can), made from the 2022–2023 "Wali" tiles of the Washington Department of Natural Resources Lidar Portal, whose terms of reuse are not yet confirmed;
  - the soil-map image (an RGB rendering in EPSG:3857 at 7.8 m, masked to the park), whose source, legend and licence are not documented.
- **Rebuilding the excluded layers.** Users rebuild them with their own access. Two open archives need a free login to download, NASA Earthdata for the GEDI L2B, L3 and L4B products and the Copernicus Data Space for the Sentinel-2 leaf area index. Their derived layers are published, since neither NASA nor Copernicus data carry a redistribution restriction.

`docs/data_policy.md` gives the tier of every source.

3.4 Code written with an AI assistant

The code, the configuration and the first drafts of this paper were written with Claude (Anthropic, model Claude Opus 5.5), used as a coding agent in Claude Code under the direction of the first author. The work was divided as follows.

- **The authors** designed the model; the assistant implemented it. The project started from D. Yao's three-dimensional mockup of the Rainier seismic network (terrain, summit lidar and earthquake layers; the first commits of the repository). From there the first author directed the whole design (the domain, the input data sets, the model units and their rules, the rock-physics law, the fusion and calibration strategy and the acceptance tests), collected verbal feedback from prospective users, and turned it into instructions for the assistant.
- **The assistant** wrote the Python package `rainier3d`, the stage scripts and the tests from instructions given in conversation, ran them, and reported the results. It also drafted the text, tables and figure scripts of this paper, and resolved the DOIs of the registry.
- **Aggregation into stages.** The assistant turned each task into a numbered script (`scripts/00_data_manifest.py` to `scripts/30_terrain_geometry.py`, stages S0 to S30) with a `pixi run` task. Shared code lives in the package, and each stage reads and writes the store through `rainier3d.io.store`. A new layer is therefore one script, one task and one test, not an edit of earlier stages.

Table 4. The development record, counted from the git history and from `docs/citations.csv`.

Record (git history of `main` at commit 70b2e72, 23–26 September 2026)	Count
Commits, excluding merges	125
Commits made in an assistant session (trailer `Co-Authored-By: Claude`)	90
Merged pull requests	29
Merged pull requests reviewed automatically by GitHub Copilot	28
Commits that answer an automated GitHub Copilot review of a pull request	29
Lines of Python in `src/rainier3d`, `scripts` and `tests`	8882, 5172 and 1501
Entries in the citation registry `docs/citations.csv`	129
Registry DOIs recorded as taken from the assistant's memory, not resolved	3
Registry sources with no identifier, recorded as not verified	4

The rules that the assistant followed are those of `AGENTS.md` at the root of the repository, which apply to human and agent contributors alike: one locked environment, one domain loader, a registry key for every number, nearest-neighbour resampling for categorical rasters, and `pixi run all` and `pixi run test` before every pull request, with tolerances never loosened to make a test pass. Two of the principles of Sect. 3 exist to catch errors of the kind a language model makes. The registry turns an invented number into a missing key or an `m1_placeholder` that can be listed. The invariant tests reject a built model that violates physical bounds or disagrees with the map or the regional model, whoever wrote the code.

The trailer marks commits made in an assistant session. It does not measure how much of a commit the assistant wrote, and commits without it may also contain assisted code. The three-dimensional viewer was written by D. Yao; one of his 25 commits carries the trailer. The three DOIs from memory are those of `brocher_2005`, `glathida` and `rgi60`; the four sources without an identifier are `pnsn_1d_wa`, `canopy_lidar_chm`, `sentinel2_lai_2023` and `soil_map_image`. Every other entry cited in this paper was resolved at doi.org, Crossref or DataCite, or read from a local file, except `synoptic_catalog`, whose verification field is empty.

The repository was created on 23 September 2026, and its full history, including every pull request and its review comments, is public. [TODO: the Claude Code session transcripts (four sessions, 23–29 September 2026) are the AI logs proper; state here whether and where they are deposited.]

The authors checked the assistant's work in three ways.

- **Rendering.** Each new layer was inspected in the three-dimensional viewer and in the figures, and layers derived from published maps and models were compared visually with the figures of their source publications.
- **Seismology.** The first author checked the event locations and the travel-time residuals of the calibration and relocation (Sects. 5.3.4, 9.2).

– **Code review.** After the first five commits (23–24 September 2026), every change reached `main` through a pull request. GitHub Copilot reviewed 28 of the 29 merged pull requests automatically; the first author read its comments, asked the assistant to address them, and merged after the assistant resubmitted.

4 Surface layers

4.1 Elevation, geology and glaciers

Three surface layers set the top of every model column.

– **Elevation.** USGS 3D Elevation Program (U.S. Geological Survey, 2026c) at 1 arc-second (about 30 m), fetched by S1 with py3dep over the box plus 0.02° and block-averaged onto the 100 m surface grid in EPSG:32610; S1 stops if any cell is left empty. It ranges from 24 to 4380 m in the box. The highest cell lies below the 4392 m of Columbia Crest because each cell is a 100 m average.

– **Geology.** The Washington Geological Survey 1:100,000 surface geology in the GeMS format (Washington Geological Survey, 2026a), queried by S1 from the Washington Department of Natural Resources feature service (layer 11, map-unit polygons intersecting the geographic box; layer 13, Description of Map Units) and cached as `data/raw/geology/dnr_gems_100k_map_units.gpkg`.

  – All 149 map symbols present in the box are assigned to 14 surface model units by 14 ordered regular-expression rules on the symbol (the first match wins), checked against each unit's full name in the map's Description of Map Units. Every cell receives a unit: S1 stops if a symbol matches no rule, and the polygons are rasterised at cell centres as uint8 without interpolation. The Ohanapecosh unit covers the largest share of the box (187,565 of 525,000 cells, 35.7%).

  – The rules are in `configs/units.yaml` (`crosswalk_rules`), and the generated crosswalk (`configs/crosswalk_geo` one entry per map symbol with its model unit and the age and full name from the Description of Map Units) is committed for review.

  – The park map of Fiske et al. (1963) (USGS I-432, sheet 1, a single 10,000 × 10,000 pixel KMZ ground overlay listed in `configs/overlays.yaml`) is carried as a georeferenced image for display in the viewers. It is read from a local copy, not fetched by script, and its units are not digitised.

– **Faults.** The same map gives 134 fault traces in the box (220 km), 106 of them high-angle dip-slip faults and 46 concealed. The Washington Quaternary fault layer adds four features (Washington Geological Survey, 2026b): the Western Rainier and Goat Rocks seismic zones, drawn as geophysical lineaments, the Devils Dream reverse fault and an unnamed oblique reverse fault. Neither source gives a dip, and no fault section of the 2023 National Seismic Hazard Model lies in the box (Hatem et al., 2021); the nearest, the Olympia and Tacoma faults, are 17 and 28 km outside it. Faults are therefore mapped traces (Fig. 24a) and are not surfaces in the 3D model.

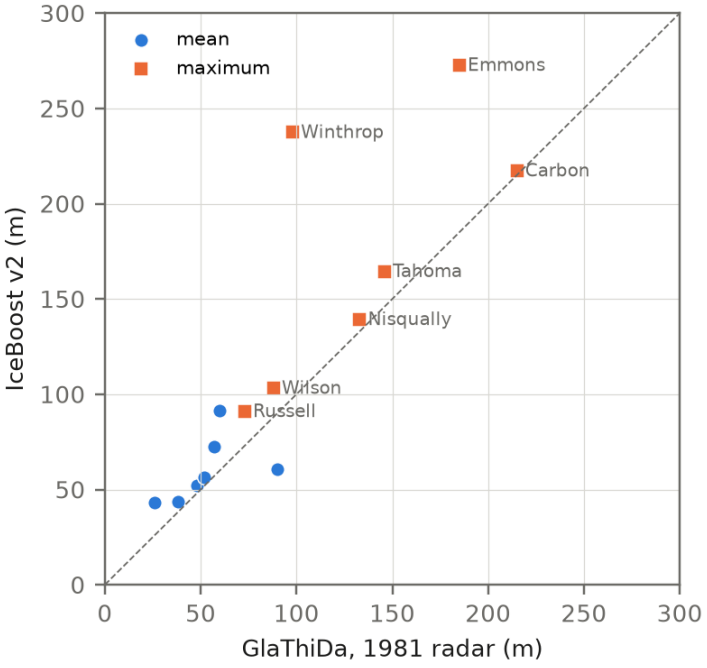


Figure 3. Mean (circles) and maximum (squares) ice thickness from IceBoost v2 against the 1981 ground-penetrating radar summaries in GlaThiDa, for seven Rainier glaciers. The dashed line is 1:1.

– **Glacier ice thickness.** IceBoost v2 per-glacier grids (Maffezzoli, 2025) for the 219 glaciers of the Randolph Glacier Inventory 6.0 (RGI Consortium, 2017) whose centroids fall in the box, 97.3 km^2 in all. The grids are area-averaged onto the 100 m grid, and each glacier is rescaled so that its volume matches the IceBoost total for it. This removes an 18% overestimate from counting partly glacier-covered cells as full. The total is 5.50 km^3, of which 5.32 km^3 is on the cone.

The glacier bed is the ground elevation minus the ice thickness. Against the 1981 radar surveys of Driedger and Kennard (1986) as archived in GlaThiDa (World Glacier Monitoring Service (WGMS), 2020), maximum thickness agrees for Carbon, Nisqually and Tahoma glaciers (Fig. 3). IceBoost is thicker on Emmons (273 vs 185 m) and Winthrop (237 vs 98 m) glaciers. Part of any difference reflects thinning since 1981 (Sisson et al., 2011).

4.2 Soil, water, land cover and imagery

Script S2 reads each source over the model box only, resamples it onto the 100 m grid and records its source (Table 5, Fig. 4). Continuous layers are averaged over each cell; categorical layers take the most common class.

Notes on individual layers:

– **SOLUS100.** It predicts soil depth only to 2.01 m (201 cm is the largest value of its centimetre grid over the box), so that value is read as “at least 2 m”; 3.5% of the valid cells reach 2.0 m. SOLUS100 does not document this cap.

Table 5. Environmental surface layers (S2); percentiles are over the cells of the box.

Layer	Source	Native resolution	Median (5th–95th percentile)
Soil thickness	SOLUS100 depth to a lithic contact (Nauman et al., 2024), read from its cloud-optimised GeoTIFF over the box and block-averaged; 98.8% of cells valid	100 m	1.41 m (0.54–1.97 m)
Water-table depth	Random-forest estimate for the conterminous US, mean (Ma et al., 2026b), block-averaged; not redistributed	about 24 m	10.1 m (4.8–20.4 m)
Water-table depth	Global groundwater model, annual mean (Fan et al., 2017), read through OPeNDAP, stored positive down and block-averaged	$30''$ (about 1 km)	23 m (0–306 m)
Streams	NHDPlus High Resolution, 60,174 flowlines, Strahler orders 1–8, with mean annual flow (U.S. Geological Survey, 2026b); each 100 m cell takes the highest order of any flowline touching it (0 = none)	1:24,000	–
Canopy height	ETH global canopy height 2020 (Lang et al., 2023), block-averaged, values outside 0–120 m masked; 98.4% of cells valid	10 m	30 m (9–45 m)
Land cover	NLCD 2021 (U.S. Geological Survey, 2023), most common class in each cell	30 m	evergreen forest in 72.5% of cells
NDVI	Sentinel-2 L2A median composite, 1 August – 30 September 2025 (European Space Agency, 2025); (B08 − B04)/(B08 + B04) from band reflectances averaged onto the grid; 99.8% of cells valid	20 m	0.85 (0.26–0.92)
NDSI	same composite; (B03 − B11)/(B03 + B11)	20 m	−0.48 (−0.60 to −0.24)

– **Sentinel-2 composite.** Scenes are searched in the Microsoft Planetary Computer catalogue, with Element84 Earth Search as fallback. It uses the least-cloudy scenes (at most 25% cloud), up to six per Sentinel-2 tile, so that every part of the box is covered; the cached composite holds 10 acquisition dates. The scene classification keeps vegetation, bare soil, water, snow and unclassified pixels (classes 4, 5, 6, 11 and 7), and the median is taken per pixel on a 20 m grid in EPSG:32610 (S2).

– **Reflectance offset.** Since processing baseline 04.00 (all scenes since January 2022), Level-2A reflectance is stored as reflectance × 10,000 + 1000, and neither distribution service removes the offset. S2 removes it before computing indices and stops if a window mixes baselines. Without the correction, median NDVI over this forest is 0.46 instead of 0.85 [TODO: record the command or log that gives 0.46].

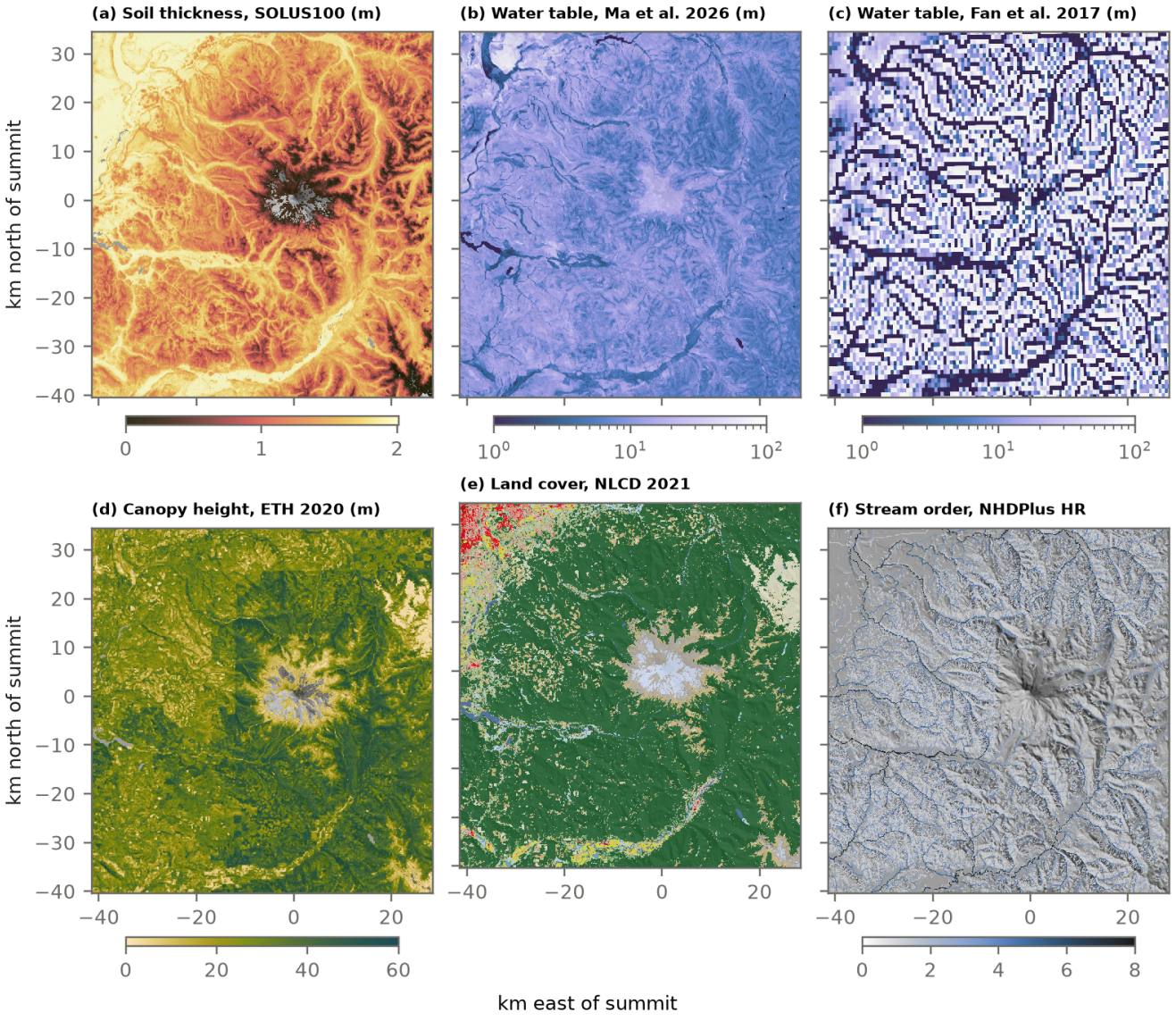


Figure 4. Environmental surface layers on the model grid, over a hillshade: (a) soil thickness, (b, c) water-table depth from two estimates on the same logarithmic scale, (d) canopy height, (e) land cover, (f) Strahler order of the NHDPlus HR flowlines. The rectangular step in (d) near 12 km west and 20 km north of the summit comes from the canopy product itself.

– **Snow and ice.** NDSI above 0.4 covers about 65 km^2 at the end of the 2025 melt season, against 97.3 km^2 of glacier in the Randolph inventory outlines (about 2000) of the 219 glaciers in the box. The difference is consistent with debris-covered glacier tongues (Carbon, Emmons, Winthrop), which read as rock in NDSI, and with glacier retreat.

Table 6. Soil, regolith and Vs30 layers (S2). Percentiles are over the cells of the box; the soil and regolith layers are empty on glacier ice.

Layer	Source	Native resolution	Median (5th–95th percentile)
Clay, sand and silt, 0–1 m	SOLUS100 (Nauman et al., 2024), depth-weighted mean over the soil column within 0–1 m; 97.1% of cells valid	100 m	clay 7.4% (3.0–18.7%), sand 62% (37–69%), silt 31% (27–47%)
Texture class	USDA class of the 0–1 m clay, sand and silt, rescaled to sum to 100%	–	sandy loam in 71.0% and loam in 27.0% of the cells with soil
Bulk density, 0–1 m	SOLUS100, oven dry	100 m	0.98 g cm^{-3} (0.88–1.33)
Rock fragments, 0–1 m	SOLUS100, fragments larger than 2 mm, by volume	100 m	22% (5–36%)
Organic carbon, 0–1 m	SOLUS100	100 m	2.2% (1.6–3.6%)
Depth to a restrictive layer	SOLUS100, any restriction	100 m	1.32 m (0.67–1.73 m)
Saturated hydraulic conductivity, 0–1 m	POLARIS v1.0 layer means (Chaney et al., 2019), block-averaged in log space (a geometric mean over the cell), then the thickness-weighted harmonic mean of the layers within 0–1 m; 97.4% of cells valid	30 m	1.6×10^{-5} m s^{-1} (4.8×10^{-6} to 5.6×10^{-5})
Porosity, 0–1 m	POLARIS saturated water content θs, thickness-weighted mean	30 m	0.65 (0.49–0.70)
Depth to bedrock	SoilGrids250m 2017 absolute depth to bedrock (Shangguan et al., 2017), interpolated bilinearly; not redistributed until the licence of the 2017 release is confirmed	250 m	19.6 m (11.4–26.8 m)
Vs30	USGS global hybrid map (Heath et al., 2020), interpolated bilinearly	30″ (about 900 m)	685 m s^{-1} (392–686 m s^{-1})

4.3 Soil properties, regolith and Vs30

S2 also reads the properties that a hydromechanical description of the critical zone needs: texture, density and coarse fragments from SOLUS100, hydraulic properties from POLARIS, the depth to bedrock from SoilGrids, and the Vs30 map of the USGS (Table 6). The depth profiles keep their depth axis in a `/soil` node of `model.zarr`: SOLUS at its seven prediction depths (0, 5, 15, 30, 60, 100 and 150 cm) and POLARIS in its six layers (0–5, 5–15, 15–30, 30–60, 60–100 and 100–200 cm). The `/surface` node holds their means over 0–1 m (`configs/soil.yaml`, `rainier3d.surface.soil`). S2 writes the numbers quoted below to `outputs/soil_layers_summary.json`.

The two estimates of depth to rock disagree by an order of magnitude (Fig. 5). SOLUS100 places the lithic contact at a median of 1.41 m (5–95%: 0.54–1.97 m) and cannot go deeper than 2.01 m (Sect. 4.2). SoilGrids places bedrock at a median

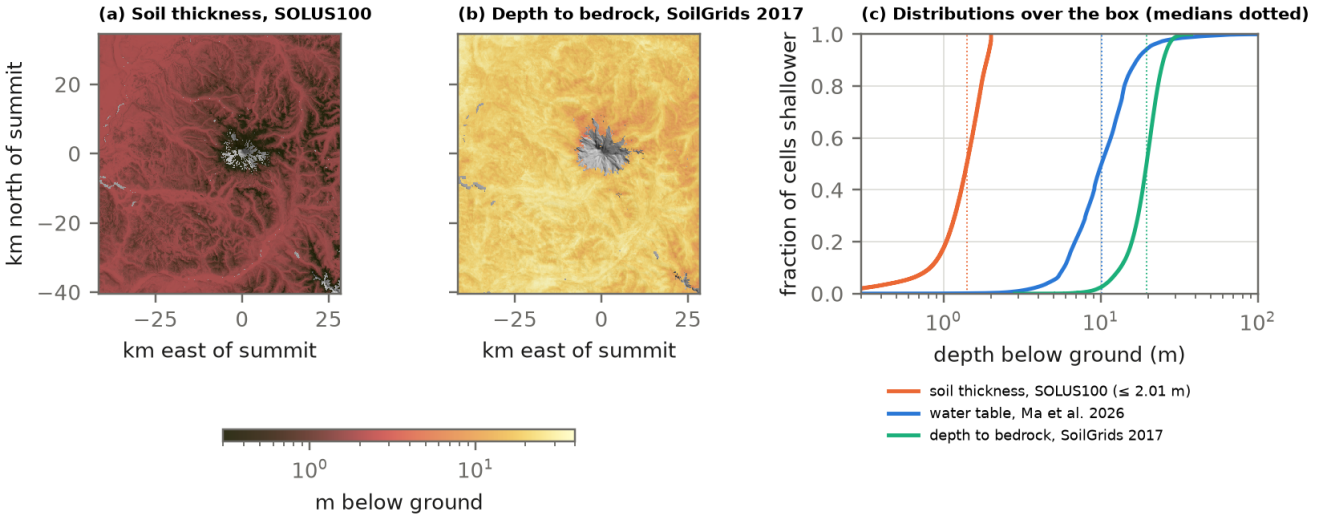


Figure 5. Depth to rock from two sources on one logarithmic scale: (a) SOLUS100 soil thickness (depth to a lithic contact, capped at 2.01 m); (b) SoilGrids250m 2017 depth to bedrock; (c) distributions over the box of the two depths and of the Ma et al. (2026) water-table depth, with medians dotted. Glacier cells are left empty. The order-of-magnitude gap between the two depths contains the water table.

of 19.6 m (11.4–26.8 m), deeper than 5 m in every cell, and 13.9 times the SOLUS depth at the median. The two products measure different things: SOLUS the soil profile described by the soil surveys, SoilGrids an absolute depth to the R horizon from a global model trained mainly on well logs, which are sparse on volcanic edifices. Neither is checked against local data on the volcano; in the lowland, the 23 bedrock picks of the borehole database lie deeper than both (median 40.5 m, Sect. 7). The disagreement matters because the water table falls between them: the Ma et al. (2026) water table (median 10.1 m) lies below the SOLUS soil and above the SoilGrids bedrock in 92.8% of the cells. The unsaturated zone and the top of the saturated zone are thus in the layer that no data set in the box constrains, between 2 m and the 50 m cells of L1. The model keeps both estimates and chooses neither.

Notes on these layers:

- **Restrictive layers.** In 42.6% of the cells the restrictive layer lies more than 10 cm above the lithic contact.
- **Porosity from two sources.** Porosity from the SOLUS bulk density and a grain density of 2.65 g cm^{-3} ($1 - \rho b/\rho s$, an `m1_placeholder` value) has a median of 0.63, against 0.65 for POLARIS θs; the two correlate at r = 0.68 over the cells with soil.
- **Vs30.** Two values cover 56.9% of the box: 686 m s^{-1} (50.0%) and 464 m s^{-1} (6.9%). These are class values of the hybrid map, not local estimates, so the layer is a reference for comparison with the near-surface velocities of the model and not a constraint.

Table 7. Vegetation layers (S19). The lidar is the 2022–2023 "Wali" acquisition of the Washington Department of Natural Resources Lidar Portal, whose terms of reuse are not yet confirmed; the provenance of a soil-map image delivered with these products is undocumented. Both are displayed in the 3D viewer and excluded from the downloadable products.

Layer	Source	Native resolution	Median (5th–95th percentile)
Canopy height	airborne lidar, digital surface minus terrain model (can); 10 m grid block-averaged per cell, values outside 0–110 m masked	10 m	14.6 m (0.3–32.4 m), on 49% of the box
Vegetation cover	same lidar, vegetation cover fraction block-averaged per cell	10 m	0.88 (0.01–1.00), on 49% of the box
Leaf area index	Sentinel-2 L2A, SNAP biophysical processor, 1 July – 15 August 2023 (European Space Agency, 2023); 10 m mosaic block-averaged; 99.8% of cells valid	10 m	2.36 (0.40–3.68)
Plant area index	GEDI L2B version 2, quality-filtered footprints gridded to 1 km (mean per cell) (Dubayah et al., 2021b); bilinear resampling; 95.9% of cells valid	1 km	2.72 (0.86–3.79)
Canopy height	GEDI L3 version 2, mean relative height 100, 18 April 2019 – 9 July 2025 (Dubayah et al., 2021a); bilinear resampling	1 km	25.5 m (13.3–39.9 m)
Aboveground biomass	GEDI L4B version 2.1, with standard error (median 21.4 Mg ha^{-1}) (Dubayah et al., 2025); bilinear resampling	1 km	198 Mg ha^{-1} (58–436)

– **Glaciers.** On the glacier cells of S1 (2.6% of the box), SOLUS returns surface values on 68%, POLARIS on all of them, and SoilGrids a median depth to bedrock of 11 m. S2 leaves the soil and regolith layers empty there (`mask_glaciers` in `configs/soil.yaml`, an author choice).

4.4 Vegetation structure

Script S19 adds the vegetation products of the canopy-storage project (M. Köpfli, University of Washington) on the same grid (Table 7, Fig. 6). They describe the canopy that intercepts precipitation and loads the ground: height, cover, leaf and plant area, and aboveground biomass.

The two canopy heights differ by design. The lidar value is the mean of 10 m cells in each 100 m cell, including gaps, while the GEDI value is a 1 km mean of the tallest return per footprint. That is why the lidar median (14.6 m) is lower than the GEDI median (25.5 m) and the ETH median (30 m).

The project's download and gridding code is part of this repository, unmodified and under its MIT licence (`third_party/canopy-sto`). Script S28 runs it inside the raw-data cache: it fetches the GEDI L3 grids and L2B footprints through NASA Earthdata and

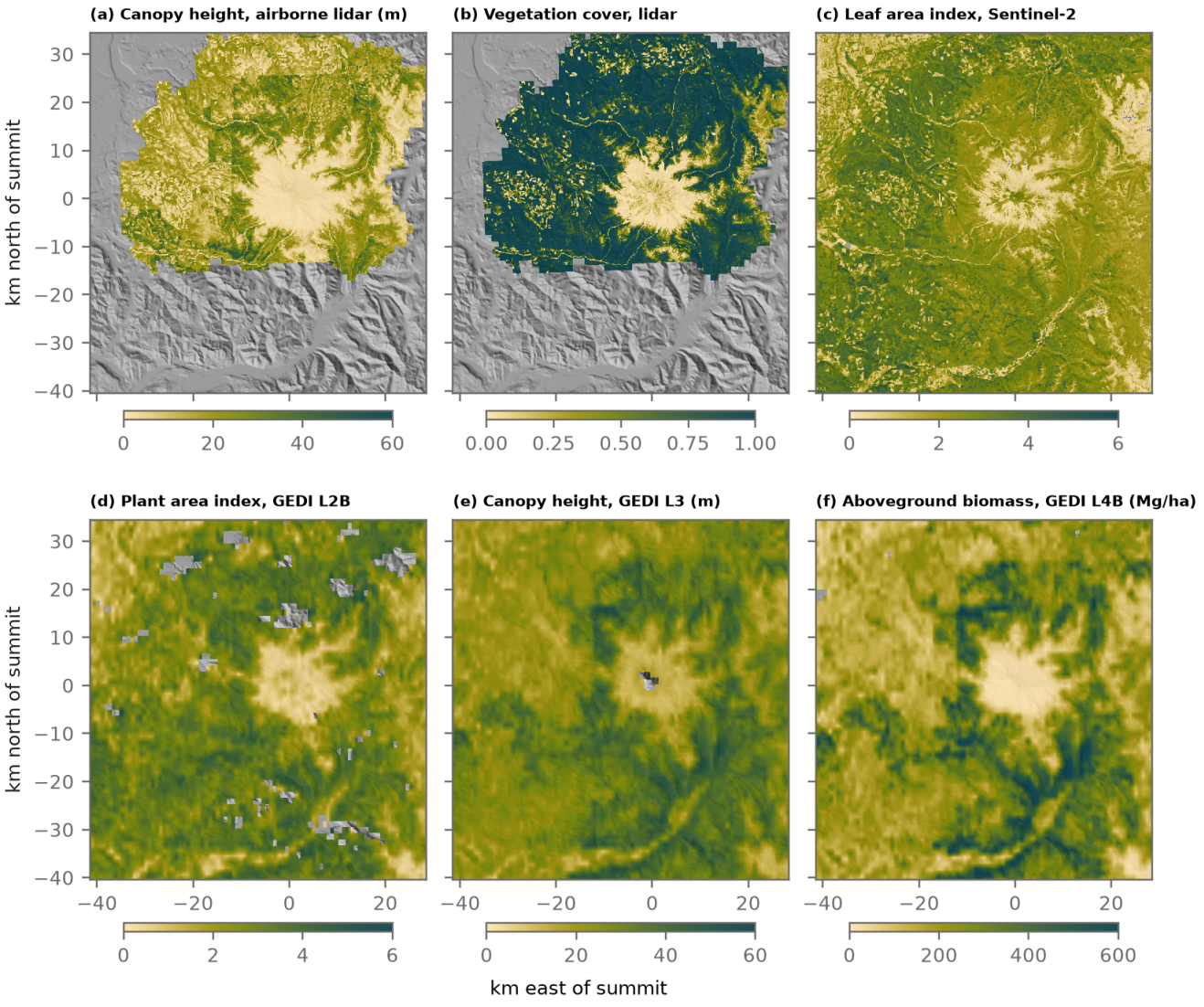


Figure 6. Vegetation layers on the model grid, over a hillshade: (a) lidar canopy height and (b) vegetation cover, (c) Sentinel-2 leaf area index, (d) GEDI plant area index, (e) GEDI canopy height, (f) GEDI aboveground biomass. Straight edges in (a) and (b) are lidar tile boundaries.

the Sentinel-2 leaf area index through the Copernicus Data Space, and S19 reads its products in place of the delivered files. The GEDI L3 canopy height it produces is identical to the delivered grid (33,701 cells). The L2B plant area index, the L4B biomass, the lidar layers and the soil map are still read from the delivered files: the vendored gridder writes the maximum plant area index rather than the mean, and the others were made outside that code. The project's lidar scripts compute canopy height only at its stations; the 10 m lidar grids were made from the same tiles in QGIS. `docs/canopy_pipeline.md` lists each service call.

4.5 Terrain geometry

Script S30 derives four layers of terrain geometry that control surface instability, as specified by co-author S. Han (`configs/terrain.yaml`, Fig. 7). They have their own 30 m grid in UTM zone 10N, 2,333 × 2,500 cells from the southwest corner of the box. The box is 70 km wide, so a 10 m strip at its east edge is not a whole cell and is left out.

- **Inputs.** Surface elevation z is USGS 3DEP fetched at 30 m (U.S. Geological Survey, 2026c) and block-averaged onto the grid. Over glaciers it is the elevation of the ice surface. Ice thickness H is the IceBoost v2 layer of the model surface (Maffezzoli, 2025), interpolated bilinearly from 100 m to 30 m; the ice volume is unchanged (5.50 km^3). Bedrock elevation is $b = z - H$.
- **Surface slope.** $\arctan |\nabla z|$, with the gradient from central differences between the two neighbours of each cell, a 60 m baseline. Over glaciers it is the slope of the ice surface.
- **Bedrock slope.** The same operator applied to b. Where neither the cell nor its eight neighbours carry ice, it equals the surface slope exactly.
- **Local relief.** The maximum minus the minimum of z within a disk of radius 510 m (17 cells) centred on each cell.
- **Valley depth.** The black top-hat of z: the grey-scale closing of z (a dilation, then an erosion, with a flat disk of radius 990 m, 33 cells) minus z. It measures incisions narrower than the disk, about 2 km; wider valleys are not filled and read as shallow.
- **Edges.** Every filter mirror-reflects the grid at its edges.

The layers, the ice thickness and the bedrock elevation are written to `data/processed/terrain_geometry.zarr`, and the four layers appear in the 3D viewer under "Terrain geometry". Three limitations follow from the inputs. The effective resolution of the 3DEP product at 30 m is 35–49 m, which smooths cliffs and headwalls and lowers their slope; its reprojection also leaves a faint diagonal striping in the slope maps. The bedrock slope inherits the errors of the IceBoost thickness and the blur of its 100 m grid, and the ice surface of 3DEP and the IceBoost thickness do not refer to the same year. The radii are parameters and can be changed in `configs/terrain.yaml`.

Table 8 samples the layers at the 1,649 event points of the mass-movement catalogue (Sect. 8), which are the crowns of mapped landslides and the locations of the other events, and in the cells covered by its flow polygons. The reference is every cell of the grid. Events sit on steeper ground than the box as a whole: their median slope is 27.9° against 18.5°, and 42% of them lie on slopes above 30° against 22% of the cells. Rock falls and rock and ice avalanches are the steepest class, with a

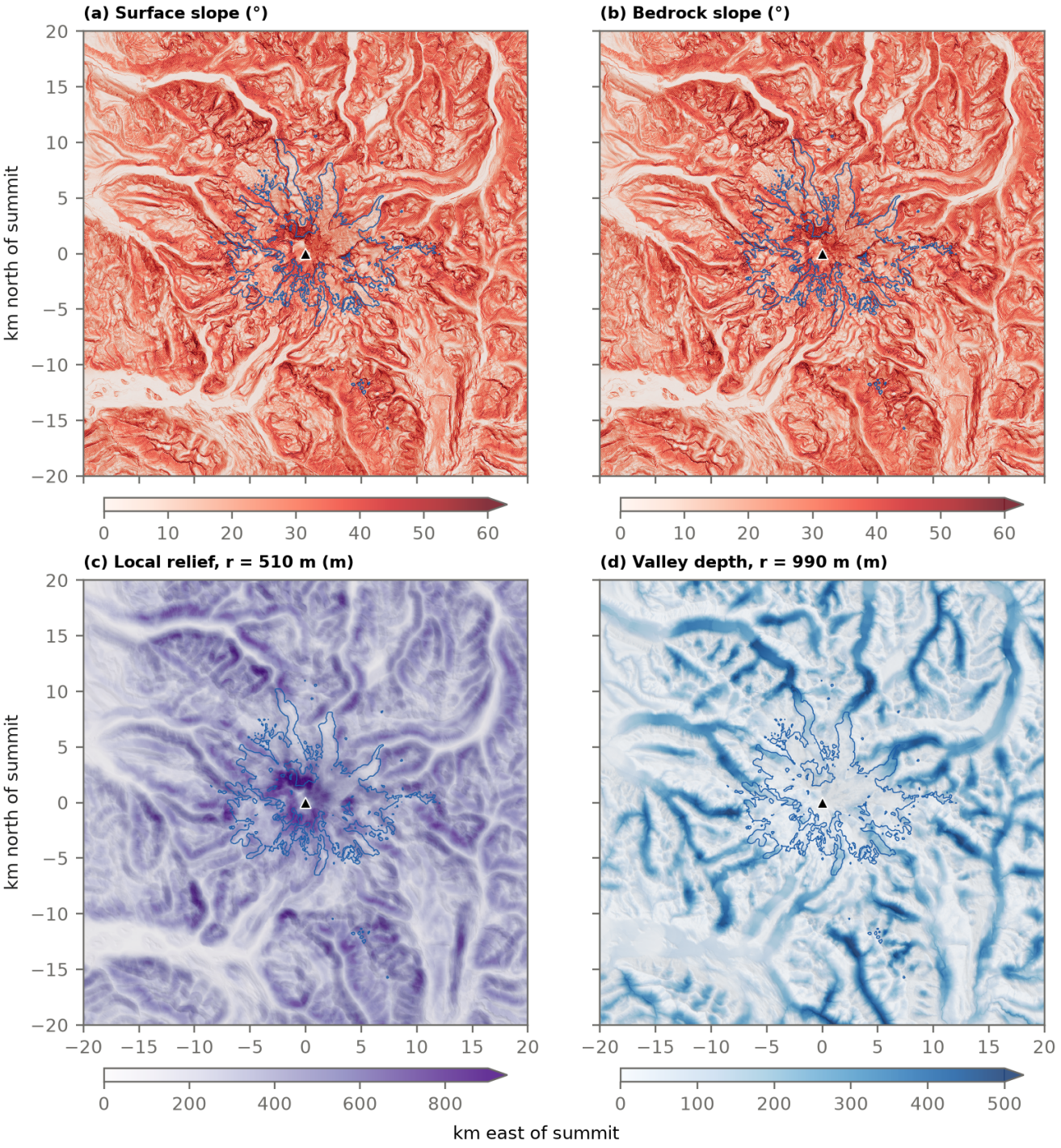


Figure 7. Terrain geometry over 40 × 40 km centred on the summit (triangle), on the 30 m grid (S30): (a) surface slope, arctan |∇z| from central differences between adjacent cells (60 m baseline), the slope of the ice surface on glaciers; (b) bedrock slope, the same operator on the 3DEP elevation minus the IceBoost v2 thickness (bilinear from 100 m), identical to (a) off the ice; (c) local relief, maximum minus minimum elevation within a disk of radius 510 m; (d) valley depth, the grey-scale closing of the elevation with a disk of radius 990 m minus the elevation. Filters mirror-reflect at the grid edges. The blue line outlines ice thicker than 10 m. The 35–49 m effective resolution of the source smooths cliffs, the bedrock slope inherits the errors of the ice thickness, and the radii are adjustable.

Table 8. Terrain geometry at the mass movements (S30, `outputs/terrain/mass_movement_terrain.csv`): median (10th–90th percentile) and the fraction of sites above 30°. The fractions above 300 m of local relief and 50 m of valley depth are in the file; these thresholds are author judgement. The two snow or ice avalanches and the single seismically recorded debris flow are in the file.

Sites	Count	Surface slope (°)	> 30°	Local relief (m)	Valley depth (m)
All cells of the grid	5,832,500 cells	18.5 (2.2–36.0)	22%	297 (78–505)	54 (1–237)
All events	1,649	27.9 (14.9–39.9)	42%	334 (176–467)	50 (0–203)
Rock fall, rock and ice avalanche	56	38.8 (22.3–51.0)	75%	380 (98–787)	89 (4–276)
Slide, debris slide	323	29.3 (16.1–40.2)	48%	340 (143–462)	40 (0–161)
Complex or unknown type	1,267	27.2 (14.5–38.9)	40%	331 (185–459)	52 (1–208)
Seismically recorded	19	48.2 (36.5–48.2)	100%	787 (646–787)	64 (0–98)
Debris flows (mapped)	463 polygons, 12,292 cells	12.6 (3.6–26.2)	6%	190 (124–329)	52 (7–140)
Osceola Mudflow	88,559 cells	2.1 (0.3–21.7)	4%	98 (8–288)	44 (1–216)
Electron Mudflow	20,833 cells	0.7 (0.3–4.6)	0%	45 (7–114)	9 (0–78)
Other lahar deposits	17,777 cells	5.3 (2.0–16.0)	1%	196 (88–307)	182 (54–337)

median of 39.8° and 77% above 30°, and the highest local relief, 380 m. Local relief separates the events from the box less clearly than slope does (median 334 m against 297 m), and valley depth does not separate them at all (50 m against 54 m): crowns lie on ridges and valley walls as often as in incised channels. The flow polygons record runout rather than source. The mapped debris flows have a median slope of 12.6°, the lahar deposits 0.7–5.3°, and the deposits other than the Osceola and Electron mudflows fill valleys, with a median valley depth of 182 m. Only 0.8% of the events lie on ice thicker than 10 m, so the bedrock and surface slopes agree at almost every event. They differ at the two snow and ice avalanches (median 37.8° on the ice surface, 34.8° on the bed). The 19 seismically located events fall on 7 distinct points, 9 of them on one, so their percentiles repeat.

5 Subsurface model

The subsurface model has three parts that share one water table, one pore pressure and one set of rock laws. The petrophysical models (Sect. 5.1) map a unit, its depth, the effective stress and the saturation to Vp, Vs and density. The critical-zone model (Sect. 5.2) resolves the top 150 m below the ground or the glacier bed in fine columns. The crustal model (Sect. 5.3) gives the three-dimensional geology and alteration of the volcano and its basement, fused with the regional velocity models and calibrated on PNSN travel times.

The model is built in four stages, each recorded in the repository with its parameters and sources (`docs/model_construction.md`). First, mapped geology is extended to depth by stated column rules (Sect. 5.3.1), which produce flat-lying contacts and vertical unit boundaries by construction. Second, every cell receives Vp, Vs and density from one crack-closure rock law at its effective pressure, and the top 150 m below the ground or glacier bed are built as a critical zone from the same law blended with a granular frame at the same effective stress (Sects. 5.1, 5.2), so that the near surface is petrophysically consistent with the rock beneath it. Third, the geology-based model is fused in ln V with the USGS Cascadia velocity model and CRESCENT Gen0, which supply wavelengths longer than 6–20 km; before fusion the rule contacts below the critical zone are low-passed with cutoffs that grow with depth, and three global rock-physics factors and a regional correction are calibrated against PNSN travel times (Sects. 5.3.3, 5.3.4). Fourth, a local-earthquake tomography perturbs the calibrated model at wavelengths of about 12 km and larger below the critical zone, which it leaves unchanged (Sect. 5.3.7, Supplement S1); its result is distributed beside the model described here, not in place of it.

5.1 Petrophysical models

Two laws map a unit, its depth, the effective stress and the water content to Vp, Vs and density: crack closure for rock under confining pressure (Sect. 5.1.1) and a granular frame for soil, Quaternary cover and weathered rock (Sect. 5.1.2). Both use one water table and one pore pressure, so they meet without a jump where weathering ends; their overburden is not yet the same (Sect. 5.1.3).

5.1.1 Rock under confining pressure

Each rock unit receives a P-wave speed that increases with effective pressure as cracks close:

$$V_P(P) = V_\infty - (V_\infty - V_0)\, e^{-P/P^*}, \qquad P = \rho_b\, g\, d - \rho_w\, g\, \max(d - z_w,\, 0), \tag{1}$$

with bulk density ρ_b = 2500 kg m^{-3}, water density ρ_w = 1000 kg m^{-3}, d the depth below the local ground and z_w the depth of the water table of Sect. 5.1.2 (hydrostatic pore pressure below it, none above).

- **Vp/Vs and density.** Ice, open water and the three unconsolidated deposits carry their own Vp/Vs and density in `configs/petrophysics.csv`, partly because the Brocher relations are not valid below Vp = 1.5 km s^{-1}. Where a unit has no ratio or density of its own, both follow Brocher (2005): Vs from his regression fit (his Eq. 6), $V_S = 0.7858 - 1.2344V_P + 0.7949V_P^2 - 0.1238V_P^3 + 0.0064V_P^4$, and density from the Nafe–Drake fit (his Eq. 1), $\rho = 1.6612V_P - 0.4721V_P^2 + 0.0671V_P^3 - 0.0043V_P^4 + 0.000106V_P^5$, with V in km s^{-1} and ρ in g cm^{-3} (valid for 1.5 < Vp < 8.5 km s^{-1}; S4, `rainier3d.petro.relations`, `rainier3d.properties.assign`).
- **Attenuation.** $Q_S = 0.05\,V_S$ (Vs in m s^{-1}) and $Q_P = 2\,Q_S$ (`q` block of `configs/perturbations.yaml`, `m1_placeholder`); S5 recomputes both from the fused Vs. Because $Q_P = 2Q_S$ implies a negative bulk quality factor where Vp/Vs < $\sqrt{(8/3)} \approx 1.63$, the 18,544 of 2.24 million non-ice fused cells below that ratio carry an inconsistent Q. The rule has the form proposed by Olsen (2003) for the Los Angeles basin.

- **Parameters.** Table 9 lists the unit parameters, chosen within published ranges. Every row of `configs/petrophysics.csv` has the source key `m1_placeholder` and a one-line basis (for example, granodiorite and quartz diorite for the plutons). The calibration of Sect. 5.3.4 scales them with three global multipliers that act on the eleven rock units, the magma body included, and leave ice and the unconsolidated deposits unchanged (block `geology` of `configs/velocity_calibration.yaml`). The contrasts between units therefore come from the table, and their overall level comes from the travel times. [TODO: the published range each table value was taken from, per unit]
- **Magma body.** Inside it, the middle-crust parameters apply (V_0 = 6.0 km s^{-1}, $V\infty$ = 6.5 km s^{-1}, P* = 50 MPa; row `magma_mush` of `configs/petrophysics.csv`). Its anomaly is not a property of the rock law: Vp, Vs and density of the fused model (Sect. 5.3.3) are reduced by up to 10%, 15% and 3% (`magma_mush` block of `configs/perturbations.yaml`, `m1_placeholder`), which raises Vp/Vs by about 6%, with a weight of 1 in the core of the body and a cosine taper to 0 at its edge (Sect. 5.3.1). It is applied after the fusion: applied before it, as in earlier versions of the model, the anomaly was largely removed by the regional low-pass and kept only at the edges of the body.
- **Alteration.** The alteration intensity a (0 to 1, from S22 in Sect. 5.3.2; zero in air and ice) scales Vp by $(1-0.30a)$, Vs by $(1-0.35a)$ and density by $(1-0.12a)$, within the ranges reported for altered volcanic rock (Heap and Violay, 2021). Fully altered rock is thus 30% slower in Vp, 35% slower in Vs and 12% less dense, with Vp/Vs about 8% higher. S4 applies the factors after the crack-closure law (`alteration` block of `configs/perturbations.yaml`, `m1_placeholder`; the file notes the values are still to be sourced).

5.1.2 Granular and partly saturated media

Soil, Quaternary cover and weathered rock are described by a granular frame whose stiffness depends on effective stress and water content. It is weighted against the rock law by the weathering index W of the critical-zone columns (Sect. 5.2).

Water and effective stress. Pressure head h is hydrostatic below the water table of Fan et al. (2017), which is placed at the glacier bed under ice. (The estimate of Ma et al. fits the wells better, Sect. 7, but its licence forbids derivatives.) Above the water table the suction head $\psi = -h$ sets the effective saturation by the retention curve of van Genuchten (1980),

$$S_e = [1 + (\alpha\psi)^n]^{-(1-1/n)}, \tag{2}$$

with α and n from POLARIS in soil and placeholder classes below. The effective stress is

$$\sigma' = \sigma_v - u, \qquad u = \begin{cases} \rho_w g h, & h > 0, \\ -\rho_w g \psi S_e W, & h \le 0, \end{cases} \tag{3}$$

where σ_v is the overburden of the column, including the weight of glacier ice, and the second line is the suction stress of Lu et al. (2010) with $\chi = S_e$ (Lu and Likos, 2006), which stiffens the unsaturated granular frame. The crack-closure pressure P of Eq. 1 in every L1 to L3 cell uses the same water table.

Table 9. Unit parameters of the crack-closure law (`configs/petrophysics.csv`). For the rock units (Rainier andesite to middle crust, and the magma body), the calibration multiplies V_0 by 1.44 (capped at 0.98 $V\infty$), P* by 1.42 and Vs by 0.951. Ice and the unconsolidated deposits keep their table values.

Unit	V_0 (km s^{-1})	$V\infty$ (km s^{-1})	P* (MPa)	Vp/Vs	Density (g cm^{-3})
Ice	3.80	3.80	–	1.98	0.917
Glacial drift	1.60	2.40	5	2.50	1.95
Alluvium, colluvium	1.50	2.20	5	2.80	1.90
Lahar deposits	1.80	2.80	8	2.30	2.00
Rainier andesite	2.80	5.60	30	Brocher	Nafe–Drake
Young volcanic rocks	3.00	5.60	30	Brocher	Nafe–Drake
Miocene intrusive rocks	4.50	6.20	40	Brocher	Nafe–Drake
Miocene volcanic rocks	3.20	5.80	40	Brocher	Nafe–Drake
Ohanapecosh Formation	3.40	5.90	40	Brocher	Nafe–Drake
Eocene volcanic rocks	3.50	6.00	40	Brocher	Nafe–Drake
Puget Group, Eocene sedimentary rocks	2.60	5.20	40	Brocher	Nafe–Drake
Mashel Formation	2.20	4.50	30	Brocher	Nafe–Drake
Russell Ranch Formation	4.00	6.00	40	Brocher	Nafe–Drake
Middle crust	6.00	6.50	50	Brocher	Nafe–Drake

Granular frame. The dry frame of soil, cover and weathered rock follows the soft-sand model of Dvorkin and Nur (1996): at the critical porosity ϕ_c = 0.40 a random pack of spheres has the Hertz–Mindlin moduli (Mindlin, 1949; Walton, 1987)

$$K_{\mathrm{HM}} = \left[\frac{C^2(1-\phi_c)^2\mu_s^2}{18\pi^2(1-\nu_s)^2}\,\sigma'\right]^{1/3}, \qquad \mu_{\mathrm{HM}} = \frac{2+3f_s-\nu_s(1+3f_s)}{5(2-\nu_s)}\left[\frac{3C^2(1-\phi_c)^2\mu_s^2}{2\pi^2(1-\nu_s)^2}\,\sigma'\right]^{1/3}, \tag{4}$$

with coordination number C = 9 and slip fraction f_s = 0.5, which reduces the shear stiffness that the pure Hertz–Mindlin model overpredicts (Makse et al., 1999). Below ϕ_c the frame joins the mineral along the modified Hashin–Shtrikman lower bound (Dvorkin and Nur, 1996; Mavko et al., 2020). The grain moduli K_s, μ_s and Poisson's ratio ν_s are the Hill average (Hill, 1952) of clay (21 and 7 GPa) and quartz–feldspar (40 and 25 GPa) for the SOLUS100 clay fraction. The pore fluid has the patchy-mixing modulus of Brie et al. (1995), $K_f = (K_w - K_a)S_w^b + K_a$, with water saturation S_w and b = 3, and the saturated bulk modulus follows Gassmann's equation (Mavko et al., 2020),

$$K_g = K_d + \frac{(1-K_d/K_s)^2}{\phi_g/K_f + (1-\phi_g)/K_s - K_d/K_s^2}, \qquad \mu_g = \mu_d, \tag{5}$$

valid at seismic frequencies, which lie in the low-frequency limit of Biot's theory (Biot, 1956). All rock-physics parameters are `m1_placeholder` values within the ranges of Mavko et al. (2020).

One medium from soil to rock. At each depth the rock end member is the crack-closure law of the unit beneath, evaluated at the same effective stress σ'. The bulk and shear moduli of the medium are the Hill average of the granular and rock moduli, weighted by W,

$$M = \tfrac{1}{2}\left[WM_g + (1-W)M_r\right] + \tfrac{1}{2}\left[\frac{W}{M_g} + \frac{1-W}{M_r}\right]^{-1}, \qquad \rho = W\rho_g + (1-W)\rho_r, \tag{6}$$

for $M = K, \mu$, so that velocity varies continuously with effective stress and saturation from soil to fresh rock, and where W = 0 the medium is exactly the rock law of Sect. 5.1.1. Permeability blends in the same way, from the conductivity of POLARIS and the cover and weathered classes to the crustal permeability–depth curve of Ingebritsen and Manning (1999) in rock, capped at 10^{-12} m^2.

5.1.3 Shared inputs of the two laws

The two laws are built to agree where they meet, and the parts of the model share their inputs:

- **One pore pressure.** Both laws use the same water table, placed at the glacier bed under ice, so the crack-closure pressure P of Eq. 1 in every L1–L3 cell and the effective stress σ' of Eq. 3 in the columns share one pore pressure. The overburden is not yet shared: the columns integrate their own densities, ice at 917 kg m^{-3} included, while Eq. 1 uses a constant ρ_b = 2500 kg m^{-3} from the ground or ice surface, so ice is loaded as rock in the crustal cells.
- **One rock law.** Where W = 0 the medium is the rock law of Sect. 5.1.1, unchanged. The calibration multipliers of Sect. 5.3.4 act on that law, so they change the crustal cells and the rock end member of every column together.
- **Continuity.** At the base of the columns (150 m), Vs is within a median 0.5% (90th percentile 1.4%) of the L1 cell below.

Table 10. Layers of the critical-zone columns. The factor f carries the weathering drivers listed below.

Layer	Base	Source
Soil	z_1: SOLUS100 soil thickness, at most 2.01 m	Sect. 4.3
Quaternary cover (drift, alluvium, lahar deposits)	z_2: the deposit thickness of the map unit	Sect. 5.3.1
Weathered rock	$z_3 = \max(z_2 + 2\ \mathrm{m},\ f\,D_{\mathrm{SG}})$	SoilGrids 2017 depth to bedrock D_{SG}
Fractured rock	$z_4 = z_3 + 15\ \mathrm{m}$	`m1_placeholder`

– **What is not yet consistent.** The overburden of the crustal cells (above). The quality factors follow a rule of their own (Sect. 5.1.1). Cell averages over a saturated weathered layer reach Vp/Vs above 3 in the top L1 cell (Sect. 5.2). The water table is that of Fan et al. (2017), not a solution of a groundwater model on these columns.

5.2 The critical-zone model

The crack-closure law of Sect. 5.1.1 describes rock under confining pressure. In the top tens of metres the medium is soil, glacial and volcanic deposits and weathered rock, partly saturated, whose stiffness depends on the water content as much as on depth. The model resolves this critical zone with fine columns and joins them to the rock below with one rock-physics law, so that the near surface and the volcano share one effective stress and one water table (S4, `rainier3d.cz.medium`, `rainier3d.cz.level`, `configs/cz.yaml`). The construction follows the geophysical studies of the critical zone in granitic and volcanic terrain (Holbrook et al., 2014; St. Clair et al., 2015; Flinchum et al., 2018; Riebe et al., 2016; Karlstrom et al., 2025) and the rock-physics chain used for surface waves in partly saturated soil (Solazzi et al., 2021; Shi et al., 2026).

Columns and layers. Every L1 cell (250 m) carries a column from the ground, or from the glacier bed, down to 150 m, in 52 layers growing geometrically from 5 cm to 16 m. Each column has four layers whose bases are $z_1 < z_2 < z_3 < z_4$:

Weathering index. A weathering index W is 1 in soil, cover and weathered rock and falls linearly to 0 across the fractured zone:

$$W(z) = \min\left[1,\ \max\left(0,\ \frac{z_4 - z}{z_4 - z_3}\right)\right]. \tag{7}$$

It sets the porosity, $\phi = W\phi_g + (1 - W)\phi_r$, with granular porosity ϕ_g from the SOLUS100 bulk density in soil, 0.35 in cover and 0.30 to 0.15 down the weathered layer, and a fracture porosity ϕ_r = 0.03 in rock (`m1_placeholder`).

Weathering drivers. The factor f on the SoilGrids depth encodes three controls, all `m1_placeholder` values to be replaced by calibration:

– glaciation: f = 0 under present ice, where the column starts with fractured rock at the bed; young glacial drift and alluvium have f = 0.5 and lahar deposits 0.3, since deposition restarts the weathering clock (Wang et al., 2020);

- lithology: consolidated rock at the surface (Rainier andesite, the other volcanic rocks, the plutons and the supracrustal units) has $f = 0.5$, because the SoilGrids prior is trained largely on deeply weathered profiles;
- forest: f is multiplied by 1.2 under canopy taller than 20 m, for root-driven weathering.

On the 23 borehole bedrock picks of the Puget lowland (S35), none of three alternative estimators (a terrain regression, the curvature regression of Flinchum et al. (2025), and SoilGrids itself) beats a constant median, so SoilGrids remains the prior.

Upscaling into the levels (Fig. 8). An L1, L2 or L3 cell that reaches into the columns takes the travel-time average of their Vp and Vs and their mean density; the top rock cell of a column averages from the ground (or the glacier bed) to its base, including the part of the column that lies in the air cell above it, and an invariant test requires the vertical S travel time of the top two rock cells to equal that of the columns to 10^{-3}; the columns are stored as the `/cz` node of `model.zarr` with the van Genuchten parameters, saturation, effective stress, porosity and permeability, the state on which a groundwater model and a data-assimilation scheme can later run. In the top 50 m of L1 the median Vs is 0.49 of the rock value without the critical zone, and at the base of the columns Vs is within 1.5% (median 0.5%, 90th percentile 1.4%) of the L1 cell below. Because the travel-time average is dominated by the slowest layers, the top L1 cell of a column whose weathered layer is saturated reaches Vp/Vs above 3: 2,542 cells in unconsolidated units (at most 4.70, slightly above the 4 to 4.5 measured in saturated loess by Pasquet et al. (2015)) and 9,331 cells (0.56% of the consolidated-rock cells, at most 3.97) in consolidated units.

Calibration with the critical zone. The rock calibration of Sect. 5.3.4 was refitted with the critical zone in every trial model. It fits the PNSN picks as well as without it (held-out RMS 0.0930 s for P and 0.1878 s for S, against 0.0924 and 0.1877 s for the calibration without the critical zone, which also predates the smoothing of the geology in the fusion), with the zero-pressure Vp multiplier rising from 1.31 to 1.44 to compensate for the slower near surface (Sect. 5.3.4). The critical-zone parameters themselves are not calibrated: the travel times barely constrain the top 50 m, and their calibration waits for near-surface observations (surface-wave dispersion, H/V and dv/v).

Synthetic observables. As a demonstration of what the columns predict, and not as data, the model gives Rayleigh-wave dispersion, depth kernels and dv/v at the 46 permanent stations, the 191 nodes of the 2025 array in the domain and every 25th channel of the Paradise fibre (S33), and dv/v through the December 2025 storm from a mass-conservative Richards solver (Celia et al., 1990) driven by MRMS rain (S34). These are simulations from the model; the nodes and the fibre were not recording during the storm. At 40 Hz half of the Vs sensitivity lies within the top 4 to 5 m, and the simulated storm lowers dv/v at 40 Hz by a median of 3.2% at the 40 permanent stations where the solver ran (`outputs/cz/storm_summary.json`), the order of the seasonal and rain-driven changes measured by noise interferometry on permanent stations and fibre (Clements and Denolle, 2023; Shen et al., 2024; Shi et al., 2026). The columns have no lateral drainage or snow yet, so these numbers illustrate the coupling, not a prediction.

5.3 The crustal model

Below the critical zone the units of the geologic map are extended to depth, their properties come from the rock law of Sect. 5.1.1, and the result is fused with the regional velocity models and calibrated on travel times.

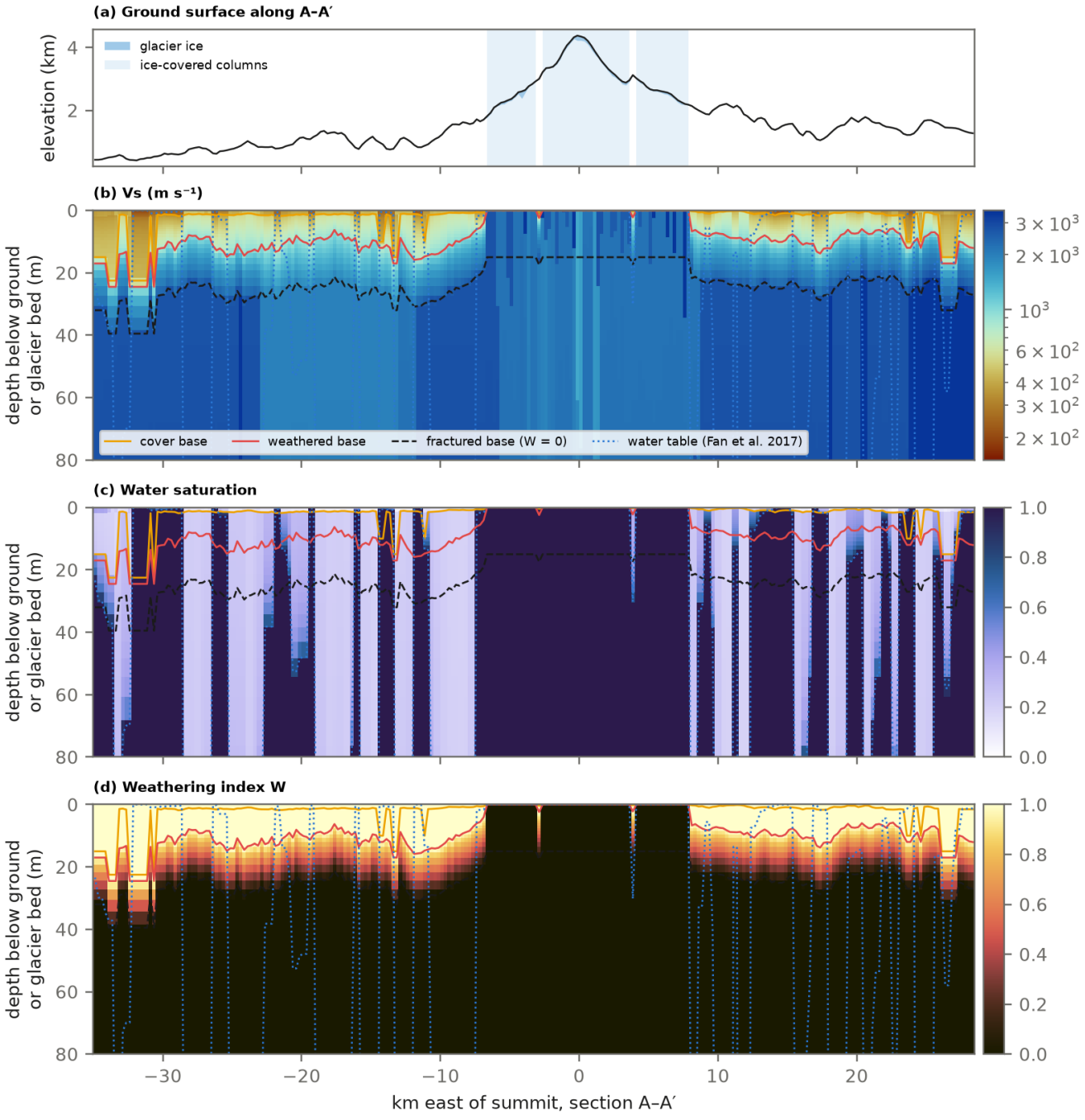


Figure 8. The critical zone along A–A′ (west–east through the summit; the `/cz` node of `model.zarr`, S4). (a) Ground surface, with glacier ice and the ice-covered columns shaded. (b) Vs, (c) water saturation and (d) weathering index W against depth below the ground or the glacier bed, with the bases of the Quaternary cover, the weathered layer and the fractured zone, and the water table of Fan et al. (2017). Under ice the columns start in fractured rock saturated from the bed (W = 0). Depths below 80 m are not shown; Vs joins the L1 cells at 150 m.

Table 11. Column rules for the 3D units (S3; `configs/units.yaml`).

Condition	Assigned unit
$d < 0$	air
$d <$ ice thickness (IceBoost v2 (Maffezzoli, 2025))	ice
$d <$ ice + deposit thickness (from the map unit description where it states one; else water 5 m, glacial drift 15 m, alluvium and colluvium 10 m, lahar deposits 15 m)	mapped surface deposit, beneath any ice
inside the edifice footprint, above the edifice base	Rainier andesite
young volcanic rocks, or andesite outside the footprint, within 300 m below the base of the ice and deposits	that cap unit
above the base of the bedrock unit (-4 km NAVD88 for supracrustal units, -10 km NAVD88 for the Miocene plutons)	bedrock unit (mapped, or nearest mapped basement unit)
otherwise (below -4 or -10 km NAVD88)	middle crust

5.3.1 Three-dimensional geology

Units are extended to depth by explicit column rules, in the manner of the San Francisco Bay region model (Aagaard and Hirakawa, 2021): the geology comes first and the velocity rules second. For a cell at elevation z and depth d below the ground, the rules of Table 11 are applied from the top down.

- **Edifice.** The edifice footprint is the Rainier andesite and ice mapped within 12 km of the summit (Columbia Crest, `configs/domain.yaml`) on the 100 m surface grid, closed morphologically over 500 m (a 5×5 cell element) with holes filled, keeping the connected region that contains the summit (S3, `rainier3d.geomodel.rules.edifice_footprint` `geometry.edifice` in `configs/units.yaml`, `m1_placeholder` values). Its base is the pre-volcanic surface: the DEM elevations of the plutonic and supracrustal outcrops in a two-cell (200 m) ring around the footprint, interpolated linearly across it (nearest value where the linear interpolant is undefined) and kept at or below the glacier bed. The base lies at 804–2234 m NAVD88 over a 252 km^2 footprint (`/surface` node of `data/processed/geomodel.zarr`).
- **Basement.** Beneath the deposits and the edifice, the basement unit is taken from the nearest outcrop: every map cell takes the plutonic or supracrustal unit mapped there, or else that of the nearest such outcrop on the 100 m surface grid (nearest-neighbour fill, `rainier3d.geomodel.rules.bedrock_units`, S3). Map symbols of the 1:100,000 state geologic map (Washington Geological Survey, 2026a) reach the model units only through the ordered regular-expression rules of `configs/units.yaml` (`crosswalk_rules`).
  - The supracrustal units (the Ohanapecosh, Fifes Peak and Stevens Ridge formations, the Eocene volcanic rocks and the Puget Group, and also the pre-Tertiary Russell Ranch Formation) extend to 4 km below sea level

Table 12. Deposit thicknesses taken from the Description of Map Units of the Washington 1:100,000 geologic map (Washington Geological Survey, 2026a). The largest unconsolidated units by area (Qs, Qa, the Vashon till Qgt and outwash Qgo) state no thickness and keep the unit defaults.

Map symbol	Unit	Stated in the map unit description	Thickness used
Qap(h)	Hayden Creek Drift	15 to 30 m thick	22.5 m
Qvl(o)	Osceola Mudflow	as much as 75 ft	11.4 m
Qvl(lc)	Lily Creek Formation, mudflows	maximum known thickness 273 ft	41.6 m
Qgos	Vashon Drift, outwash sand	commonly 2 to 50 m	26.0 m
Qao(e)	Evans Creek Drift, outwash	generally 3 to 8 m	5.5 m
Qapo(h)	Hayden Creek Drift, outwash	generally 3 to 5 m	4.0 m
Qvl(e)	Electron Mudflow	a few millimetres to at least 26 ft	4.0 m
Qp	peat	less than 1 ft to as much as 23 ft	3.5 m

(`supracrustal_base_z` = −4000 m NAVD88 in `configs/units.yaml`, an `m1_placeholder`); the middle crust fills the column below.

- The Miocene plutons (Tatoosh, White River, Carbon River and Nisqually; map symbols starting `Mi`, `MOi`, `PLOi`, `PLMi`) extend to 10 km below sea level (`pluton_base_z` = −10000 m NAVD88 in `configs/units.yaml`, an `m1_placeholder`). They therefore continue through L2 into the top four 1 km layers of L3, where they cover about 700 km^2 in plan view (2,706 L3 cells).
- The Mashel Formation (Miocene sedimentary rocks, map symbols starting `Mc`) is capped at 300 m thickness below the ice and deposits (`max_thickness_m: 300` in `configs/units.yaml`, an `m1_placeholder`); below that, the column takes the nearest mapped basement unit other than the Mashel (`rainier3d.geomodel.rules.build_level`).

– **Unconsolidated deposits.** Their thickness beneath any ice is set per map symbol where the Description of Map Units of the 1:100,000 map states one (`geometry.symbol_thickness_m` in `configs/units.yaml`), taking the midpoint of the stated range, from zero when only a maximum is given (an `m1_placeholder` rule). Eight symbols state a thickness (Table 12); they cover 7.0% of the box and 26.3% of its unconsolidated cells. Elsewhere the thickness is the default of the model unit (`geometry.unconsolidated_thickness_m`, all `m1_placeholder`): glacial drift 15 m, alluvium and colluvium 10 m, lahar deposits 15 m, and 5 m for mapped open water, which is given the properties of saturated alluvium. S1 writes the result as `unconsolidated_thickness` on the surface grid. Most deposits are thinner than half a 50 m L1 cell, so they appear only where a cell centre falls inside them: 6618 of the 7.39 million L1 cells (6189 with the unit defaults alone).

– **Magma body.** A slow body below the summit is a sphere centred beneath Columbia Crest at 11.5 km below sea level (`center_z` = −11500 m NAVD88) with a radius of 6.5 km, so it spans 5–18 km below sea level (`geometry.magma_body` in `configs/units.yaml`, source key `moran_1999`). As unit `magma_mush` it replaces every non-air unit it contains: 305 L2 cells and 1168 L3 cells. The weight of its anomaly (`magma_weight`) is 1 within 4.5 km of the centre (389 L3 cells) and falls to 0 at the radius as a squared cosine over 2 km (`radius_m`, `taper_m`, `m1_placeholder`). It follows the low-Vp body imaged 5–18 km beneath the summit by Moran et al. (1999). [TODO: confirm whether the 5–18 km of Moran et al. (1999) is below sea level or below the summit; `configs/units.yaml` line 59 marks the depth reference as still to verify]

5.3.2 Hydrothermal alteration from the helicopter electromagnetic survey

Clay-bearing altered rock is electrically conductive, while fresh lava and ice are resistive. The 1996 helicopter electromagnetic (EM) and magnetic survey of Rainier (Rystrom et al., 2000) measured apparent resistivity at four frequencies (33 kHz, 4737 Hz, 4341 Hz and 837 Hz) on a 50 m grid. Finn et al. (2001) used it to map collapse-prone altered zones. Script S22 turns these grids into an alteration field in three steps.

1. **Registration.** The grids are converted from NAD27 to the model datum. Their orientation is checked against the data, not assumed: the magnetic anomaly correlates with high-passed topography at 0.60 as read, and at 0.08 if flipped north–south.
2. **Intensity.** The survey saturates at a different resistivity at each frequency, so each frequency f is referred to its own fresh level L_f. L_f is the median log resistivity over the surveyed edifice lavas: 4.35, 3.89, 3.74 and 3.00 ($\log_{10}\ \Omega$ m). The intensity is
$$a_f = \mathrm{clip}\left(\frac{L_f - 0.3 - \log_{10}\rho_a}{0.7}, 0, 1\right), \tag{8}$$
which starts at half the fresh resistivity and reaches 1 at a tenth of it.
3. **Depth.** Each frequency senses to a depth of investigation of half its skin depth, $503\sqrt{\rho_a/f}$ m, capped at 150 m. The three-dimensional field is the maximum over frequencies of a_f tapered below that depth. Depth is measured below the glacier bed, and the field is restricted to the edifice lavas and young volcanic rocks.

Over the 126 km^2 of surveyed edifice lava, 10.6 km^2 reaches an intensity of 0.5 or more (Fig. 9). That corresponds to 1.9 km^3 of equivalent fully altered rock in the model grid. Table 13 gives the distribution around the summit.

The concentration east of and around the summit agrees with the exposed east–west belt of altered rock described by Finn et al. (2001) and John et al. (2008). The EM field sees only the top ~150 m. It cannot see the altered rock buried beneath fresh cover on the upper west flank, which Finn et al. (2001) inferred from magnetic modelling. The survey's terrain-correlated apparent magnetisation is published as a separate map.

5.3.3 Fusion with the regional models

Regional models. The regional model has two parts:

Table 13. Alteration of the surface rock from the EM survey (S22, `docs/alteration.md`).

Area (edifice lavas within 6 km of the summit)	Mean intensity at the surface	Share with intensity ≥ 0.5
West flank	0.08	8%
Summit (within 1.5 km)	0.21	19%
East flank	0.27	25%

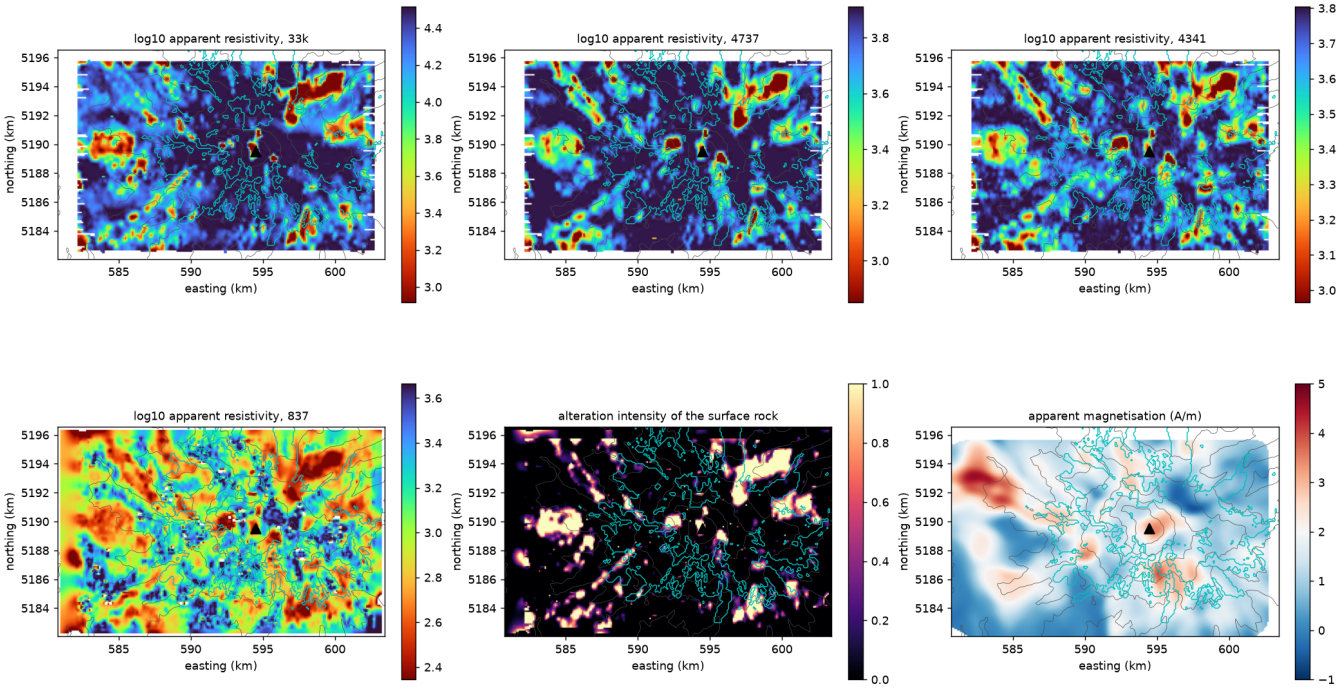


Figure 9. Apparent resistivity at the four EM frequencies, alteration intensity of the surface rock and terrain-correlated apparent magnetisation (S22). Contours: elevation every 500 m; cyan: glaciers thicker than 10 m; triangle: summit.

- **Down to 9.9 km below the ground:** the USGS Cascadia velocity model v1.7 (Wirth et al., 2025), for Vp and Vs, from its levels L01 (0–100 m every 10 m, then every 100 m to 1.2 km) and L2 (1.5–9.9 km every 300 m) on a 200 m horizontal grid, with L2 interpolated onto the L01 grid; its level L3 (10.8–59.4 km) is not used (`rainier3d.fusion.regional.load_cvm`). Its depth axis is below the ground surface. For the shallow level this is confirmed by a median top-sample Vs of 194 m s^{-1} over the model domain (`data/raw/regional/cvm17_domain.nc`; 206 m s^{-1} over the larger area of `configs/sources.yaml`), and for the deeper level it is inferred from continuity: at the summit, Vs is 2841 m s^{-1} at 1.2 km in L01 and 2870 m s^{-1} at 1.5 km in L2 (`configs/sources.yaml`, key `cvm17`).
- **Below 9.9 km:** CRESCENT Gen0 Vs (He et al., 2026), with Vp from Brocher (2005) through his regression $V_P = 0.9409 + 2.0947V_S - 0.8206V_S^2 + 0.2683V_S^3 - 0.0251V_S^4$ (km s^{-1}). The CRESCENT depth axis (-4 to 100 km) is taken as kilometres below sea level; the model is resampled onto a 2 km UTM grid padded by 5 km around the box and interpolated linearly to the cells, and its Vs uncertainty is carried as `vs_unc_regional`. The switch from the Cascadia model is at 9.9 km below the ground, so it follows the topography (`rainier3d.fusion.regional`).

Below 1 km, both are multiplied by a static, depth-dependent correction fitted to the PNSN arrivals (Sect. 5.3.4). The corrected regional Vs is capped at Vp/1.6, so that its Vp/Vs is at least 1.6 (Christensen, 1996); the geology side is not bounded this way.

Strategy. Neither model is right at every wavelength. The regional models resolve the crust at wavelengths of several kilometres and are constrained by teleseismic, ambient-noise and local-earthquake data; they carry no information on the mapped units. The geology model carries the mapped units and their contrasts, but its absolute level rests on laboratory-style rules. The fusion keeps each where it is informative: the regional model supplies the long wavelengths and the geology model the short ones. The combination is done in the logarithm of velocity, so contrasts are relative:

$$\ln V = \mathrm{LP}_{\lambda_c}(\ln V_{\mathrm{reg}}) + [\ln V_{\mathrm{geo}} - \mathrm{LP}_{\lambda_c}(\ln V_{\mathrm{geo}})]\,, \tag{9}$$

where LP is a horizontal Gaussian low-pass filter with half amplitude (a quarter of the power) at the cutoff wavelength λ_c, that is $\sigma = \sqrt{2\ln 2}\,\lambda_c/2\pi$.

Implementation details.

1. **Fused quantities.** Vs and Vp/Vs are fused rather than Vp and Vs separately, so the fused Vp/Vs always lies between the two input ratios.
2. **Cutoff wavelength.** λ_c grows with depth below the ground: 6 km at the surface, 10 km at 2 km, and 20 km from 10 km down. The values approximate the lateral resolution of the regional models.
3. **Air cells.** Cells above the ground are excluded from the filters by normalised convolution.
4. **Overshoot.** A Gaussian high-pass overshoots at sharp contrasts and puts fast rims around slow bodies. Six alternating projections remove them: each cell is clamped between its geology and regional values, then the regional low-pass is restored.

Table 14. Departure of the calibrated fused model from the corrected regional model at the regional wavelengths below 1 km depth, and mean offset over all cells (`outputs/fusion_report.csv`). The tolerance is 0.03. The departure is measured after the magma anomaly of item 7, which the regional models do not contain; without it the L3 values are 0.0004 (Vp) and 0.0003 (Vs) (`outputs/paper/numbers.json`).

Level	RMS of LP(ln V) − LP(ln V_reg), Vp	Same, Vs	Mean ln(V / V_reg), Vp	Same, Vs
L1	0.011	0.011	+0.052	+0.070
L2	0.001	0.002	+0.001	+0.002
L3	0.006	0.008	−0.001	−0.002

5. **Top kilometre.** The top 300 m keep the geology model unchanged, and a linear taper reaches the fused model at 1 km. The regional models do not resolve this depth range; their shallowest layer is a generic near-surface model.
6. **Rule contacts.** The column rules of Sect. 5.3.1 make flat-lying contacts (the edifice base, the base of the cap and of the Mashel Formation) and vertical unit boundaries that would reach the model as steps of up to a factor of two between neighbouring cells; the horizontal low-pass of Eq. 9 does not act on flat contacts, and the geology is kept as it is in the top 300 m. Below the 150 m critical-zone columns, the geology is therefore low-passed before the fusion, in the vertical with a cutoff that grows to 1 km 250 m below the column base and horizontally with a cutoff that grows as the depth below that base, to 2 km (`lambda_z_m`, `lambda_h_m` in `configs/domain.yaml`, `m1_placeholder`; `rainier3d.fusion.blend.smooth_geology`). The critical zone is neither smoothed nor smoothed into the cells below, and mapped contacts stay sharp at the surface and fade with depth, where the extrusion rule is least constrained.
7. **Magma body.** Its anomaly (Sect. 5.1.1) is applied to the fused model, so neither the low-pass nor the clamping of item 4 removes it.
8. **Density.** Density follows Vp through the ratio of Nafe–Drake densities, which preserves the density contrasts between units.

Fig. 10 shows the three models along a section through the summit. The fusion is checked by an invariant (Table 14): below 1 km depth, the low-passed fused model must equal the low-passed regional model within an RMS of 0.03 in ln V.

5.3.4 Calibration on P and S−P travel times

Data. The data are the analyst picks of the 88 PNSN earthquakes with M $\geq$ 2 in the box between 2015 and 2026 that have at least six picks, four of them P. They comprise 1823 P and 1280 S picks at 50 stations, with origins and picks from the USGS ComCat catalogue (U.S. Geological Survey, 2026a). The event list is committed (`configs/validation_events.csv`), so every run uses the same set. Picks are weighted by the RMS of PNSN's own location residuals: 0.14 s for P and 0.23 s for S.

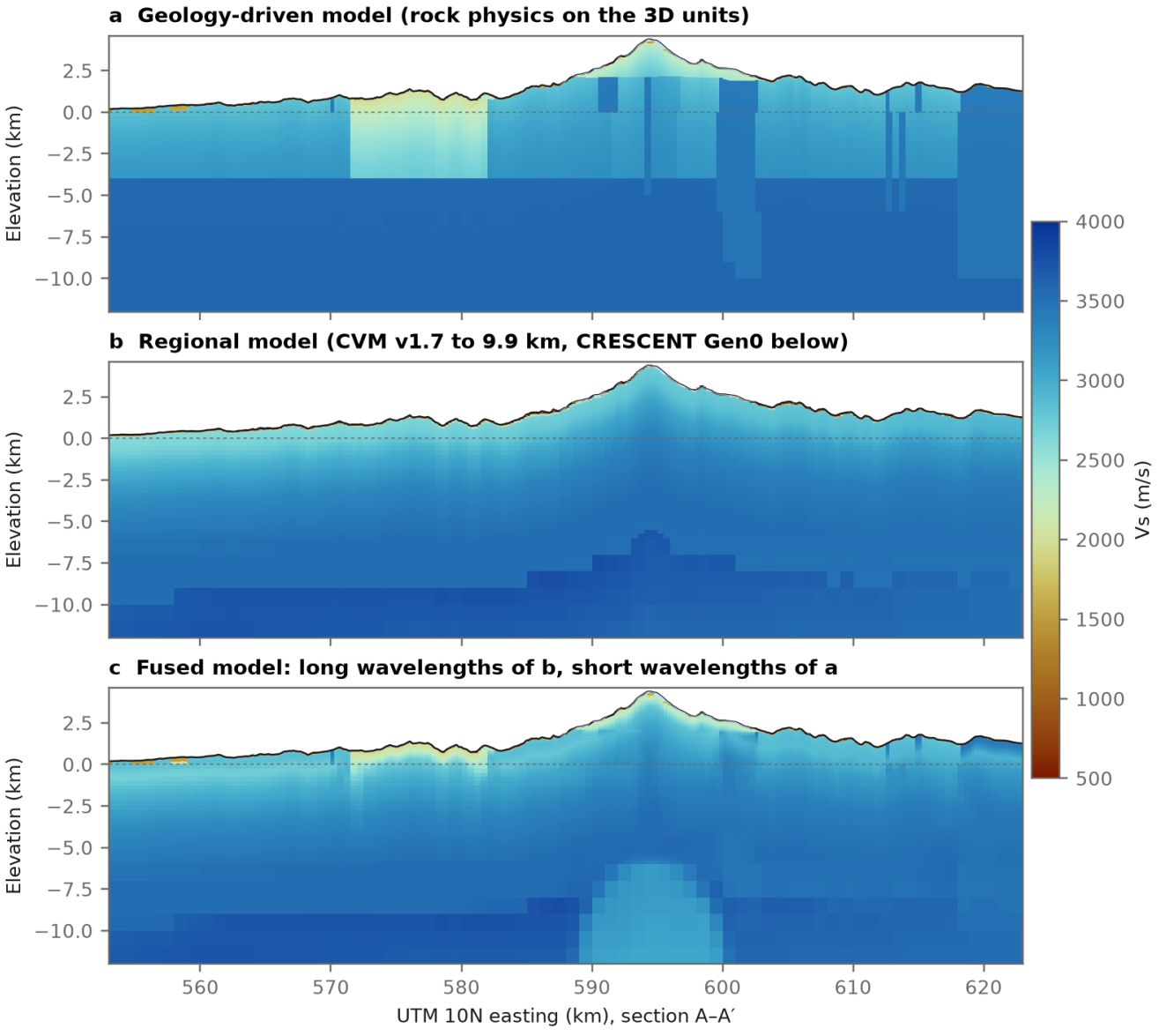


Figure 10. Vs along section A–A′ (west–east through the summit): (a) geology model, (b) regional model, (c) fused model. The fused model keeps the regional long wavelengths and the geological contrasts: the Puget Group sedimentary block near 572–582 km, the Tatoosh and White River plutons, and the slow edifice.

Travel times. They are computed on a 500 m grid with pykonal's point-source solver (White et al., 2020), one solve per station and phase by reciprocity.

- **Solver accuracy.** Against analytic travel times on a grid of the same size (70 × 75 × 24 km) and spacing (0.5 km), for an off-grid source and 360 receivers, its RMS error is 8 ms in a uniform 6 km s^{-1} model and 11 ms with the gradient 4 + 0.1z km s^{-1}; fteikpy (second-order fast sweeping) gives 22 and 46 ms. At 0.25 km spacing the pykonal errors fall to 4.7 and 5.7 ms at about nine times the cost (`scripts/bench_eikonal.py`, `docs/eikonal_benchmark.md`, pykonal 0.4.1, run on 24 September 2026).
- **Topography.** The travel-time grid is the S6 grid (`validation` block of `configs/domain.yaml`): 141 × 151 × 70 nodes at 500 m, from 4.5 km above to 30 km below sea level. Cells more than one cell (500 m) above the ground carry the speed of sound in air, 343 m s^{-1}, so rays follow the rock and cannot cut across valleys (`rainier3d.validate.locate.with_topography`). The one-cell skin keeps every station in rock; station elevations match the DEM within ±70 m (median +5 m). Against rock-filled air, the air handling changes the station-mean residuals by a median of 0.2 ms.
- **Cross-check.** For station OBSR, NonLinLoc's Grid2Time P times on the exported NonLinLoc grids (S9, `rainier3d_nll_500m.tar.gz`) agree with the pykonal times to 12 ms on average (RMS 31 ms, 85 events). This number comes from a separate session's record and has not been rerun.

What is estimated. Fifteen parameters $\boldsymbol{\theta}$ scale the two models that are fused, and the hypocentres and origin times of the 88 events are re-estimated in every model tried. The travel-time data therefore constrain the velocity parameters only through the part of the misfit that relocation cannot absorb. This avoids the circularity of scoring a model with catalogue hypocentres that were located in a 1D model with station corrections.

Parameters.

- **Rock physics (3).** Log-multipliers on three quantities shared by all rock units: the zero-pressure velocity V_0 and the crack-closure pressure P* of Eq. 1, and Vs. The Vs multiplier acts at fixed Vp, so it is a change of Vp/Vs. Ice and unconsolidated deposits are not scaled.
- **Regional correction (12).** A static depth profile $m(d)$ of the log-factor that multiplies the regional velocity, $V_{\mathrm{reg}} \to V_{\mathrm{reg}}\, e^{m(d)}$, one profile for Vp and one for Vs. Here d is depth below the ground. Each profile is piecewise linear between knots at 0, 1, 2, 4, 7, 11, 16 and 25 km and constant below 25 km. The values at 0 and 1 km are fixed at zero, because the fused model takes the top kilometre from the geology model (Sect. 5.3.3); the six deeper values are free.

Forward problem. For a trial $\boldsymbol{\theta}$, the model is rebuilt from its inputs by the same code that builds the published model. At $\boldsymbol{\theta} = 0$ it returns the uncalibrated model cell for cell, which checks that the calibration fits the model that is delivered and not an approximation of it. The steps are:

1. Vp of every rock-unit cell from Eq. 1 with the scaled V_0 and P*, and Vs from that Vp through the unit's Vp/Vs ratio, or the regression of Brocher (2005) where the unit has none, times the Vs multiplier;

2. the regional Vp and Vs multiplied by $e^{m(d)}$;
3. the fusion of Eq. 9, including the geology-only top 300 m and the taper to 1 km;
4. P and S travel times from every station on the 500 m grid, with air above the ground, as described under Travel times;
5. every event relocated in that model (see Relocation below).

Objective. With t_{ij} the pick of phase at station j for event i, T_j the travel-time field, and $(\mathbf{x}_i, \tau_i)$ the hypocentre and origin time, the normalised residual is

$$r_{ij} = \frac{t_{ij} - \tau_i - T_j(\mathbf{x}_i; \boldsymbol{\theta})}{\sigma_{\mathrm{ph}}}, \tag{10}$$

where σ_{ph} is the pick uncertainty of the pick's phase: $\sigma_P = 0.14$ s for P picks and $\sigma_S = 0.23$ s for S picks. The calibration minimises

$$\Phi(\boldsymbol{\theta}) = \sum_{ij} w_{ij}\, r_{ij}^2 + \boldsymbol{\theta}_g^{\mathsf{T}} \mathbf{C}_g^{-1} \boldsymbol{\theta}_g + \lambda_s \|\mathbf{D}\mathbf{m}\|^2 + \lambda_d \|\mathbf{m}\|^2, \tag{11}$$

where the hypocentres minimise the first term for each $\boldsymbol{\theta}$. The weights w_{ij} are the Huber weights of the relocation (threshold 1.5σ), so outlying picks count less. $\boldsymbol{\theta}_g$ holds the three rock-physics log-multipliers, with prior standard deviations of 0.3, 0.7 and 0.1 on the diagonal of $\mathbf{C}_g$ (factors of 1.35, 2 and 1.1). $\mathbf{m}$ holds the regional log-factors at all eight knots, with the two fixed at zero, and $\mathbf{D}$ takes second differences along depth. The smoothing weight λ_s and the damping weight $\lambda_d = 0.01$ are relative: both are multiplied by the mean diagonal of the data term for the regional parameters.

Update. Each iteration linearises the residuals about the current model and hypocentres, $\mathbf{r}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}) \approx \mathbf{r} - \mathbf{G}\,\delta\boldsymbol{\theta}$, with $\mathbf{G} = \partial\mathbf{T}/\partial\boldsymbol{\theta}$ scaled like the residuals.

- **Separating hypocentres from velocity.** For each event, the columns $\mathbf{H}_i$ of partials with respect to $(\mathbf{x}_i, \tau_i)$ are formed at its current location. The event's rows of $\mathbf{G}$ and $\mathbf{r}$ are projected onto the orthogonal complement of $\mathbf{H}_i$, through a QR factorisation of $\mathbf{H}_i$ (Pavlis and Booker, 1980). What remains is the part of the misfit that no shift of that hypocentre can remove. Events with five or fewer usable picks keep at most one degree of freedom after the projection and are left out of the update.
- **Step.** With the projected $\tilde{\mathbf{G}}$, $\tilde{\mathbf{r}}$ and the regularisation matrix $\mathbf{L}$ of Eq. 11,

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \left(\tilde{\mathbf{G}}^{\mathsf{T}}\tilde{\mathbf{G}} + \mathbf{L}\right)^{-1} \left(\tilde{\mathbf{G}}^{\mathsf{T}}\tilde{\mathbf{r}} - \mathbf{L}\,\boldsymbol{\theta}\right). \tag{12}$$

- **Relocation.** Every event is then relocated in the updated model before the next linearisation, so velocity and hypocentres are updated in alternation, as in the minimum one-dimensional model of Kissling et al. (1994). Relocation is a grid search over nodes at or below the ground, within 15 km of the catalogue epicentre, on the sum of |r| with the median origin time, followed by Huber Gauss–Newton (Huber, 1964) on interpolated travel times with a penalty on hypocentres more than 50 m above the ground. It follows the grid-search-then-refine design of NonLinLoc (Lomax et al., 2000).

Derivatives.

– **Rock physics.** A rock-physics multiplier changes every rock-unit cell, and its effect passes through the fusion filters, so its column of $\mathbf{G}$ is a one-sided finite difference ($h = 0.05$ in log) through the whole forward problem, with the hypocentres held fixed. That costs one full rebuild and eikonal solution per parameter, and it is recomputed at every iteration on the fitting half.

– **Regional correction.** For knot k the derivative is the ray integral

$$\frac{\partial T}{\partial m_k} = -\int_{\text{ray}} \beta(d)\,\phi_k(d)\,s\,\mathrm{d}\ell, \tag{13}$$

where s is slowness, ϕ_k the linear interpolation weight of knot k at depth d, and $\beta(d)$ the share of the regional model in the fused model: 0 above 300 m, rising linearly to 1 at 1 km. Rays are traced down the gradient of each station's travel-time field (Thurber, 1983). The integral assumes that a depth-only factor on the regional model passes through the fusion unchanged. The low-pass filter of Eq. 9 passes a factor that varies slowly across the box, but the clamping of fused values between the two input models is not linear, and is the likely cause of the difference below. Against a finite difference through the full forward problem, for the Vs knot at 4 km over all 1280 S picks, the ray derivatives correlate at 0.97 with a slope of 0.81: they underestimate the sensitivity by about 20%. The residuals are always evaluated with the full forward problem, so this error does not bias the data fit. At convergence it acts, to first order, like regularising the regional correction about 20% more weakly than λ_s and λ_d state.

Validation and choice of smoothing. Events alternate between a fitting half and a held-out half in origin-time order, and held-out events are relocated in every iterate, so their misfit is an independent score. At the first iteration, λ_s is chosen from 0.01, 0.1, 1 and 10 by the normalised RMS of the held-out separated residuals predicted by the linearised step: 0.720, 0.719, 0.732 and 0.772. The adopted value is 0.1.

Iterations and uncertainties. Three iterations on the fitting half are followed by two on all events, which reuse the last rock-physics derivatives (Table 15); the run takes 29 min on a 10-core laptop. Posterior standard deviations are the square roots of the diagonal of $(\tilde{\mathbf{G}}^{\mathsf{T}}\tilde{\mathbf{G}} + \mathbf{L})^{-1}$ at the last iteration, with the covariance multiplied by the mean reduced χ^2 of the two phases, 0.574, the mean of 0.459 for P and 0.689 for S (`scripts/13_joint_calibration.py`). They are linearised. They do not include the error of the ray derivatives, or the trade-off with hypocentres beyond what the projection removes.

The fit is resolved as follows (Table 16, Table 17, Fig. 11, Fig. 12).

– **V_0.** It is the parameter the data require: $\times 1.44 \pm 0.04$ ($\times 1.31$ without the critical zone). The table rock is too slow near the surface, and the slower critical zone above it raises the multiplier further. With it, the zero-pressure Vp of the rock law is 4.0 km s^{-1} for Rainier andesite and 4.9 km s^{-1} for the Ohanapecosh Formation. Because the multiplier is global, the Miocene plutons reach the cap of 0.98 $V\infty$ = 6.08 km s^{-1} at zero pressure (4.5 × 1.44 = 6.5), so their rock law starts at the cap; the velocity drop toward the surface comes from the critical zone alone.

– **P*.** It is not resolved: its log-multiplier ranged from -0.28 to +0.52 across the iterations (final $\times 1.42$, posterior s.d. 0.19 in ln) while the held-out RMS changed by less than 1 ms after the second iteration, and it trades off against the regional correction at 2 km.

Table 15. Gauss–Newton iterations of the calibration (S13 with the critical zone of Sect. 5.2 in every trial model, `outputs/joint_calibration/history.json`): RMS after relocation and the geology parameters. In the final row the fitting column is the RMS over all 88 events. Most of the misfit reduction happens in the first iteration.

Iteration	P, fitting (s)	P, held out (s)	S, fitting (s)	S, held out (s)	ln V_0	ln P*	ln Vs
0 (uncalibrated)	0.124	0.120	0.318	0.325	0	0	0
1	0.099	0.095	0.195	0.190	0.183	0.520	−0.008
2	0.097	0.094	0.193	0.188	0.251	0.004	−0.040
3	0.097	0.093	0.193	0.188	0.299	−0.284	−0.024
final (all events)	0.095	0.093	0.191	0.188	0.362	0.348	−0.050

Table 16. Geology multipliers (`configs/velocity_calibration.yaml`, block `geology`), fitted with the critical zone in every trial model; the calibration without it is kept as `configs/velocity_calibration_v2.yaml` (V_0 ×1.31, P* ×0.91, Vs ×0.976). Posterior standard deviations are linearised and scaled by the reduced χ^2.

Parameter	Multiplier	Posterior s.d. (ln)	Resolved
V_0, zero-pressure Vp	1.44	0.027	yes
P*, crack-closure pressure	1.42	0.19	no
Vs of rock units	0.951	0.013	yes

- **Rock Vs.** It falls by 4.9 ± 1.2% (ln multiplier −0.050 with posterior s.d. 0.013, against a prior s.d. of 0.1). Because it acts at fixed Vp, it raises the Vp/Vs of every rock unit by about 5%.
- **Regional correction.** It changes Vp by less than 4% at every depth. It makes Vs 5–8% faster at 2–7 km and 4–8% slower at 16–25 km; with the rock multipliers, the median Vp/Vs at 2–4 km below the ground falls from 1.83 in the regional model to 1.74 in the fused model.
- **Agreement between the two models.** After calibration, the geology model and the corrected regional model agree to within 3.4% between 0.3 and 4 km depth (largest difference +0.033 in ln Vs at 0.3–1 km). Uncalibrated, the geology is up to 20% slower (−0.227 in ln Vp at 0.3–1 km), and the fusion invariant exceeds its tolerance in L1 (0.031 for Vp, 0.032 for Vs). These are mean ln(V_geology / V_regional) by depth below ground over L1–L3 without air and ice, from `model.zarr` and from the uncalibrated model rebuilt in memory by the same code with the critical zone (`rainier3d.report.paper_numbers`, `outputs/paper/numbers.json`). In the top 300 m, which the fused model takes from the geology alone, the calibrated geology, critical zone included, remains 18% faster in Vp and 23% faster in Vs than the regional model (mean ln ratios 0.16 and 0.21).

Why this parameterisation. Two simpler parameterisations were tested against the same data (Table 18).

Table 17. Static correction of the regional models (block `regional_bias`). Formal standard deviations of the log-factors are 0.003–0.016. The deepest knots rest on few rays: 30 events are deeper than 11 km and 8 deeper than 16 km below sea level (36 and 11 below the ground, the depth axis of the table).

Depth below ground (km)	2	4	7	11	16	25
Factor on regional Vp	1.016	0.999	0.964	0.978	0.987	0.984
Factor on regional Vs	1.047	1.081	1.061	0.994	0.955	0.925

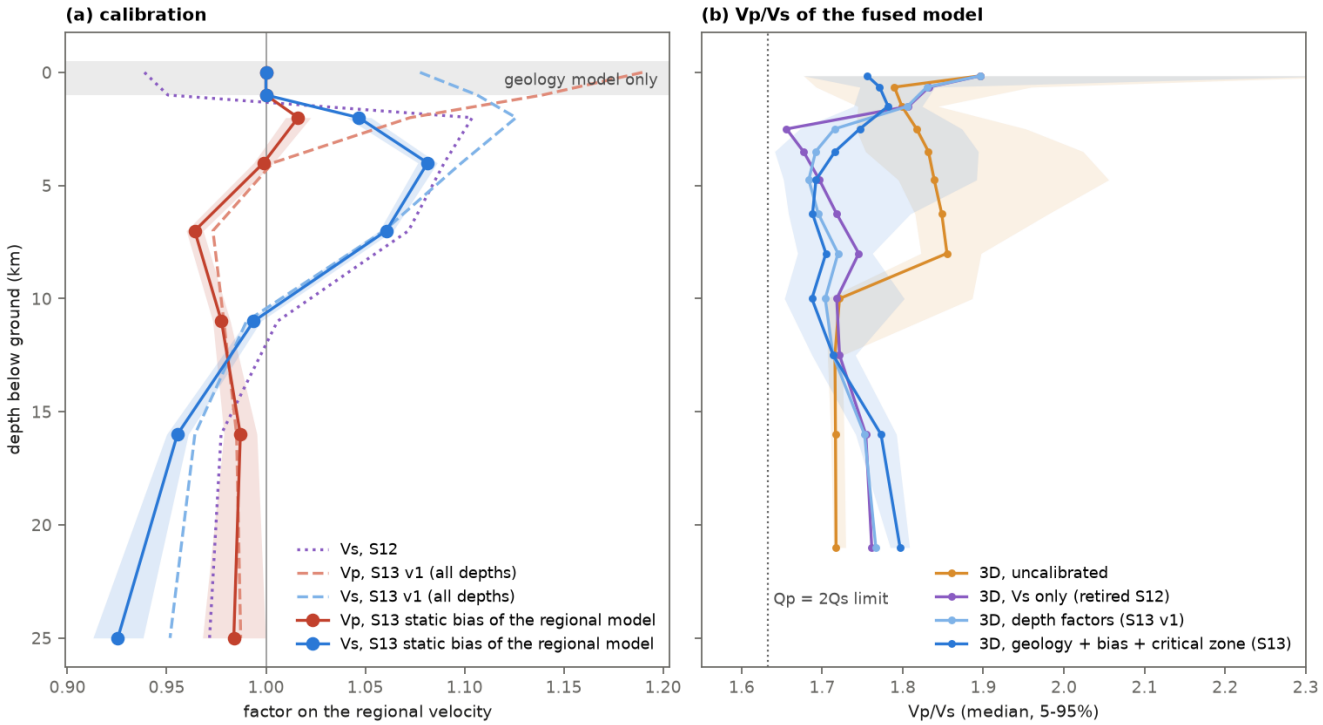


Figure 11. (a) Static correction of the regional Vp and Vs (solid, ±1 standard deviation). The dashed and dotted lines are the two alternative parameterisations of Table 18. The grey band is the geology-only zone. (b) Median and 5–95% range of Vp/Vs of the fused model against depth below ground.

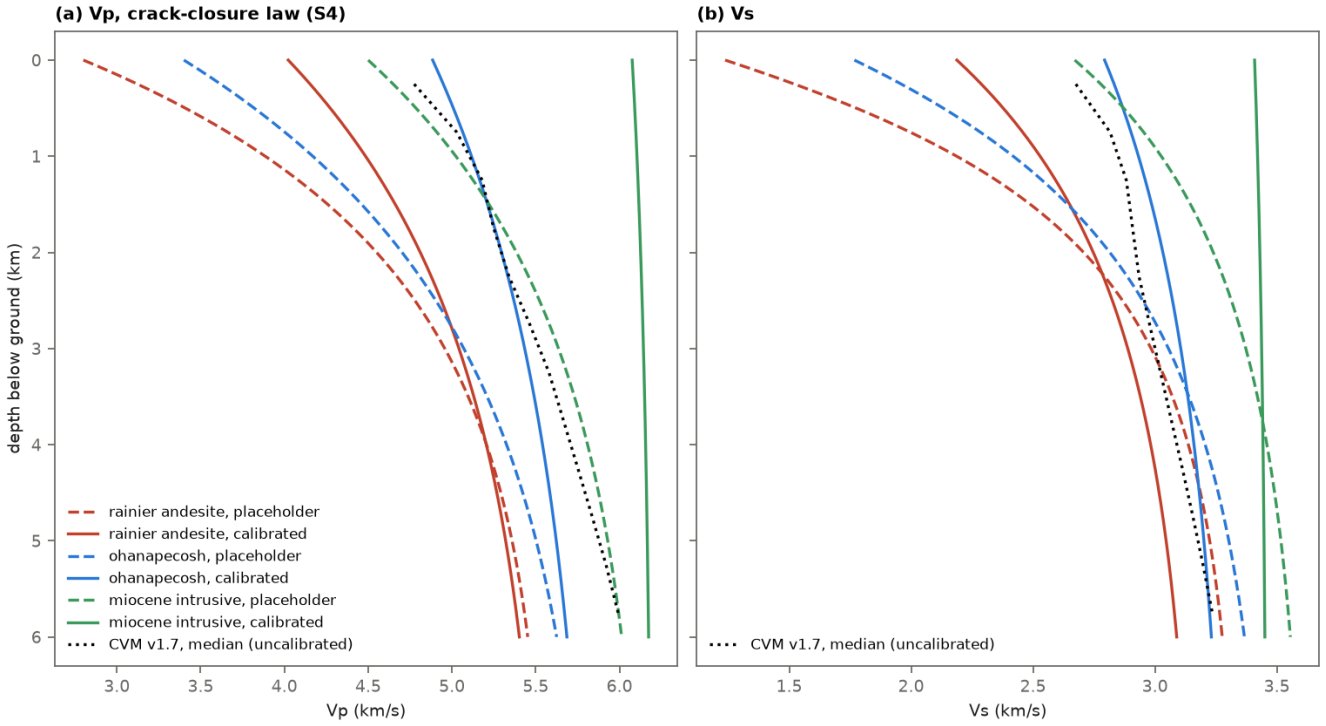


Figure 12. Crack-closure Vp (a) and Vs (b) against depth below ground for three rock units, with table (dashed) and calibrated (solid) parameters. Dotted: median of the Cascadia velocity model over the domain.

- **Vs-only depth factor, catalogue hypocentres fixed.** It fits S as well as the adopted model, but pushes Vp/Vs to 1.60–1.67 at 2–4 km. That is low for crustal rock and lower still than expected beneath a volcano with a hydrothermal system: the S−P misfit of fixed hypocentres is partly a location error. This is the earlier S12 calibration (`scripts/12_calibrate_vs.py`, retired and kept in the git history; `configs/vs_calibration.yaml`), which raised Vs by up to 10% at 2–7 km; it left 30,086 fused cells below Vp/Vs = $\sqrt{(8/3)}$ in the model of that time, against 18,544 for the adopted model.
- **Depth factors on the regional Vp and Vs at all depths, with relocation.** It fits as well as the adopted model, but places the correction in the regional model where the fusion does not use it. It requires Vp 7–19% faster in the top 2 km, where the fused model follows the geology, and it breaks the fusion invariant (0.050). This is the first S13 run (`configs/velocity_calibration_v1.yaml`): Vp factors of 1.19, 1.14 and 1.07 at 0, 1 and 2 km, and an L1 invariant of 0.050 for Vp and 0.044 for Vs (`outputs/relocation/s5_joint.log`).
- **Adopted parameterisation.** Fitting the geology's rock physics and correcting the regional model only below 1 km fits equally well, satisfies the invariant (0.011 for Vp and for Vs, L1 in `outputs/fusion_report.csv`) and keeps the unit contrasts in physical parameters.

Alteration and the calibration. The first calibration without the critical zone was run with a conduit-centred alteration field; replacing it with the EM-based field of Sect. 5.3.2 changed the relocated RMS from 0.095 to 0.093 s for P and left S at 0.190 s. The calibration with the critical zone was run directly on the EM-based field.

5.3.5 Validation

Held-out events. The 44 events of the held-out half are relocated in every iterate but enter no fit until the last two iterations (Table 15). After the three iterations on the fitting half, their RMS has fallen from 0.120 to 0.093 s for P and from 0.325 to 0.188 s for S, while the fitting half ends at 0.097 and 0.193 s; the held-out misfit is not larger than the fitted one. These are the independent scores of the calibration. The smoothing weight λ_s is chosen on the same half (see Validation and choice of smoothing), so they are not fully blind to it. The two final iterations use all 88 events, and the published model (critical zone and EM-based alteration, `data/processed/model.zarr`) relocated with S14 gives 0.093 s for P (949 picks) and 0.188 s for S (655 picks) on the same 44 events, against 0.131 and 0.263 s in the PNSN 1D model; these events are no longer independent of that model. The split is in time, not in space: held-out and fitting events share source regions and ray paths (`outputs/paper/numbers.json`, scored from `outputs/relocation/<model>/picks.csv` with the split of S13: event identifiers in origin-time order, every second one held out).

All models scored the same way. Table 18 relocates all 88 events in each model with the same locator and topography. Relocation alone does not rescue the uncalibrated three-dimensional model: its S residuals stay larger than those of the 1D model. The S−P misfit therefore lies in the velocity model, not in the catalogue hypocentres. With the ground bound, no event is placed at the top of the grid, and two of the 88 events end more than 50 m above the ground in the published model, at the limit of the ground penalty (`outputs/relocation/fused_joint/stats.json`, `above_ground_gt_50m` = 2; three in the conduit-alteration run).

Independent tomography. The iMUSH project imaged Vp and Vs around Mount St. Helens from local earthquakes and explosions recorded by a 70-station array (Ulberg et al., 2020; Incorporated Research Institutions for Seismology, 2020).

- **Coverage.** Its grid spans the whole model box. Its Vp is resolved over 82–90% of the box. Its Vs is resolved over 83–93% of the southern strip (south of 46.65° N) but only 11–19% of the northern half. Script S21 samples our models at its 51,300 resolved nodes below ground. The model is the EarthScope EMC file `iMUSH-localEQ-Ulberg-2020.r0.0-n4c.nc`, with 1.2 km nodes masked where checkerboard tests do not recover 20 km features; its depth axis is read as kilometres below sea level, as the EMC page states, although the netCDF long_name says below the surface. Coverage is counted over 0–14 km below sea level, and north of 46.75° N between 188 and 1354 Vs nodes fall in each depth bin.
- **Beneath Rainier, north of 46.75° N (Table 19).** The iMUSH model finds the Cascadia model's Vs 5–8% too slow at 1–8 km and 3–4% too fast at 11–20 km, the shape of Table 17. With the correction, the difference in Vs is 2% or less below 2 km. Vp/Vs moves from 1.83–1.85 towards the iMUSH value of 1.75–1.77; at 4–8 km our Vp/Vs (1.70) is about 0.05 below it.

Table 18. Residuals after relocating all 88 events in each model (S14, `outputs/relocation/<model>/stats.json`; the conduit-alteration row from `outputs/relocation/s14_fused_joint.log`). The published model was relocated on 14 October 2026; the other rows are the runs of 24 September 2026, before the critical zone and the smoothing of the geology in the fusion, and are kept as the comparison of parameterisations. Depth shifts are relocated minus ComCat, positive deeper. [1] Fitted on these events; the held-out scores are the fair measure.

Model	P RMS (s)	S RMS (s)	Epicentre shift from ComCat, median (m)	Depth shift, median (m)
PNSN 1D	0.132	0.268	1157	+511
3D, uncalibrated	0.122	0.316	1285	+257
3D, Vs-only depth factor[1]	0.102	0.189	847	+306
3D, depth factors on regional Vp and Vs[1]	0.094	0.190	835	+727
3D, geology multipliers and regional correction, conduit-centred alteration[1]	0.095	0.190	856	+852
3D, published model: recalibrated with the critical zone, EM-based alteration[1]	0.095	0.191	824	+947

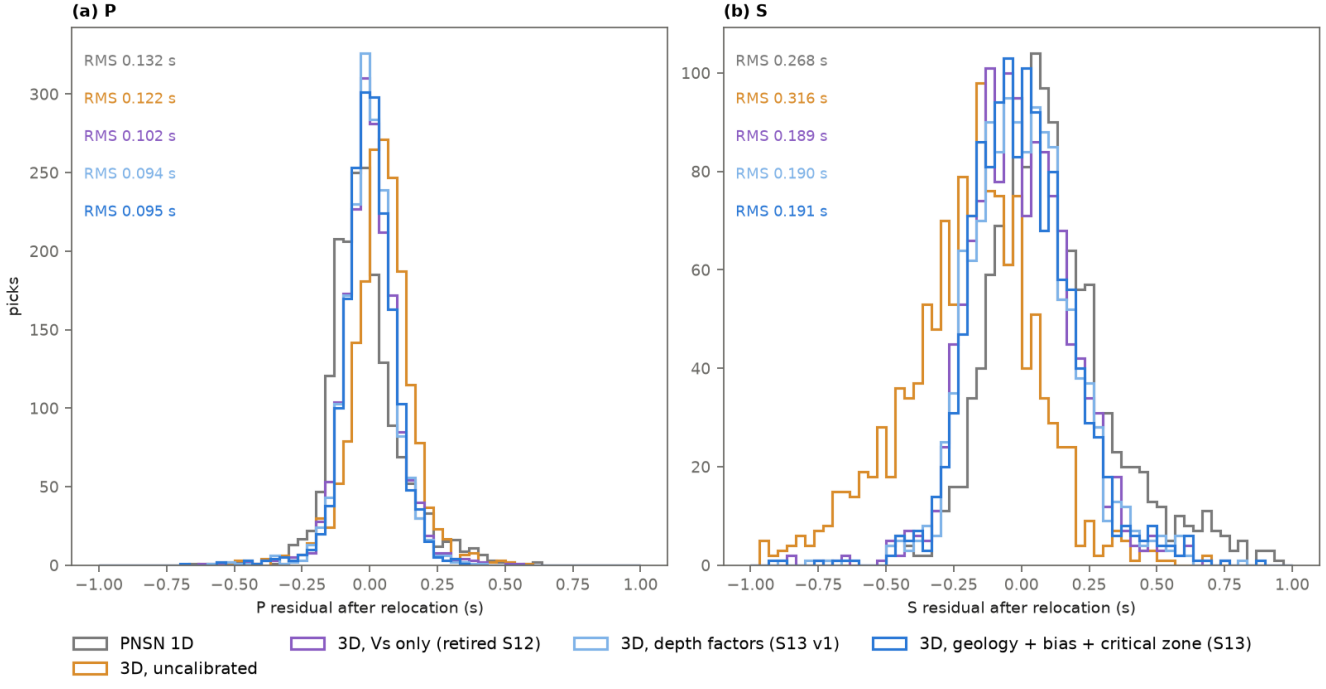


Figure 13. P (a) and S (b) residuals after relocating all events in each model; the RMS of each model is printed in its colour.

Table 19. North of 46.75° N: mean $\ln(V_{iMUSH}/V_{model})$ for Vs and median Vp/Vs (S21, `outputs/model_comparison/ulberg2020_by_depth.cs`

Depth below ground (km)	1–2	2–3	3–4	4–6	6–8	11–15	15–20
Vs, regional as distributed	+0.084	+0.073	+0.064	+0.055	+0.046	−0.030	−0.035
Vs, regional with the correction	+0.062	+0.019	−0.006	−0.016	−0.010	−0.009	+0.014
Vp/Vs, iMUSH	1.770	1.759	1.754	1.757	1.755	1.744	1.765
Vp/Vs, regional	1.810	1.828	1.850	1.851	1.851	1.720	1.723
Vp/Vs, rainier3d	1.792	1.760	1.733	1.701	1.699	1.724	1.782

- **South of 46.65° N, inside the iMUSH array.** The iMUSH Vs is within 1% of the Cascadia model at 1–6 km, and the correction makes our Vs 5–6% too fast at 2–6 km. This suggests that the correction is local to Rainier, which a laterally varying correction would test.
- **Independence.** Both studies use PNSN arrivals from 2015–2016. The iMUSH-only stations and explosions make the comparison largely independent in the south, less so in the north.

Catalogue hypocentres. At the ComCat hypocentres, which were located in a 1D model, the three-dimensional model predicts later arrivals than the 1D model at every station: by 0.31 s for P (mean over 50 stations of the station-mean difference) and 0.49 s for S (45 stations with at least five picks); per pick, the means are 0.26 and 0.44 s (`outputs/pnsn_residuals.csv`). With each event's mean residual removed, the RMS is 0.133 s (P) and 0.228 s (S) for the 3D model, against 0.159 and 0.287 s for the 1D model. The per-station 3D − 1D delay correlates with the mean 1D residual at 0.67 for P and 0.65 for S (Fig. 15). The 3D structure therefore explains part of what station corrections absorb.

5.3.6 The fused model

Figs. 16, 17 show the fused model along two sections through the summit, and Fig. 18 compares vertical profiles.

- **Level L1 (the ground to sea level).** L1 is on average 5% faster in Vp and 7% faster in Vs than the regional model. The calibrated rock is stiffer than the shallowest layer of the Cascadia model, a generic near-surface model with a median top-sample Vs of 194 m s^{-1} over the model domain. The travel times constrain this difference only beneath the stations. The averages are the mean ln(V_fused / V_regional) over all non-air L1 cells, from 0 to 4.4 km NAVD88: +0.052 for

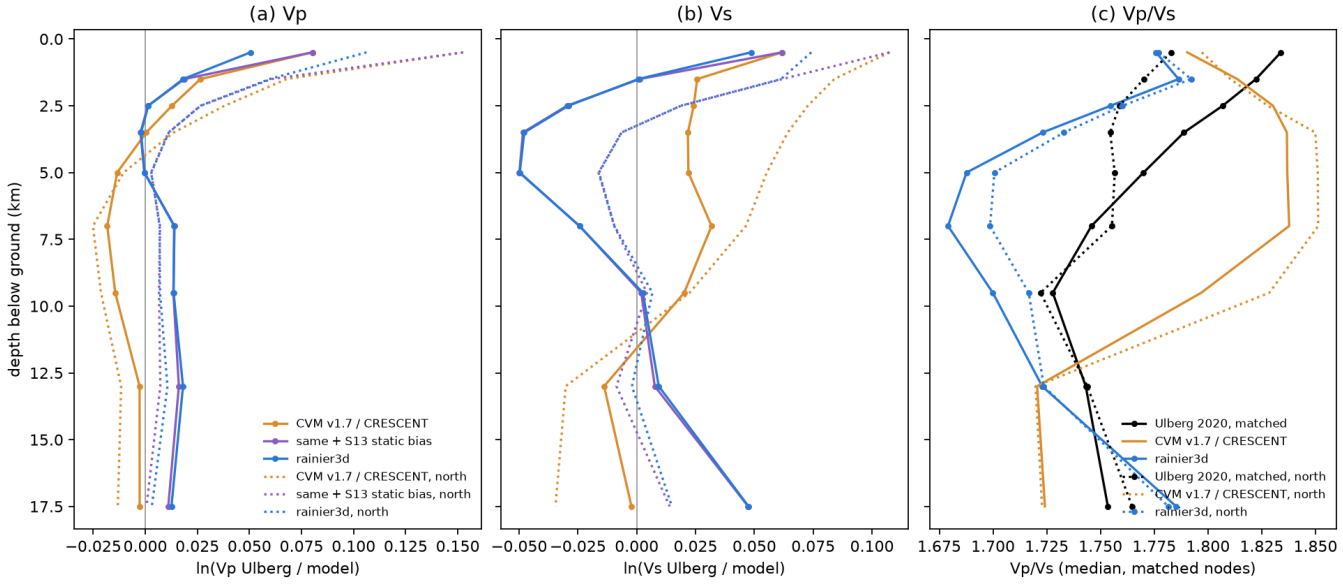


Figure 14. rainier3d and the regional models against the iMUSH tomography. (a, b) Mean $\ln(V_{iMUSH}/V_{model})$ against depth below ground, over the box (solid) and north of 46.75° N (dotted). (c) Median Vp/Vs where the iMUSH P and S models are both resolved.

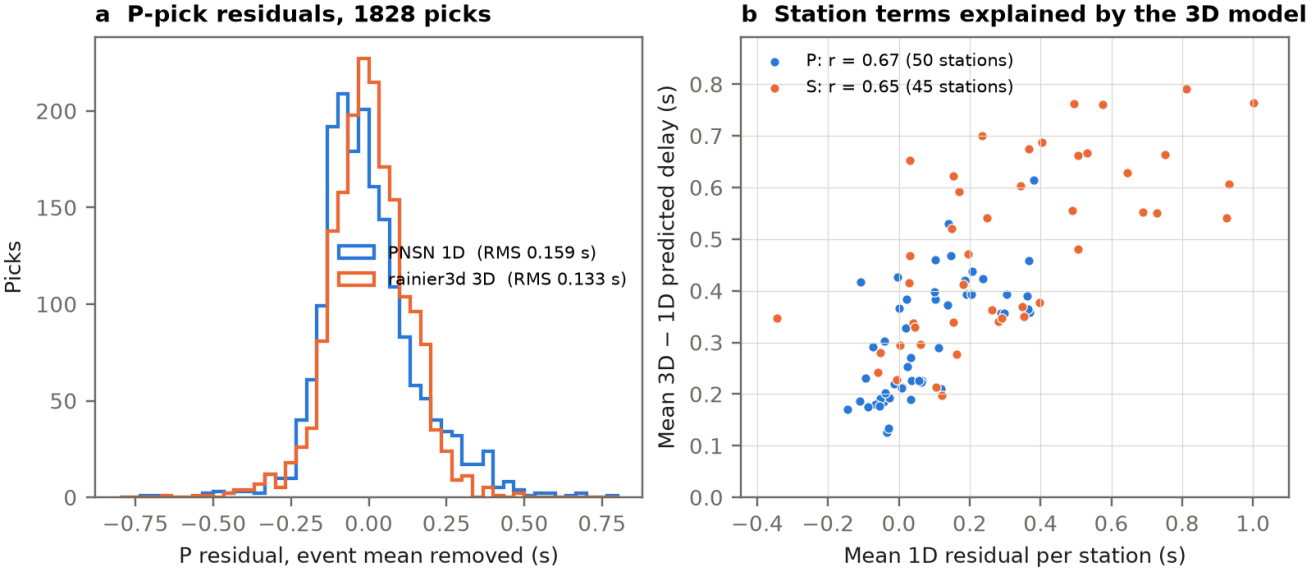


Figure 15. At the ComCat hypocentres (S6: all 89 events of the box with picks, 1828 P picks, one more event than the calibration set): (a) distribution of P residuals after removing each event's mean, for the PNSN 1D model and the fused model; (b) mean 3D − 1D predicted delay against mean 1D residual for each station with at least five picks.

Vp and +0.070 for Vs (`outputs/fusion_report.csv`). The difference is largest in the top 300 m, where the calibrated geology, critical zone included, is 18% faster in Vp and 23% faster in Vs than the regional model.

- **Plutons.** The Miocene plutons stand out as fast columns to 10 km below sea level, the base set by `pluton_base_z`. With the calibrated V_0 their geology Vp is at the cap of 6.08 km s^{-1} just below the critical zone (200–300 m) and 6.13 km s^{-1} at 2 km depth (L1 medians in `data/processed/properties_geology.zarr`).
- **Puget Group.** The Puget Group block west of the summit is slow down to 4 km below sea level. It is unit `eocene_sedimentary` (V_0 = 2.6 km s^{-1}, $V\infty$ = 5.2 km s^{-1} in Table 9), whose base is the supracrustal base at -4 km NAVD88, and on the summit row it extends from 571.6 to 581.9 km easting (UTM 10N, cell centres).
- **Seismicity.** Summit earthquakes form a column from the edifice to about 3 km below sea level. WRSZ earthquakes concentrate 4–12 km below sea level, 12–18 km west of the summit. The sections show 4369 PNSN events of M 0.5–3.4 from 2015 to 2026 (`web/atlas/data/events.geojson`, depths taken as below sea level). Of the 1559 events within 2 km of A–A′ and 3 km of the summit, 90% lie between 2.3 km above and 1.9 km below sea level and 99% above 2.6 km below. Of the 329 West Rainier Seismic Zone (WRSZ) events within 2 km of A–A′ and 8–25 km west of the summit, 90% lie at 4.5–11.8 km below sea level and 10.8–18.2 km west.
- **Magma body.** The slow body (5–18 km below sea level, Sect. 5.3.1) is applied to the fused model after the fusion, so it is the one structure of the model that the regional models do not constrain at its own wavelength. In L3, where it lies, its core (weight 1) is 15.4% slower in Vs than the middle crust at the same elevations, and the whole body, taper included, 9.4% slower (mean ln ratios -0.168 and -0.098; `outputs/paper/numbers.json`); L2 holds only cells of its taper. Whether a body of the size imaged by Moran et al. (1999) and Pang et al. (2025) belongs in the model is a question for data that resolve it: the tomography of Sect. 5.3.7 has almost no rays below 12 km.

5.3.7 Local-earthquake tomography

The calibration adjusts global factors and a depth profile, so it cannot correct lateral errors of the geological rules between the wavelengths that the geology encodes and those that the regional models resolve. A local-earthquake tomography (S37, `rainier3d.validate.tomo`; Supplement S1) perturbs ln Vp and ln Vs of the calibrated model on nodes 3 km apart, using 99,716 analyst picks of 3770 PNSN earthquakes from 2015 to the data freeze, with travel times on a 250 m eikonal grid, every event relocated in every trial model, and earthquakes in 10 km squares withheld from the fit to choose the smoothing and stop the iterations. The perturbation is zero in the top 150 m below the ground or glacier bed, so the critical zone keeps its rock physics, and all invariant tests of the model hold in the result. Two steps reduce the RMS residual of the held-out earthquakes from 0.0909 to 0.0725 s for P and from 0.1796 to 0.1209 s for S, and the RMS of the held-out residuals averaged per station from 0.051 to 0.023 s for P and from 0.152 to 0.067 s for S, without station terms. Checkerboard tests recover 12 km structure between sea level and 6 km depth with correlations of 0.63–0.75 and about half of its amplitude; below 12 km, and beneath the summit below 5 km, ray coverage is nearly absent and the perturbation is set by the regularisation. Within the sampled volume the update has a mean close to zero and an RMS of about 5%. The result is distributed as `model_let.zarr` beside

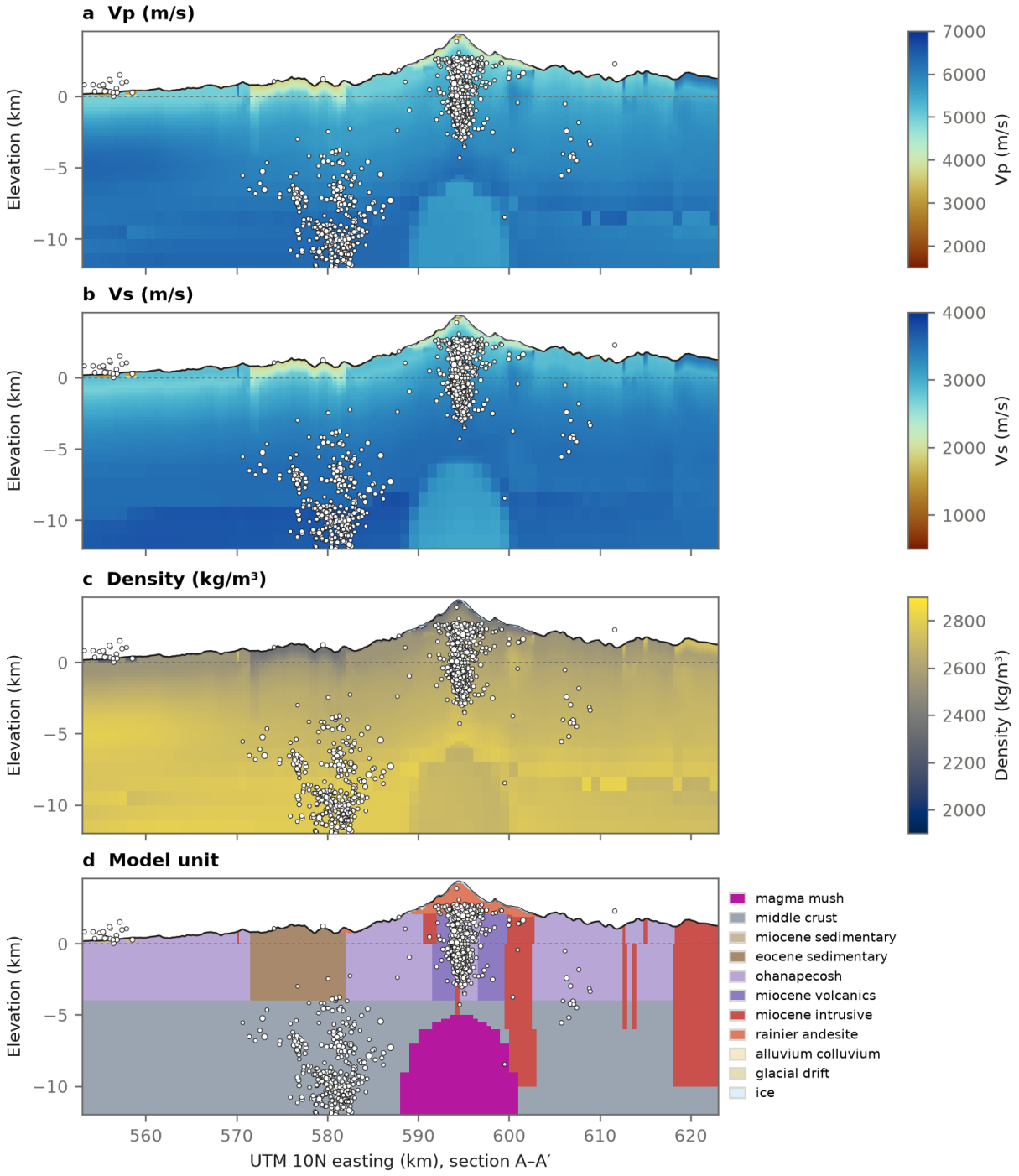


Figure 16. Fused model along A–A′ (west–east through the summit): (a) Vp, (b) Vs, (c) density, (d) model units. Light blue at the surface is glacier ice. White dots are PNSN earthquakes within 2 km of the section. The dashed line is sea level.

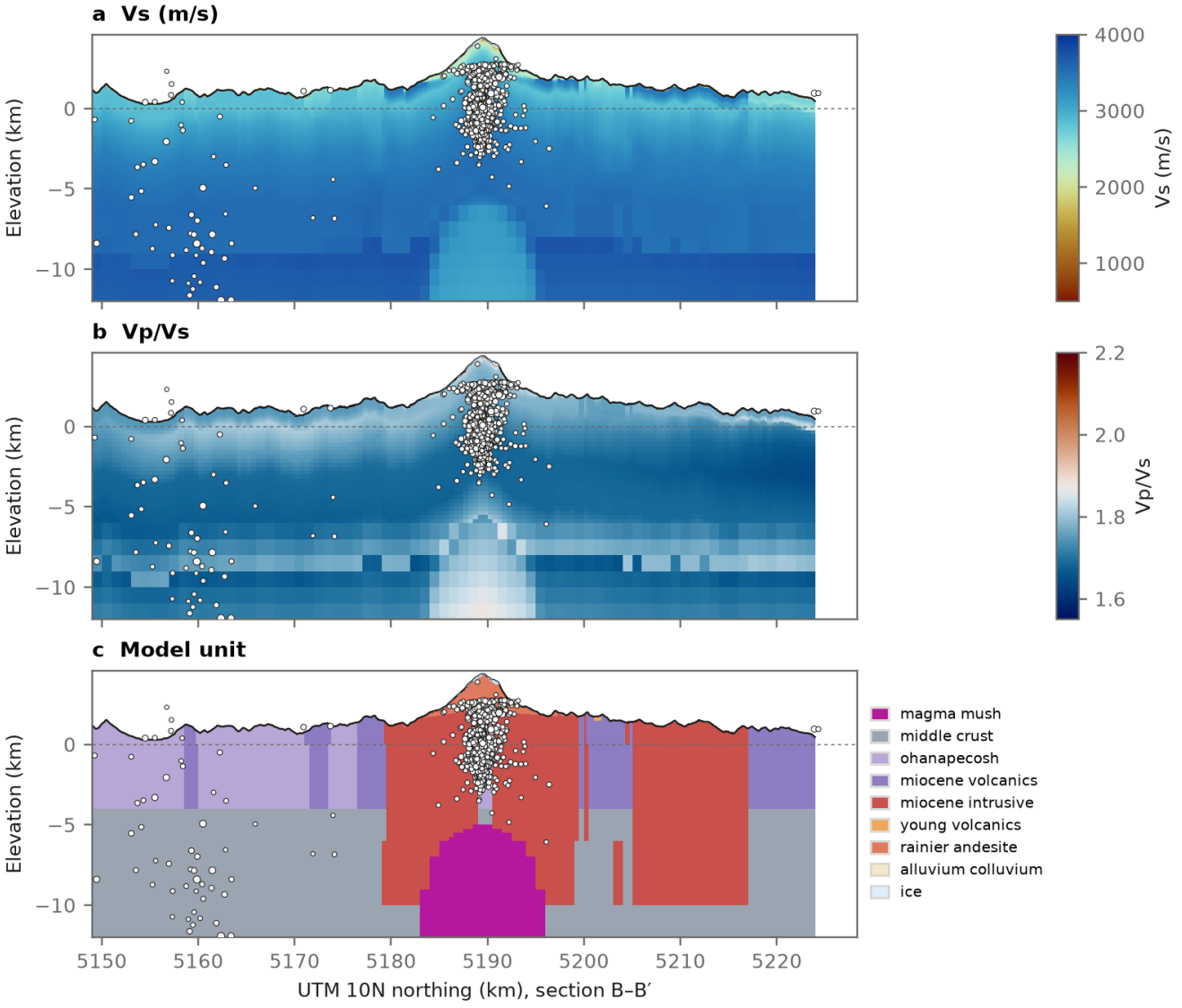


Figure 17. Fused model along B–B′ (south–north through the summit): (a) Vs, (b) Vp/Vs, (c) model units. The step in Vp/Vs at 9.9 km below the ground (the base of the Cascadia model used, `CVM_MAX_DEPTH` in `rainier3d.fusion.regional`) marks the change from the Cascadia model to CRESCENT with Brocher's Vp.

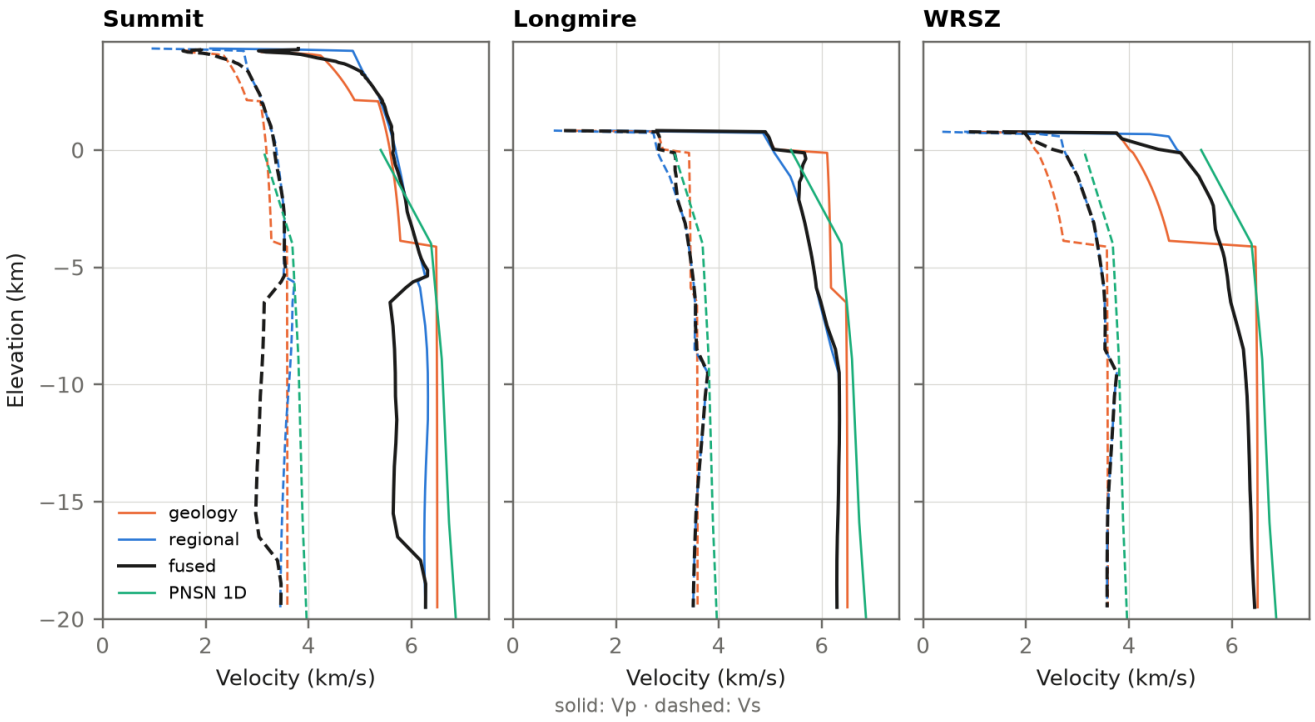


Figure 18. Vertical profiles of Vp (solid) and Vs (dashed) at the summit, at Longmire and in the WRSZ, for the geology, regional and fused models and the PNSN 1D model.

the model product; it does not replace `model.zarr` until the resolution tests and the station coverage of Supplement S1 have been extended.

6 Geodetic strain and stress at depth

6.1 GNSS data and velocities

Daily GNSS positions come from two archives:

- **PANGA.** The Pacific Northwest Geodetic Array (PANGA, Central Washington University) is the regional analysis centre (Pacific Northwest Geodetic Array, 2026): S17 (`scripts/17_gnss_fetch.py`) reads its daily GIPSY positions (`panga_raw.zip`, outlier-screened, not detrended) and its NA20 horizontal and vertical velocity fields for all 188 sites in the network box of 124.0–119.8° W, 45.6–48.2° N (`configs/gnss.yaml`, `network_bbox`), in its North-America-fixed frame; the PANGA series run from 1 January 2008 (`start`) to 18 July 2026, and 281 step epochs come from the headers of its DQRFIT trajectory fits.

– **UNR.** The Nevada Geodetic Laboratory (NGL, University of Nevada, Reno) supplies IGS20 North-America-fixed daily series (`.NA.tenv3`) for the 54 sites in the box that PANGA lacks (Blewitt et al., 2018), plus 20 sites that both archives process (the 20 longest PANGA series that UNR also holds), used only for frame alignment; the UNR series run to 1 September 2026 and the PANGA series to 18 July 2026, which is also the end of the regional daily strain series, and the NGL steps database adds 162 equipment and processing steps (code 1) for these sites. Every downloaded file is cached under `data/raw/gnss/` and listed with its URL, retrieval time and SHA-256 in `data/raw/gnss/manifest.csv`.

Together they give 1,050,643 site-days in 256 series.

– **Frame alignment.** The 20 sites that both archives process align UNR to PANGA (S18, `align_frames` in `scripts/18_gnss_strain.py`): the UNR minus PANGA differences of horizontal MIDAS velocity are fitted by least squares with a translation and a rotation about the centroid of the shared sites in the local UTM plane, with a residual RMS of 0.10 mm yr^{-1}, and the fitted correction is removed from the velocities and trends of the UNR-only sites; where both archives have a site, only the PANGA series is used.

– **Velocities.** Station velocities come from MIDAS (Blewitt et al., 2016), implemented in `rainier3d.geodesy.velocity`: the median of slopes between positions one year apart ($\pm$0.001 yr), skipping pairs that span a known step, trimmed at two scaled median absolute deviations (MAD) and taken again, with an uncertainty of 1.2533 $\times$ 1.4826 MAD / $\sqrt{(\mathrm{N}/4)}$ floored at 0.2 mm yr^{-1}. Series shorter than 365 days or 2.5 years (`min_years`) are not used, which leaves 203 sites (169 PANGA, 34 UNR). Equipment steps are taken from the PANGA fit headers and the UNR steps database. Each component is also fitted with a trajectory model (trend about 2015.0, annual and semiannual terms and steps; Huber-weighted least squares, k = 1.345, five iterations), whose annual terms and residuals feed the seasonal and daily strain.

– **Quality control.** Sites are flagged (S18) when their MIDAS horizontal velocity differs by more than 1 mm yr^{-1} from PANGA's published NA20 velocity (PANGA sites only), or by more than 2.5 mm yr^{-1} from the median velocity of at least three neighbours within 40 km (`configs/gnss.yaml`, `qc`; thresholds are author choices, `m1_placeholder`). This flags 22 of 203 sites (`outputs/gnss/summary.json`, `qc_flagged`): 11 on the PANGA test alone, 9 on the neighbour test alone and 2 (MUIR, V096) on both. They include the high edifice sites MUIR, CSHR, SNRS, PNHG and PNHR; PNHG moves 44 mm yr^{-1} east, which is not tectonic, and PNHR departs from its neighbours by 266 mm yr^{-1}. Flagged sites are kept in `outputs/gnss/velocities.csv` with their flag and left out of the strain fits.

The data are fetched again every week, and the products are recomputed (Sect. 9). The numbers below are for the series ending on 1 September 2026, which are frozen in `configs/gnss.yaml`.

6.2 Strain rate

The horizontal strain-rate tensor $\dot{\varepsilon}_{ij}$ is estimated from the velocities.

Table 20. Secular strain rates from uniform-strain fits to MIDAS velocities (S18, `outputs/gnss/summary.json`).

Region	Sites	Dilatation (nanostrain yr^{-1})	Maximum shear (nanostrain yr^{-1})	Azimuth of $\dot{\varepsilon}_1$
WRSZ (polygon + 25 km)	17	-16.6 ± 3.5	11.1	136°
Edifice (within 25 km of the summit, quality-controlled)	6	-27.9 ± 12.4	4.3	171°
Edifice, all sites	11	$+120.5 \pm 11.8$	61.2	103°

- **Grid.** On a 5 km grid over the network box (64 × 59 nodes in EPSG:32610, 3502 with a solution), a uniform velocity-gradient field is fitted at each node by weighted least squares to the quality-controlled MIDAS velocities, with Gaussian distance weights $\exp(-(d/s)^2)$ times inverse-variance weights (Shen et al., 2015). The smoothing scale s is tried in the order 5, 7.5, 10, 15, 20, 30, 40 and 60 km until the summed weight reaches 6 and at least four stations have a weight above 0.05 (`configs/gnss.yaml`, `strain`; author choices); inside the model box the scale is 30 km at 134 of the 210 nodes and 40–60 km elsewhere. The result is `data/processed/gnss/strain_grid.nc` (S18).
- **Regions.** For the two regions of interest it is a uniform-strain fit (the same gradient fit with equal distance weights) to the MIDAS velocities of the quality-controlled stations around them: for the West Rainier Seismic Zone (WRSZ), the stations within 25 km of the polygon 122.10–121.88° W, 46.60–47.00° N, drawn around the band of Pacific Northwest Seismic Network seismicity 12–18 km west of the summit (U.S. Geological Survey, 2026a); for the edifice, the stations within 25 km of the summit (121.7603° W, 46.8523° N). The regions are defined in `configs/gnss.yaml` (`regions`); the radii are author choices (`m1_placeholder`).

The reported components are the dilatation $\dot{\varepsilon}_{xx} + \dot{\varepsilon}_{yy}$, the maximum shear $\sqrt{(\dot{\varepsilon}_{xx} - \dot{\varepsilon}_{yy})^2/4 + \dot{\varepsilon}_{xy}^2}$, the principal rates and their azimuth, the rotation and the second invariant. The grid also carries the amplitude and peak day of the annual areal strain, the seasonal signal of hydrological loading.

Both regions contract at a few tens of nanostrain per year (Table 20, Fig. 20a), as the rest of the model box does (median -19 nanostrain yr^{-1}). The edifice result depends on quality control: with the flagged summit sites included, it changes sign. On the grid, the summit cell has a dilatation rate of -14.6 nanostrain yr^{-1} at a smoothing scale of 30 km, because the flagged summit sites are excluded. The annual areal-strain amplitude has a median of 15 nanostrain over the box (5–95%: 4–39), peaking in late August.

6.3 Daily strain and the summit swarms

The daily uniform strain of each region's stations is fitted to trajectory residuals, with trend, annual and semiannual terms and steps removed per site. Stations joining or leaving the network therefore do not shift the series (Fig. 19). Across the two summit swarms, the 60-day changes are small (Table 21).

Table 21. Median absolute deviation (MAD) of the daily strain and 60-day change in areal strain across the summit swarms, in nanostrain (S18).

Region	Daily scatter (MAD)	13 February 2009 swarm	8 July 2025 swarm
WRSZ	20 [TODO: not reproduced; 23 from `outputs/gnss/strain_wrsz.csv`, no script computes it]	+6	+7
Edifice	106 [TODO: not reproduced; 133 from `outputs/gnss/strain_edifice.csv`, no script computes it]	no quality-controlled coverage (fewer than four quality-controlled sites before 9 September 2012)	−166, inside the 411-nanostrain scatter of the 30-day medians

Neither swarm shows a transient above about 10 nanostrain in the WRSZ. On the edifice, winter excursions of up to 1500 nanostrain [TODO: the 30-day median of `outputs/gnss/strain_edifice.csv` reaches 2335 nanostrain; state the period and statistic] come from the close-spaced high sites. With stations a few kilometres apart, a few millimetres of snow-related antenna error become about 1000 nanostrain of apparent strain. Resolving strain of the edifice or of a magma body needs other instruments: tiltmeters, borehole strainmeters or interferometric synthetic-aperture radar.

6.4 Stress from the edifice load

The weight of the edifice is a static load on the crust beneath it.

- **Load.** The edifice is taken as everything above the pre-volcanic surface of the geology model (surface node `edifice_base`, inside the edifice footprint; Sect. 5.3.1). Its 100 m columns of height h are summed onto 500 m cells as vertical Boussinesq point loads $P_k = \rho g h_k A_k$ on an elastic half-space (Johnson, 1985): 1131 cells with a total weight of 2.34 $\times$ 10^{15} N (95 km^3 of rock at 2500 kg m^{-3}, g = 9.81 m s^{-2}), on a flat reference plane at 1789 m, the mean elevation of the edifice base under the load (S18, `rainier3d.geodesy.load`; `configs/gnss.yaml`, `load`; ρ and Poisson's ratio ν = 0.25 are `m1_placeholder` values).
- **Output.** The six stress components (Pa, tension positive), the mean stress, the maximum shear (half the difference of the extreme principal stresses) and the most and least compressive principal stresses are computed at every node of the model levels L1 (250 m horizontal, 50 m vertical, 25 to 4375 m elevation), L2 (500 m, 250 m, −125 to −5875 m) and L3 (1000 m, 1000 m, −6500 to −19,500 m) and written to `data/processed/edifice_load.zarr`; nodes less than 250 m below the reference plane (above 1539 m) are left empty, where the point-load field is singular.
- **Checks.** Two unit tests on a single 10^{12} N point load check the implementation (`tests/test_geodesy.py`): the vertical stress integrated over a plane 2 km below the load (200 m sampling over ±60 km) balances the load within 1%,

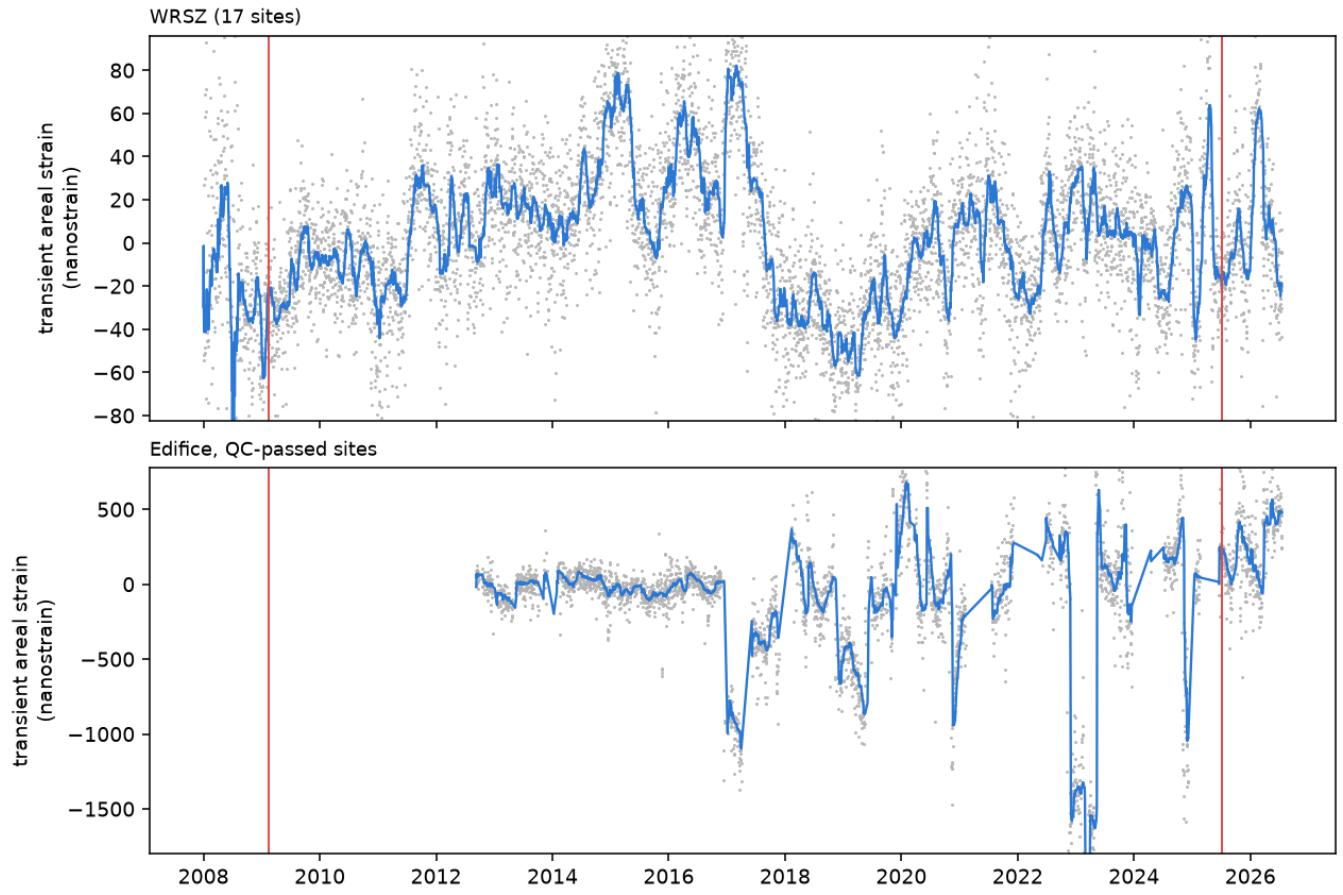


Figure 19. Daily areal strain of the stations around the WRSZ and the edifice, from trajectory residuals (grey), with 30-day medians (blue). Red lines mark the 2009 and 2025 summit swarms.

Table 22. Stress from the edifice load beneath the summit (compression negative; S18, `edifice_load.zarr`).

Elevation (m)	Vertical stress (MPa)	Mean stress (MPa)	Maximum shear (MPa)
0	−38.5	−18.7	15.2
−2000	−24.0	−9.4	11.2
−5000	−13.0	−4.4	6.6
−11,500 (magma-body centre)	−4.9	−1.5	2.6

and the divergence of the stress, by 1 m central differences at a point off the axis, is below 10^{-7} of the local vertical stress per metre; the on-axis branch matches the $r \to 0$ limit to 10^{-4}.

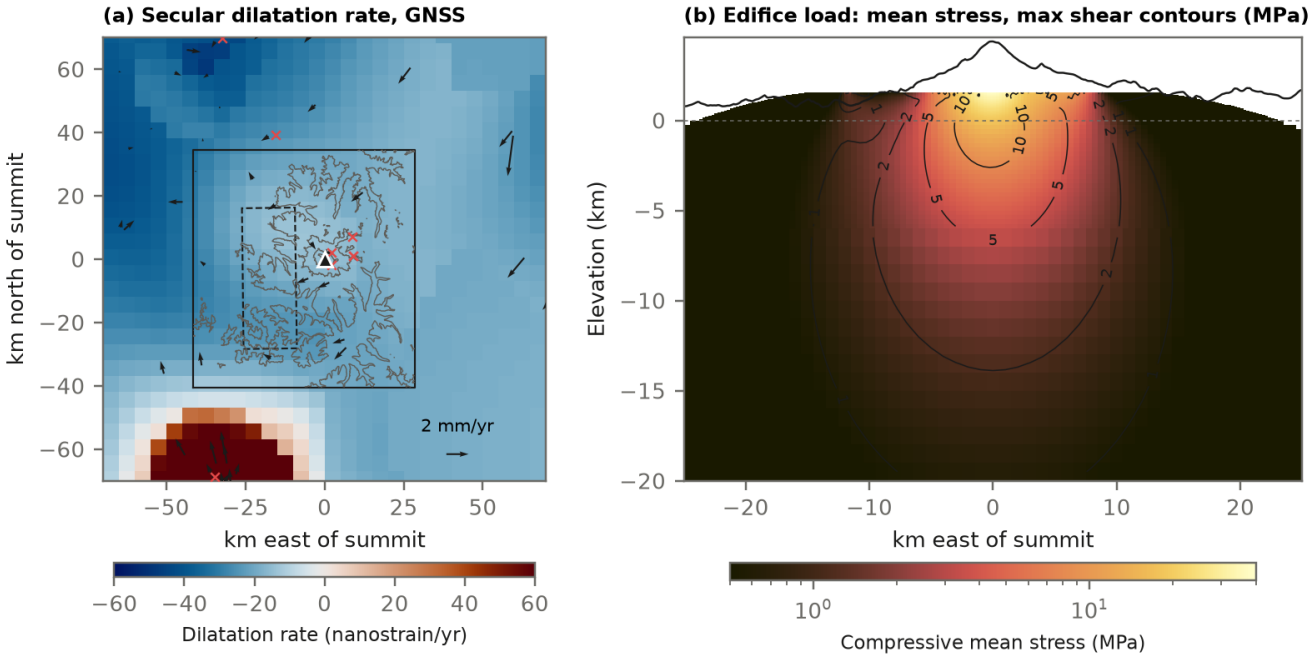


Figure 20. (a) Secular dilatation rate from the GNSS velocities, with the velocities of quality-controlled sites relative to the network median (arrows) and flagged sites (red crosses). The solid box is the model domain and the dashed polygon the WRSZ region. (b) Compressive mean stress from the edifice load along a west–east section through the summit, with contours of maximum shear (MPa).

The load stress decays from tens of megapascals beneath the summit to a few megapascals at the depth of the magma body (Table 22, Fig. 20b). At the tectonic stress rate implied by the geodetic strain rates (about 1 kPa yr^{-1} for a shear modulus of 30 GPa and 3×10^{-8} yr^{-1}), the load stress beneath the summit (1.5–38.5 MPa, Table 22) corresponds to 10^3–10^4 years of tectonic loading. If the tectonic background stress is no larger than this accumulated loading, the load stress dominates the orientation of stresses in the upper crust beneath the edifice; the background stress is not known (Sect. 6.5.1). The model is a homogeneous half-space with a flat reference plane, and its Poisson's ratio (0.25) and density (2500 kg m^{-3}) are author choices.

6.5 Strain in the model volume

The volume step of S18 (formerly script S25) puts two strain fields on the 500 m × 250 m grid of the three-dimensional viewer, from the surface to 20 km below sea level. Both are written with the axes of maximum shortening, so that they can be compared with the fast directions of shear-wave splitting.

Tectonic strain rate at depth. GNSS constrains the horizontal strain rate at the surface only. The horizontal tensor of Table 20 is carried down unchanged through the elastic upper crust. The vertical component follows from plane stress, $\dot{\varepsilon}_{zz} =$

Table 23. Tectonic strain rate in the WRSZ (S18, `outputs/gnss/strain_3d_summary.json`).

Quantity	Value
Axis of maximum shortening	N48°E
Maximum shear strain rate	10.4 nanostrain yr^{-1}
Right-lateral shear strain rate on WRSZ-parallel planes	10.0 nanostrain yr^{-1}
Normal strain rate across the zone	-12.9 nanostrain yr^{-1} (contraction)

$-\nu/(1-\nu)\,(\dot{\varepsilon}_{xx}+\dot{\varepsilon}_{yy})$, with $\nu = 0.25$. Both steps are assumptions, not observations. The tensor is then resolved on vertical planes parallel to the WRSZ. The zone's strike, N174°E, is the long axis of its 1292 epicentres (elongation 3.1).

At 5 km below sea level inside the WRSZ polygon:

The planes of maximum shear strike N3.5°E and N93.5°E. The WRSZ lies 9° from the first, so the geodetic strain rate shears it in the right-lateral sense at close to the maximum shear rate (Table 23, Fig. 21). This is a statement on the strain rate at the surface, extended to depth by assumption; it says nothing on the stress state or the friction on the zone.

Static strain of the edifice load.

- **Method.** The Boussinesq stress of Sect. 6 is converted to strain with the local stiffness of the velocity model, $\mu = \rho V_S^2$ and $\lambda = \rho V_P^2 - 2\mu$, with V_P, V_S and ρ of the fused model resampled to the 500 m × 250 m viewer grid (rock cells only, down to -20 km); the full strain tensor, the volumetric and areal strain, the horizontal maximum shear strain, the SHmax azimuth and the mean stress are written to `data/processed/strain_3d.zarr` (S25).
- **Cone interior.** Inside the cone above the half-space, that is higher than 1539 m, the stress is taken as the laterally confined overburden, $\sigma_{zz} = -\rho g d$ and $\sigma_h = \nu/(1-\nu)\,\sigma_{zz}$, with ρ = 2500 kg m^{-3}, g = 9.81 m s^{-2}, ν = 0.25 and d the depth below the local ground. This approximation defines the volumetric strain but no horizontal stress direction. The level 1539 m is the reference plane (1789 m) minus the 250 m exclusion depth, and these cells are marked `load_method` = 2 in the product (1 for half-space cells, 0 for air).

Beneath the summit (Table 24), the volumetric strain is compressive throughout. It is largest at the base of the cone and decays below. Neither the load nor the geodetic field produces dilatation in the edifice above sea level, where the shallow swarms of the summit occur (Fig. 22). The GNSS areal strain rate there is -15 nanostrain yr^{-1}. Dilatation in that volume would need a source these models do not contain, such as the pressurisation of the hydrothermal system.

SHmax of the load is tangential around the summit at and above sea level and radial from 5 km below sea level down (Fig. 23). The horizontal shear strain of the load reaches 1–30 microstrain within 20 km of the summit, which is 10^2–10^3 years of accumulation at the geodetic rates (per-cell ratio of the load shear to the geodetic shear rate within 20 km of the summit: 103, 377 and 1440 years at the 5th, 50th and 95th percentiles; `data/processed/strain_3d.zarr`, variables `load_max_shear_h` and `tect_max_shear_rate`).

The two fields are published on regular grids at fixed elevations (`strain_orientation.csv`):

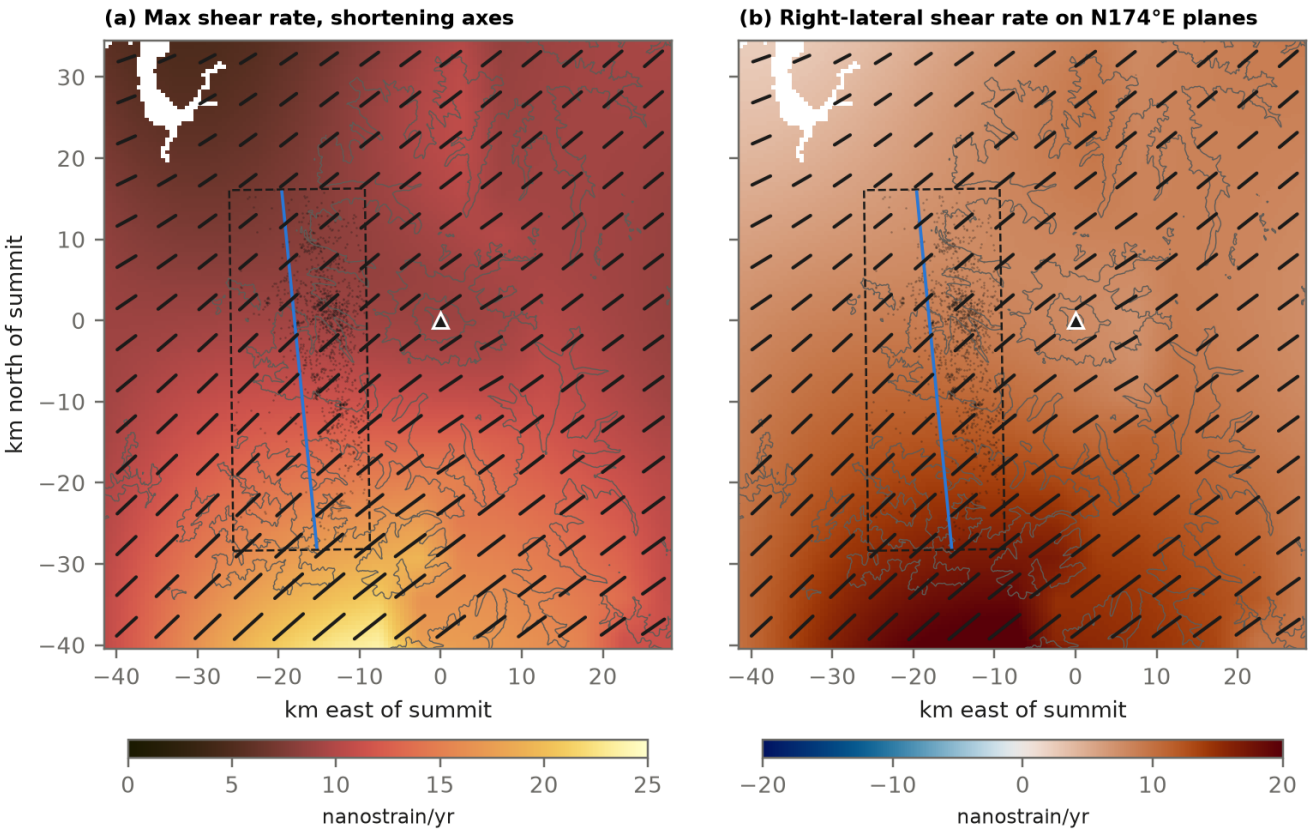


Figure 21. Tectonic strain rate from GNSS, the same at every depth under the stated assumptions: (a) maximum horizontal shear strain rate with the axes of maximum shortening (bars, length scaled with the shear rate); (b) right-lateral shear strain rate on vertical planes parallel to the WRSZ. The dashed polygon is the WRSZ region, grey dots its epicentres, and the blue line the strike fitted to them.

Table 24. Volumetric strain of the edifice load beneath the summit (S18, `outputs/gnss/strain_3d_summary.json`). Above 1539 m the stress is the confined overburden.

Elevation (m)	2500	1500	1000	500	0	−2000	−5000	−11,500
Volumetric strain (microstrain)	−568	−826	−672	−527	−428	−198	−73	−28

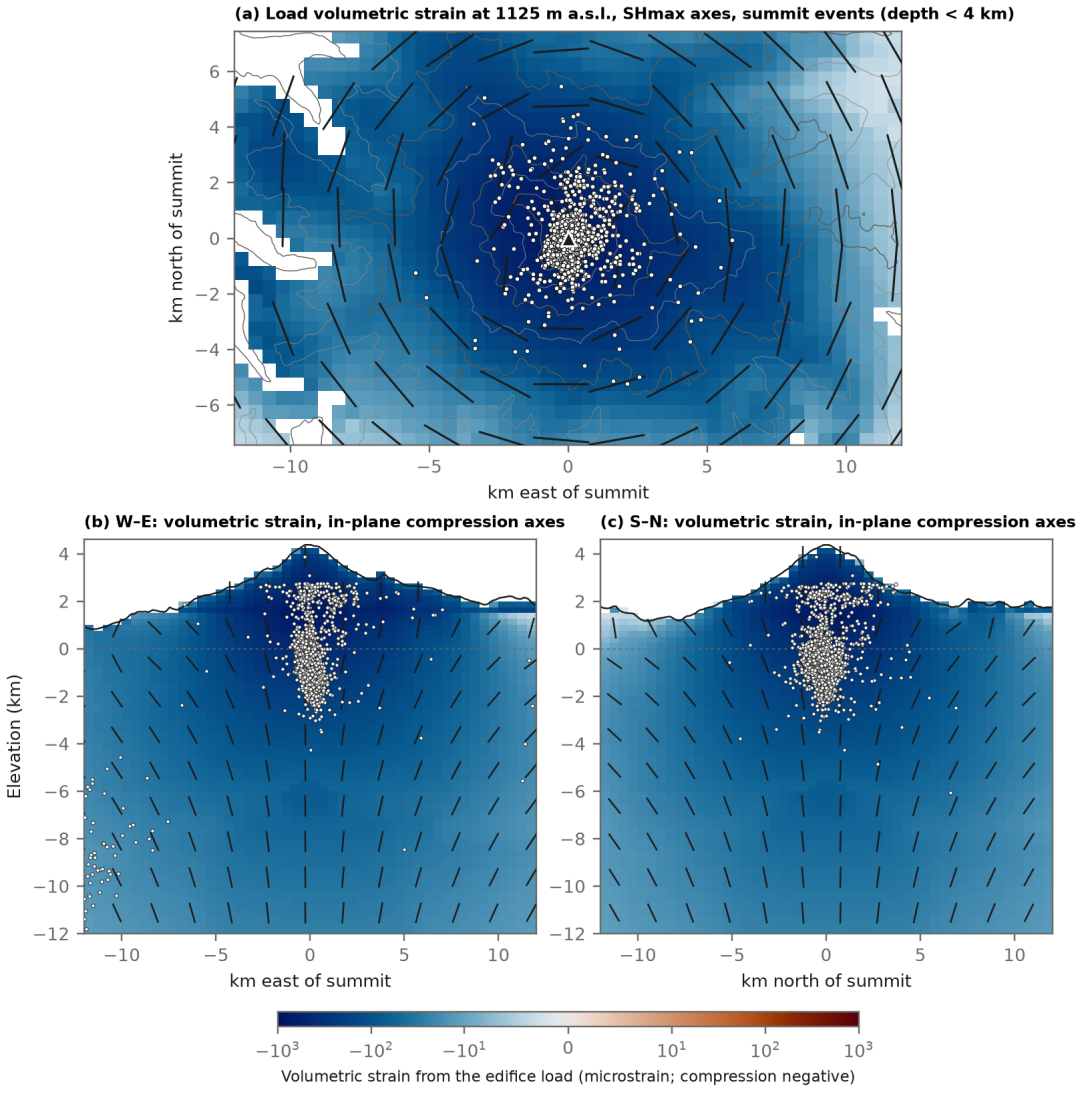


Figure 22. Static strain of the edifice load. (a) Volumetric strain at 1125 m above sea level, with the direction of the most compressive horizontal stress (SHmax, bars) and the summit earthquakes shallower than 4 km (dots). (b, c) West–east and south–north sections through the summit, with the in-plane axis of maximum compression; inside the cone, the confined overburden makes it vertical.

- the geodetic shortening axes (azimuth of maximum horizontal shortening, with the maximum horizontal shear strain rate and the right-lateral shear strain rate on WRSZ-parallel planes), every 5 km over the model box, at the grid levels nearest to 1500, 1000, 0, −2000, −5000, −10,000 and −15,000 m (1625, 1125, 125, −1875, −4875, −9875 and −14,875 m);
- the SHmax azimuth of the load, with its horizontal maximum shear strain and volumetric strain, every 2 km within 20 km of the summit at the same levels except inside the cone, where SHmax is undefined (2942 rows for both fields, `configs/gnss.yaml`, `volume.bars`).

These grids are for comparison with the fast directions of shear-wave splitting, which aligned cracks orient parallel to the most compressive horizontal stress.

6.5.1 How the strain fields are computed

Both fields are first-order estimates. Neither is computed with Green's functions of the heterogeneous model or with a rheology other than linear elasticity. This subsection gives the calculation as implemented (`rainier3d.geodesy.load`, `rainier3d.geodesy.volume`, script S18) and then lists what it leaves out.

Tectonic strain rate. The horizontal strain-rate tensor $\dot{\varepsilon}_{\alpha\beta}(x,y)$, with $\alpha,\beta \in \{x,y\}$, is the velocity-gradient fit of Sect. 6 at the surface. At depth it is not calculated but assigned:

$$\dot{\varepsilon}_{\alpha\beta}(x,y,z) = \dot{\varepsilon}_{\alpha\beta}(x,y,0), \qquad \dot{\varepsilon}_{zz} = -\frac{\nu}{1-\nu}\left(\dot{\varepsilon}_{xx}+\dot{\varepsilon}_{yy}\right), \qquad \dot{\varepsilon}_{xz} = \dot{\varepsilon}_{yz} = 0. \tag{14}$$

The vertical component follows from a vanishing vertical stress rate (plane stress). This is the thin-plate view of the upper crust: it holds when the sources of deformation are far or deep compared with the depth of interest and the crust is laterally uniform. No source (megathrust locking, fault slip, block rotation) is inverted from the GNSS velocities.

The shear on the WRSZ is resolved as

$$\dot{\gamma} = -\,\mathbf{s}\cdot\dot{\boldsymbol{\varepsilon}}\cdot\mathbf{n}, \tag{15}$$

with $\mathbf{s}$ the horizontal unit vector along the strike and $\mathbf{n}$ the horizontal normal. The sign is positive for right-lateral motion. The strike is the long axis of the covariance of the WRSZ epicentres.

Edifice-load stress. The load is the rock above the pre-volcanic surface of the geology model. Its columns of height h are summed on 500 m cells into vertical point forces $P_k = \rho g h_k A_k$ (ρ = 2500 kg m^{-3}). The forces act on a flat reference plane at the mean elevation of the edifice base (1789 m). The stress at a point is the sum over k of the Boussinesq solution for a point force on the surface of a homogeneous, isotropic, linear-elastic half-space (Johnson, 1985). At horizontal distance r and depth z below the plane, with $R = \sqrt{r^2+z^2}$ and tension positive:

$$\sigma_{zz} = -\frac{3Pz^3}{2\pi R^5}, \quad \sigma_{rz} = -\frac{3Prz^2}{2\pi R^5}, \quad \sigma_{rr} = \frac{P}{2\pi}\left[\frac{1-2\nu}{r^2}\left(1-\frac{z}{R}\right)-\frac{3r^2z}{R^5}\right], \quad \sigma_{\theta\theta} = -\frac{P(1-2\nu)}{2\pi}\left[\frac{1}{r^2}\left(1-\frac{z}{R}\right)-\frac{z}{R^3}\right]. \tag{16}$$

These stresses depend on Poisson's ratio ($\nu = 0.25$) but not on the elastic moduli. Points within 250 m below the plane are left out, where the point-force field is singular.

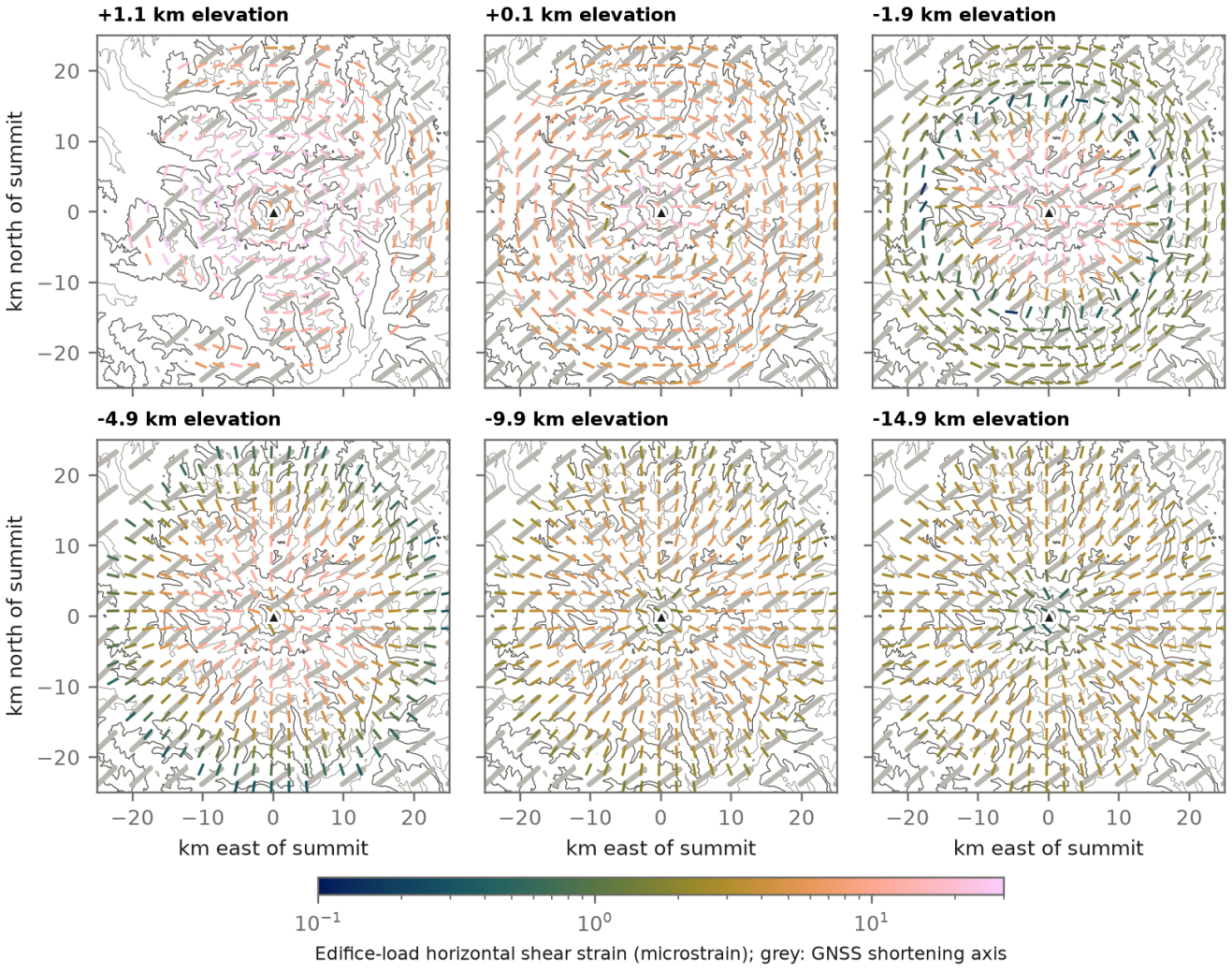


Figure 23. Axes of maximum horizontal shortening at six elevations: GNSS (grey, the same at every depth) and SHmax of the edifice load (coloured by its horizontal shear strain).

Edifice-load strain. The stress is converted to strain with the isotropic compliance of each cell:

$$\varepsilon_{ij} = \frac{\sigma_{ij}}{2\mu} - \frac{\lambda \sigma_{kk}}{2\mu(3\lambda + 2\mu)} \delta_{ij}, \qquad \mu = \rho V_S^2, \quad \lambda = \rho V_P^2 - 2\mu, \tag{17}$$

with V_P, V_S and ρ taken from the fused model.

- **Cone interior.** Above the half-space (higher than 1539 m), the stress is the laterally confined overburden, $\sigma_{zz} = -\rho g d$ and $\sigma_{xx} = \sigma_{yy} = \nu \sigma_{zz}/(1-\nu)$, with d the depth below the local ground.
- **SHmax.** It is the eigenvector of the horizontal stress tensor $(\sigma_{xx}, \sigma_{yy}, \sigma_{xy})$ with the most compressive eigenvalue. It is undefined in the cone interior, where the horizontal stress is isotropic. Azimuths are in degrees east of north, from 0 to 180°.

What the calculation does not include.

- **Green's functions of the 3D model.** The stress is that of a uniform half-space, so the stiffness contrasts of the model neither concentrate nor divert it. Applying the local compliance (Eq. 17) makes soft rock strain more under the same stress. The stress field, however, is not in equilibrium in the heterogeneous medium, and its axes do not rotate around stiff plutons or the soft edifice.
- **Topography as part of the elastic body.** The load sits on a flat plane. The cone above it is not part of the elastic domain, and its interior is approximated in one dimension.
- **Rheology beyond linear elasticity.** There is no viscoelastic relaxation of the warm lower crust or of the magma body, no plasticity, no pore-pressure coupling and no thermal stress.
- **A tectonic source model.** The GNSS rates are not explained by locking, fault slip or block motion, so their variation with depth (Eq. 14) is assumed rather than computed.
- **Absolute stress.** The load gives a static stress and GNSS a strain rate. Without the magnitude of the tectonic background stress the two cannot be summed, so their orientations are reported separately.

A consistent model would compute the same quantities in three steps:

1. a static finite-element solution on the model mesh, with gravity acting on the real topography and heterogeneous λ and μ from the fused velocities;
2. Green's functions computed in the same 3D elastic model, used to invert the GNSS velocities for their sources before computing strain rates at depth;
3. a viscoelastic lower crust for time-dependent loading.

The load pattern of Fig. 23, tangential beneath the summit and radial at depth, is the qualitative signature of any load on a cone. Heterogeneity would change its magnitudes and rotate its axes near stiffness contrasts. The depth extension of the GNSS axes is the least certain part of the product.

7 Geohydrology

Water controls much of what rainier3d describes:

- the seismic velocities of the shallow crust;
- the conductivity that maps alteration;
- the seasonal loading seen by GNSS (annual areal-strain amplitude of median 15 nanostrain over the model box, Sect. 6);
- the lahars and debris flows of Sect. 8.

The model compiles the hydrological information needed for a hydrological description of the mountain. It holds no hydrological state: it neither models groundwater flow nor assimilates hydrological observations. Table 25 lists what it holds.

The water table. The two estimates differ in level and in pattern.

- **Ma et al. (2026).** The random-forest estimate of Ma et al. (2026b) (ensemble mean on the ~24 m CONUS2 grid, block-averaged to 100 m by S2) keeps the water table within 20 m of the ground in 95% of the cells of the box (median 10.1 m, 95th percentile 20.4 m, maximum 117 m).
- **Fan et al. (2017).** The global groundwater model of Fan et al. (2017) (annual mean at 30″, about 1 km, block-averaged to 100 m) places it hundreds of metres down under ridges: median 23 m, 95th percentile 306 m, maximum 847 m, and deeper than 100 m in 27% of the box.

Water table and depth to rock. The water table also sits between the two estimates of depth to rock (Sect. 4.3, Fig. 5): below the 1.41 m median SOLUS soil and above the 19.6 m median SoilGrids bedrock in 92.8% of the cells. Whether it lies in unconsolidated regolith or in fractured rock is therefore not known from the data.

For the shallow seismic model the choice matters more than either value. A water table separates dry from saturated rock, and saturation raises Vp and Vp/Vs below it. The Ma et al. grid is used under its CC-BY-NC-ND licence for non-commercial research and is not redistributed. Its ensemble uncertainty is served through the HydroGEN platform and needs an account.

Groundwater observations. No aquifer maps exist for the park. The groundwater study of the upper White River (Fuhrig et al., 2024) is the only park-scale assessment we found. The surficial hydrothermal system described by Frank (1995) is represented only through the alteration it left.

Wells. Script S31 (`scripts/31_wells.py`, `configs/wells.yaml`) fetches the water levels and borehole logs of the box once, up to 23 September 2026, and caches them under `data/raw/wells/`: the well reports of the Washington Department of Ecology (Washington State Department of Ecology, 2026), requested without owner names or addresses; the groundwater levels of the USGS (U.S. Geological Survey, 2026d), one median per well; and the subsurface database of the Washington Geological Survey (Washington Geological Survey, 2026d), whose water wells are Ecology reports and are counted once. A static level deeper than the completed well or than 300 m is rejected. Table 26 compares the 5,591 accepted water levels with the two gridded water tables at the nearest 100 m cell; S31 writes the numbers to `outputs/wells/summary.json`.

Both grids place the water table deeper than the wells do, and both rank the wells poorly. The wells are a lowland sample: 97% lie below 600 m, so they test the grids where people drill, not on the volcano. The 23 bedrock picks of the borehole

Table 25. Hydrological information in rainier3d.

Component	Layer in rainier3d	Source	Section
Ice	glacier thickness and bed on the 100 m grid: 5.50 km^3 of ice in 219 glaciers covering 97.3 km^2, at most 363 m thick	IceBoost v2 (Maffezzoli, 2025), RGI 6.0 (RGI Consortium, 2017)	Sect. 4.1
Snow	NDSI of the Sentinel-2 L2A median composite of 1 August – 30 September 2025, the end of the melt season: median -0.48 (5–95%: -0.60 to -0.24)	Sentinel-2 (European Space Agency, 2025)	Sect. 4.2
Surface water	60,174 flowlines (1:24,000) with Strahler order 1–8 and mean annual flow (EROM QAMA), rasterised to stream order on the 100 m grid	NHDPlus HR (U.S. Geological Survey, 2026b)	Sect. 4.2
Soil storage	soil thickness (depth to a lithic contact): median 1.41 m (5–95%: 0.54–1.97 m), with the 2.01 m maximum read as $\geq$ 2 m; texture, bulk density and rock fragments at 0–1.5 m	SOLUS100 (Nauman et al., 2024)	Sect. 4.2, Sect. 4.3
Soil hydraulics	saturated hydraulic conductivity and van Genuchten θs, θr, α and n in six layers to 2 m; 0–1 m harmonic-mean conductivity median 1.6×10^{-5} m s^{-1}, porosity 0.65	POLARIS (Chaney et al., 2019)	Sect. 4.3
Regolith	depth to bedrock: median 19.6 m (5–95%: 11.4–26.8 m)	SoilGrids250m 2017 (Shangguan et al., 2017)	Sect. 4.3
Groundwater	water-table depth, two estimates block-averaged to 100 m: random forest, median 10.1 m (5–95%: 4.8–20.4 m); global model, median 23 m (0–306 m)	Ma et al. (2026) (Ma et al., 2026b); Fan et al. (2017) (Fan et al., 2017)	Sect. 4.2
Interception and transpiration	canopy height (airborne lidar and ETH 2020 at 10 m, GEDI L3 at 1 km), lidar vegetation cover, Sentinel-2 leaf area index, GEDI L2B plant area index and GEDI L4B aboveground biomass	lidar, Sentinel-2, GEDI	Sect. 4.4
Hydrothermal fluids	alteration intensity (0–1) from apparent resistivity at four frequencies, sensed to at most 150 m below the ground or glacier bed; 10.6 km^2 of the 126 km^2 of surveyed edifice lava reach 0.5 or more	1996 helicopter EM survey (Rystrom et al., 2000)	Sect. 5.3.2
Hydrological loading	annual areal-strain amplitude and peak day on the 5 km GNSS strain grid: median 15 nanostrain over the model box (5–95%: 4–39), peaking in late August	GNSS (S18)	Sect. 6

Table 26. Water levels in wells against the gridded water tables (S31). Ecology 3,500 wells (719 levels rejected), USGS 1,944, Washington Geological Survey 147 boreholes that are not Ecology reports. The 600 m split separates the Puget lowland from the upland; only 142 wells lie above it.

Water table	Wells	Observed median	Gridded median	Median of gridded − observed	Median absolute difference	Spearman r
Ma et al. (2026), all	5,591	5.8 m	16.1 m	2.7 m	7.1 m	0.27
Ma et al. (2026), at or above 600 m	142	4.6 m	9.1 m	2.7 m	3.5 m	0.27
Fan et al. (2017), all	5,591	5.8 m	16.8 m	3.3 m	12.3 m	0.27
Fan et al. (2017), at or above 600 m	142	4.6 m	6.4 m	3.6 m	8.5 m	−0.13

database, all in the lowland, have a median depth of 40.5 m, against 22.4 m for SoilGrids and the 2 m reach of SOLUS at the same cells.

How the layers can enter the seismic model.

- **Fluid substitution.** The critical-zone columns separate unsaturated from saturated material at the water table of Fan et al. (2017) and apply Gassmann substitution with a patchy fluid (Sect. 5.1.2); the crack-closure law of the crustal cells takes hydrostatic pore pressure below the same water table (Sect. 5.1.1).
- **Valley fill.** Valley fill along the mapped streams would appear as slow, high-Vp/Vs bodies; at present the unconsolidated deposits take the thickness stated in the map unit description for eight map symbols and fixed defaults elsewhere (glacial drift and lahar deposits 15 m, alluvium and colluvium 10 m, Sect. 5.3.1).
- **Hydrothermal core.** A fluid-saturated core would join the alteration field, which at present reaches only the top ~150 m sensed by the EM survey (Sect. 5.3.2).
- **Loading.** The annual strain signal (peak in late August) and the snow and water storage could be compared as a loading model; the model holds only end-of-season NDSI, not snow water equivalent.

Only the first coupling is applied, through the critical-zone columns (Sect. 5.2).

8 Mass movements: landslides, lahars, debris flows and avalanches

Mass movements are Rainier's most frequent hazard. Debris flows and outburst floods from its glaciers recur on a scale of years (Walder and Driedger, 1995; Legg et al., 2014), rock and ice avalanches fall from its steep upper slopes (Crandell and Fahnestock, 1965), and its lahars have reached the Puget Lowland (Crandell, 1971; Vallance and Scott, 1997). This section defines a catalogue of these events on the model grid: its scope, its record structure and its sources (Table 28). The catalogue adds a time-dependent layer to the model and provides labelled sources for the seismic detection of surface events.

Scope. Four classes of event:

- landslides, including rock avalanches and edifice collapses (catalogue classes `rock_avalanche` for rock falls, topples and rock or ice avalanches, `slide` for slides and debris slides, and `complex` for complex or unrecorded movement types);
- lahars, kept as the mapped outlines of the Qvl units of the 1:100,000 geologic map (Osceola Mudflow, Electron Mudflow, other lahar deposits) (Washington Geological Survey, 2026a);
- debris flows, including glacial outburst floods that bulk into debris flows (inventory polygons of type Flow, Debris flow or Hyperconcentrated flows, and seismic events whose type names a flow or an outburst; class `debris_flow`);
- snow and ice avalanches (class `snow_ice_avalanche`: types naming snow, or an ice avalanche without rock; the two such events in the box are seismic events of Allstadt et al. (2017)).

For each event, the catalogue records the following:

- class (field `cls`, one of the five classes above) and date (`date`, ISO day, 377 of the 1,649 events), or the relative age of undated deposits (`age`, for example "Pre-historic (>150 years)", 272 events);
- source area and runout on the model grid: flows as polygons clipped to the model box (466 in `flows.gpkg`), events as points in EPSG:32610 (`located`: crown 1,623, seismic location 19, mapped point 7), with `crown_dem` giving the 3DEP resolution under each crown;
- volume where published (`volume_m3`, converted from ft^3 for the inventory: 260 events, 247 inventory deposits and 13 seismic events) and failure depth (`depth_m`, converted from ft, 247 events);
- the triggering conditions where known [TODO: `events.csv` has no trigger field; the 351 landslides of the January 2009 storm are identifiable only by date; add a field or drop this item];
- the source (`source`, a key of `configs/sources.yaml`, whose entry carries the licence; the Washington inventory states none) and a confidence (`confidence`: the inventory's confidence class, or the location uncertainty in km for seismic events).

The catalogue. Script S24 fetches four machine-readable sources, clips them to the box and writes the catalogue to `outputs/mass_movements/` (Table 27, Fig. 24). `docs/mass_movements.md` lists each service call with its endpoint, parameters, paging and cache path. The catalogue has two parts.

- **Flows.** Lahar deposits and debris flows keep their mapped outlines, because their runout is the information. The lahar deposits are the Qvl units of the 1:100,000 map (Washington Geological Survey, 2026a): the Osceola Mudflow (Vallance and Scott, 1997) covers 79.7 km^2 of the box, the Electron Mudflow 18.8 km^2 and other lahar deposits 16.0 km^2. The debris flows are the flow-type polygons of the Washington State Landslide Inventory Database (Washington Geological Survey, 2026c): 198 from its lidar-protocol mapping and 265 from its compilation of older mapping.
- **Events.** Every other mass movement is a point. The Allstadt et al. compilation of seismogenic mass movements (Allstadt et al., 2017) gives 19 events in the box with a time and a seismic location: the ten 2011 rock and ice avalanches of the Nisqually headwall, five rock falls from Russell Cliff in 1989 and 1992, a 2010 snow avalanche, a 2014 ice avalanche, and the August 2015 rock fall and debris flow. The inventory gives 1,624 landslide polygons of other types and seven recent-landslide points. A polygon becomes a point at its crown, the highest point of its outline, where failure began. The crown is sampled every 2 m on a 1 m window of the USGS 3DEP elevation service around each polygon (U.S. Geological Survey, 2026c), which returns the best 3DEP source at each point: 994 crowns lie on 1 m lidar, 441 on 3 m and 186 on 10 m data. Three windows were refused by the service on every attempt; those crowns are placed on the 30 m DEM and flagged. The compilation repeats three protocol deposits; those copies are dropped.

The inventory records a failure depth for every lidar-protocol deposit (453 deposits in the box, median 14.6 m), so these deposits could later be given a volume in the model. The 1998 lahar inundation zones (Hoblitt et al., 1998; Schilling et al., 2008) are kept as a separate hazard layer, not as events: the case 1 zone covers 964 km^2 of the box. The Exotic Seismic Events Catalog version 3 (Collins et al., 2025) extends the seismic compilation to 2025 but has no scripted download yet. The slope, local relief and valley depth at each event and in each flow deposit are compared with the whole box in Sect. 4.5 (Table 8).

Other machine-readable sources. The U.S. Landslide Inventory (Belair et al., 2025; Mirus et al., 2020) compiles the same state data nationally, and the compilation layer includes the park-wide mapping of 448 mass movements over 37 km^2 by Riedel and Dorsch (2016). PNSN classifies surface events in its own catalogue, but these classes are not in ComCat: within 25 km of the summit, ComCat lists 14,288 events and none of them is a surface event. The curated PNSN dataset of Ni et al. (2023) and the classifiers of Kharita et al. (2026) give labelled waveforms for training detectors.

Sources in the literature. Other sources are reports and papers, from which events are digitised.

- **Lahars.** Holocene lahars and their deposits: Crandell (1971), Scott et al. (1995), and the Osceola Mudflow (Vallance and Scott, 1997).
- **Rock avalanches.** The 1963 rock avalanches from Little Tahoma Peak (Crandell and Fahnestock, 1965).
- **Debris flows and outburst floods.** Along Tahoma Creek (Walder and Driedger, 1994, 1995), and debris-flow initiation in proglacial gullies (Legg et al., 2014).
- **River response.** River aggradation measured by repeat lidar (Anderson and Pitlick, 2014).
- **Seismic signatures.** Seismic and acoustic signatures of mass movements at volcanoes (Allstadt et al., 2018), and the lahar-detection network of the Cascades Volcano Observatory (Kramer et al., 2024).

Table 27. The mass-movement catalogue in the model box (S24, `outputs/mass_movements/summary.csv`, run of 2026-10-07). Seismic events are from Allstadt et al. (2017); mapped and recent events from Washington Geological Survey (2026c). Of the 1,649 events, 377 are dated; 351 of these are landslides of the January 2009 storm in the compilation.

Part	Class	Source	Count	Area (km^2)
Flows	Osceola Mudflow	Washington Geological Survey (2026a)	1	79.7
	Electron Mudflow	Washington Geological Survey (2026a)	1	18.8
	Other lahar deposits	Washington Geological Survey (2026a)	1	16.0
	Debris flows	Washington Geological Survey (2026c)	463	11.2
Events	Rock fall, rock and ice avalanche	seismic 16; mapped 40	56	
	Snow or ice avalanche	seismic	2	
	Debris flow, outburst flood	seismic	1	
	Slide, debris slide	mapped 316; recent 7	323	
	Complex or unknown type	mapped	1,267	

Table 28. Sources for the mass-movement catalogue.

Class	Main sources	Machine-readable
Landslides, rock avalanches	WGS inventory (with NPS mapping); U.S. Landslide Inventory; ESEC	yes (polygons; seismic events)
Lahars	Crandell (1971); Scott et al. (1995); Vallance and Scott (1997); USGS hazard zones	zones only; events from the literature
Debris flows, outburst floods	Walder and Driedger (1994, 1995); Legg et al. (2014); ESEC; PNSN surface events	partly
Snow and ice avalanches	Northwest Avalanche Center; PNSN surface events	on request

– **Snow avalanches.** Observations are held by the Northwest Avalanche Center (Northwest Avalanche Center, 2026), with programmatic access on request.

Each record carries its source, like every other layer. Records from the literature are to be entered from their tables with a citation to the page.

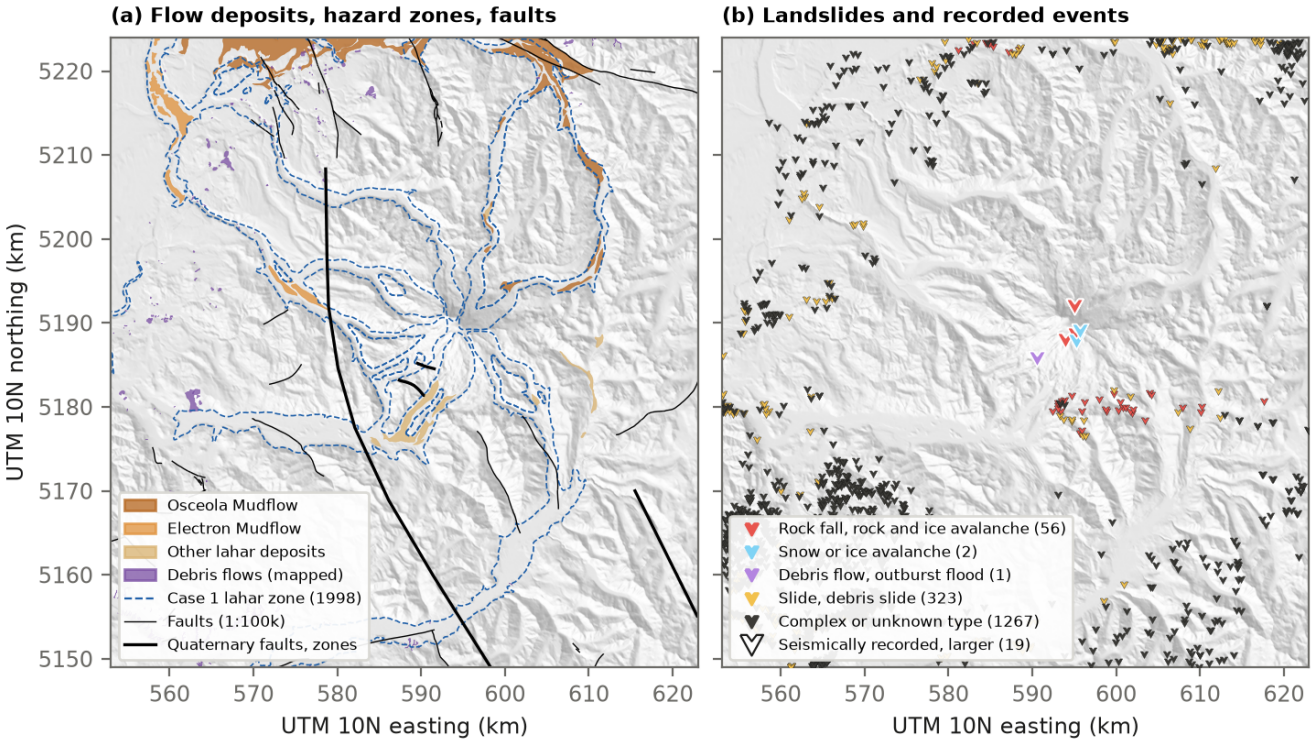


Figure 24. Mass movements and faults. (a) Lahar deposits and mapped debris flows, the case 1 lahar inundation zone of 1998 (dashed), and faults of the 1:100,000 map (thin) and of the Quaternary fault layer (thick). (b) Events by class, drawn as downward chevrons as in the viewer: landslides at their crown (small) and seismically recorded events (large, white rim).

9 Accessing the model

9.1 Products and the client

The derived products are published as assets of the release `products-v1.0.0` of the code repository, under CC-BY 4.0 (Table 29). The archive names and their SHA-256 checksums are listed in the release file `SHA256SUMS` and in the package catalogue `src/rainier3d/products.json`.

Table 30 states, product by product, what uncertainty a user receives and what is not quantified. Most gridded fields carry none per cell: their parameters are author choices within published ranges (`m1_placeholder`) or come from rule-based geometry, and the travel-time calibration constrains only three global multipliers and a depth profile. Two resampled third-party uncertainties are not in the release because the release holds only computed fields (Sect. 9); S5 and S2 rebuild them in `model.zarr`.

A products release holds only what rainier3d computes; `configs/products.yaml` classifies every variable of the model as computed, geometry or resampled, and a test fails on an unclassified one. Each release also carries `REBUILD.md`,

Table 29. Downloadable products [TODO: sizes and contents of products-v1.1.0, the first release under this rule].

Product	Size	Content
`model`	111 MB	the subsurface model, `model.zarr` (xarray DataTree, zarr v3, UTM 10N and NAVD88): levels L1–L3 (Table 2) with Vp, Vs, density, Qp, Qs, unit, alteration, and the geology and calibrated regional inputs, plus the surface layers that rainier3d computes (units, deposit thickness, ice, alteration, vegetation indices and the 0–1 m soil summaries); third-party fields that are only resampled onto the grid (water tables, soil and bedrock depths, Vs30, land cover, canopy height, streams, the `/soil` profiles) are left out and rebuilt by S2
`grids`	55 MB	the subsurface model on one uniform 500 m grid (S9): CF netCDF (vp, vs, rho, qp, qs, alteration, air, surface_elevation; UTM 10N, elevation up), EMC-style netCDF (longitude, latitude, depth), NonLinLoc P and S slowness grids
`strain_3d`	72 MB	strain in the volume (S18, Sect. 6.5) on a 500 m horizontal by 250 m vertical grid: GNSS strain rate carried down (areal, volumetric, maximum shear, shortening azimuth, right-lateral shear on planes parallel to the WRSZ) and the static strain of the edifice load (full tensor, volumetric, SHmax azimuth)
`edifice_load`	115 MB	stress from the weight of the edifice, Boussinesq half-space solution, on L1–L3 (S18)
`alteration`	1 MB	alteration from the 1996 helicopter electromagnetic survey (S22, Sect. 5.3.2; Rystrom et al. (2000); Finn et al. (2001)): 3D intensity in the top 200 m, bedrock elevation, resistivity and depth of investigation per frequency, apparent magnetisation; CF netCDF, UTM 10N
`mass_movements`	1 MB	the mass-movement catalogue (S24, Sect. 8): events, flow deposits and a summary (the fault traces and the 1998 lahar zones are copies of their sources and are left out), GeoPackage and CSV; the Washington landslide inventory states no licence
`gnss`	4 MB	station velocities (MIDAS (Blewitt et al., 2016)) with quality flags, strain-rate grid, daily regional strain series and the download manifest (S17, S18); refreshed every Monday at 09:00 UTC in the rolling release `gnss-latest`, with dated copies; `products-v1.0.0` holds the copy of 2026-09-01

Table 30. Uncertainty and fitness for use of each product. Numbers from `outputs/joint_calibration_cz/history.json`, `configs/velocity_calibration.yaml`, `model.zarr`, `data/processed/gnss/strain_grid.nc` and `outputs/catalog/catalog_relocated.csv`.

Product	Uncertainty provided	Size	Not quantified
`model`: Vp, Vs, density	travel-time fit on held-out events; posterior s.d. of the calibration multipliers; Vs uncertainty of CRESCENT Gen0 (`vs_unc_regional`, rebuilt by S5, not in the release)	RMS 0.093 s (P), 0.188 s (S); ln V_0 0.027, ln P* 0.19, ln Vs 0.013; CRESCENT Vs 5.3% (L1), 1.8% (L2), 1.5% (L3), medians	per-cell uncertainty; the unit parameters, geometry rules and alteration factors; the geology and corrected regional models differ by up to 3.4% between 0.3 and 4 km
`model`: critical zone (`/cz`)	none	depth-to-rock priors differ by a factor of 13.9 at the median (Sect. 4.3)	all granular rock-physics and hydraulic parameters (not calibrated)
`model`: surface layers	ice thickness error of IceBoost v2 (`ice_thickness_error`, rebuilt by S2, not in the release); water tables against wells (Table 26)	ice 32 m median (86% of the thickness); water table median misfit 7.1 m (Ma et al.), 12.3 m (Fan et al.)	soil and vegetation layers carry their source products' uncertainty, which is not propagated
`alteration`	none		intensity thresholds (`m1_placeholder`); depth beyond ~150 m
`gnss`	standard deviation of the dilatation rate (`dilatation_sigma` in the strain grid)	3.9 nanostrain yr^{-1}, median	shear rate and azimuth
`strain_3d`	none		the extension of the surface strain rate to depth is an assumption (Sect. 6.5.1)
`edifice_load`	none		half-space approximation, density and Poisson's ratio (author choices)
`mass_movements`	none		completeness by class and date; three crowns on the 30 m DEM
relocated catalogue	NonLinLoc formal uncertainty per event	0.69 km in depth and 0.57 km horizontally, medians over grade A and B events	velocity-model error in the locations

which gives, for every product, the commit that made it, the data freeze, the stages to run from a clean clone and the inputs that must be obtained by hand, and `inputs.csv`, which lists every cached file each product reads with its SHA-256. S20 refuses to publish from a working tree with uncommitted changes, so the commit named in `REBUILD.md` is the code that made the products.

The archives need no software from this project. The subsurface model is downloaded, checked and opened with standard tools:

```
B=https://github.com/Denolle-Lab/mt-rainier-digital-model/releases/download/products-v1.0.0
curl -LO $B/SHA256SUMS -LO $B/rainier3d_model.zarr.zip -LO $B/rainier3d_grids.zip
sha256sum -c SHA256SUMS --ignore-missing     # or: shasum -a 256 -c (macOS)
unzip rainier3d_model.zarr.zip && unzip rainier3d_grids.zip
```

```
import xarray as xr
tree = xr.open_datatree("model.zarr", engine="zarr", consolidated=False)
vs = tree["L2"].to_dataset().vs.interp(x=594500, y=5189500, z=-2000)   # m/s, 2 km below sea level
g = xr.open_dataset("grids/rainier3d_fused_500m.nc")                    # whole model, 500 m spacing
```

In `model.zarr`, cells above the ground are empty (NaN), and the unit codes are those of `configs/units.yaml`.

A client is installed with the package. It is available as the `rainier3d` command and as the Python module `rainier3d.api`. It downloads a product once, checks its checksum, caches it, and writes grids for other codes:

```
pip install "git+https://github.com/Denolle-Lab/mt-rainier-digital-model"
rainier3d list                                          # products, versions and licences
rainier3d fetch model                                   # prints the folder that holds model.zarr
rainier3d export model --format specfem --bbox -122.0 46.7 -121.6 47.0 \
    --dx 250 --dz 250 --zmin -10000 --out out/tomography_model.xyz
rainier3d export model --format nll --dx 500 --dz 500 --out out/nll/rainier3d
rainier3d export surface --layers elevation soil_thickness --out out/surface/   # GeoTIFFs
rainier3d sample --lon -121.76 --lat 46.85 --depth 5000   # values at one point
rainier3d fetch gnss                                    # latest weekly strain product
```

```
import rainier3d.api as r3
tree = r3.open_model()                                      # downloads once, then reads the cache
g = r3.grid(tree, bbox=(-122.0, 46.7, -121.6, 47.0), dx=250, dz=250, zmin=-10000)
r3.export(g, "netcdf", "out/rainier3d_250m.nc")             # also nll, emc, specfem, pylith, csv
```

The repository file `docs/products.md` gives examples for every product.

Resampled grids are interpolated linearly within each level. Cells above the ground take the value of the first rock cell beneath them, so receivers at the surface sit in rock, and a variable `air` flags them. Table 31 lists the formats.

9.2 Locating earthquakes in the three-dimensional model

Of the 15,660 PNSN earthquakes since 1980 in the box, 500 have ComCat hypocentres at or above the ground surface, mostly shallow edifice events. Those events were located in one-dimensional models that know neither the topography nor the slow

Table 31. Export formats of `rainier3d export`.

Format	Coordinates	Use
CF netCDF4	UTM 10N x, y (m); z = elevation (m, up)	eikonal solvers (pykonal, fteikpy), custom codes
NonLinLoc 3D grids (P, S), slowness × cell size	UTM 10N km, depth down below sea level, `TRANSFORM NONE`	Grid2Time, NLLoc
EMC netCDF3	longitude, latitude, depth below sea level (km)	EarthScope Earth Model Collaboration tools
SPECFEM3D `tomography_model.xyz`	UTM 10N m, elevation up, x fastest	SPECFEM3D Cartesian
PyLith spatial database (SimpleGridDB): density, Vs, Vp, with a parameter snippet	UTM 10N m, elevation up	PyLith static and quasi-static elasticity
CSV	x, y, z and variables, one row per node	spreadsheets, GIS
GeoTIFF (surface layers)	UTM 10N, 100 m	GIS

edifice. Script S26 relocates the catalogue with NonLinLoc (Lomax et al., 2000) twice: once in the PNSN 1D model and once in rainier3d. Both runs use the same picks, stations, grid geometry and settings, so the difference between the two catalogues isolates the velocity model. ComCat is kept as a third reference. The settings are:

- **Grids.** Equal spacing in all directions (500 m here; 250 m is practical near the edifice). The time grids run from the summit rounded up to a whole cell (4.5 km above sea level) to 20 km below sea level. The 1D model is written on the same grid as the 3D model. The search volume is inset 1 km from each side of the box and starts one cell below the top of the time grids.
- **Keep hypocentres in rock.** The search volume is masked below the ground with `LOCTOPO_SURFACE`, reading an ASCII GMT grid of the DEM in kilometres on the UTM frame. With `TRANS NONE`, NonLinLoc compares $-z$ with that grid. The exported velocities are left unchanged, and air cells carry the rock velocity below them, so travel times are unbiased. Slow air above a rock skin is an alternative, but the skin must be at least two cells thick: with a one-cell 500 m skin, Grid2Time's finite-difference start box reaches the slow air around summit stations and delays P times at station OBSR by 0.41 s on average.
- **Stations.** The 45 stations inside the box that carry picks, in the UTM kilometre frame, with depth = $-$elevation/1000 (`GTSRCE <sta> XYZ <x_km> <y_km> <−elev_km> 0.0`).
- **Travel times and location.** Grid2Time with the Podvin–Lecomte finite-difference eikonal solver (`GT_PLFD 1.0e-3`, `GTMODE GRID3D ANGLES_NO`) (Podvin and Lecomte, 1991); NLLoc with the equal-differential-time likelihood (`LOCMETH EDT_OT_WT`) and oct-tree search (`LOCSEARCH OCT 10 10 6 0.01 30000 10000`). Events need

Table 32. Relocation of the 2023–2025 PNSN catalogue, magnitude $\geq$ 1 (S26, `outputs/catalog/summary.json` and `catalog_relocated.csv`). Rows below "Located" are for the 340 grade A and B events located in both models. Summit events lie within 3 km of the summit and WRSZ events 8–25 km west of it, both by ComCat epicentre.

	ComCat	NonLinLoc, PNSN 1D	NonLinLoc, rainier3d
Located (of 370)	–	347	354
Rejected: base of the volume / side	–	2 / 21	10 / 6
Median RMS, grade A and B (s)	–	0.119	0.109
Median depth sd; largest horizontal uncertainty (km)	–	0.61; 0.61	0.69; 0.57
Above the ground	0	4, pinned at the mask	0
Summit (165 events): median elevation; 5–95% range (km)	−0.31; −1.29 to 2.18	−0.28; −0.90 to 0.77	−0.93; −1.89 to 0.65
WRSZ (129 events): median elevation; 5–95% range (km)	−8.8; −14.7 to −3.7	−9.6; −16.0 to −3.4	−9.6; −15.3 to −4.7

at least six picks, four of them P. Gaussian pick errors are 0.14 s for P and 0.23 s for S, as in the calibration; model errors are 0.1 s (`LOCGAU 0.1 0.0`) and grow with travel time as 2% of it, between 0.05 and 0.5 s (`LOCGAU2 0.02 0.05 0.5`).

- **Quality.** Graded from the rainier3d location. Grade A: gap < 180°, at least eight phases and depth standard deviation < 2 km. Grade B: gap < 250° and at least six phases. Grade C: the rest.

The 2023–2025 catalogue. We applied S26 to the 371 PNSN earthquakes of magnitude 1 or larger in the box from 2023 to 2025. They carry 20,425 analyst picks (12,546 P and 7,879 S) at 45 stations inside the box, and 370 have at least six picks. NonLinLoc locates 347 events in the 1D model and 354 in rainier3d. The others are rejected because their most likely location lies on the edge of the search volume (Table 32):

- in rainier3d, 10 rejected events lie at its base, at 18.2 km below sea level, the deepest node of the search grid (z_0 = 3.8 km below sea level plus 44 steps of 0.5 km in `scripts/26_relocate_catalog.py`);
- the remaining rejections, in either model, are epicentres at the edge of the box, beyond which there are few stations.

What changes. The residuals fall by 9% in rainier3d (median RMS 0.109 against 0.119 s). The events move a median 0.6 km horizontally and 0.49 km deeper than in the 1D model (10th–90th percentile −0.49 to +1.60 km).

- **Summit.** The largest change is under the summit. The 1D model gathers the events near sea level, over 1.7 km (5–95%). rainier3d places them 0.65 km deeper (median) and spreads them over 2.5 km. In the 1D model, four events rise until

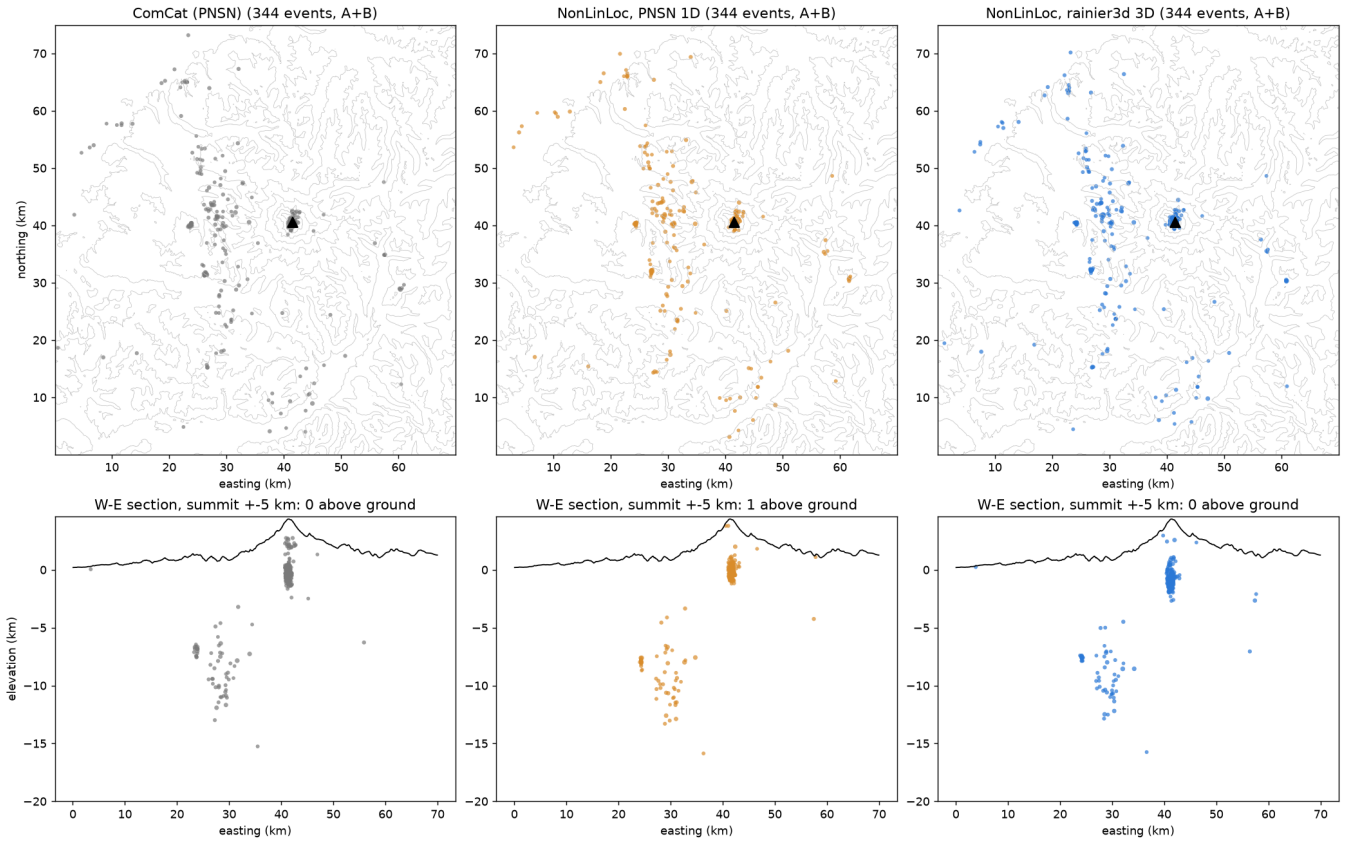


Figure 25. The 2023–2025 PNSN catalogue (magnitude ≥ 1; the 344 events graded A or B in rainier3d) as located by ComCat, by NonLinLoc in the PNSN 1D model and by NonLinLoc in rainier3d. Top: epicentres over elevation contours. Bottom: west–east sections within 5 km of the summit, with the ground profile, and the number of events of that band above the ground (the four 1D events pinned at the topography mask of Table 32 include three outside the band).

they meet the topography mask: ComCat places them 2.8–7.1 km below sea level, and rainier3d 1.3–9.6 km below it. No event ends above the ground in rainier3d.

- **WRSZ.** The median depth changes by less than 0.1 km, and the depth range narrows from 12.6 to 10.6 km.
- **Edges.** Epicentres near the northwestern and southeastern edges of the box shift outward in both models, where the station coverage ends.

In this period ComCat places no event above the ground. The problem of the 500 above-ground events therefore lies in the older part of the catalogue, which S26 can relocate once its picks are cached (Sect. 10).

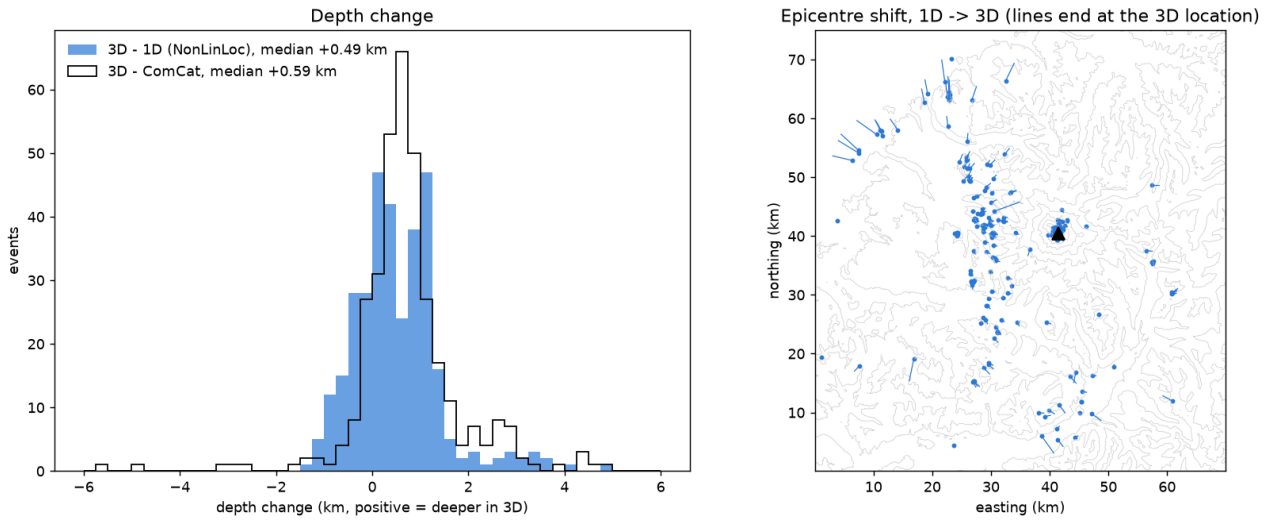


Figure 26. (a) Depth change of each event between rainier3d and the 1D model (filled) and between rainier3d and ComCat (outline). (b) Epicentre shifts from the 1D to the 3D location.

9.3 The three-dimensional viewer

The viewer runs in a web browser, including on phones. It is a React and three.js application (MIT licence) published at https://denolle-lab.github.io/mt-rainier-digital-model/, and it draws the following on the terrain:

- the 38 draped layers of `model/layers.json` in the data release (Sect. 4), from Sentinel-2 imagery, geology, ice and soil thickness to canopy, vegetation indices, land cover, water-table depth, soil texture and hydraulic properties, depth to bedrock, Vs30 and terrain geometry;
- the alteration field at the surface and the apparent magnetisation from the 1996 helicopter survey (Sect. 5.3.2);
- the 970 sensor sites of the S8 inventory (EarthScope FDSN, including the 2025 node array Z5 (Schmandt, 2025); EarthScope GNSS; Synoptic), 123 permanent and 847 temporary: seismometers, accelerometers, geophone nodes (813), infrasound, GNSS, strain and tilt meters, weather, snow and streamflow stations, and the Paradise–Nisqually Entrance distributed acoustic sensing (DAS) fibre with 3,191 channels, drawn as permanent or temporary and filtered as current or past, with the 2025 node array and the DAS fibre as surveys on their own toggles;
- the PNSN seismicity from ComCat (U.S. Geological Survey, 2026a): 15,631 events from 1980 to 23 September 2026, magnitude −1.6 to 4.9, in the overview box (46.58–47.12° N, 122.16–121.40° W), drawn at their hypocentres; the 435 that ComCat places above the viewer terrain are counted but not drawn;
- the mass movements of Sect. 8: the 466 flow deposits as a draped layer and the 1,649 events as points on the ground (crown or seismic location), filtered by class and date from the legend;
- the December 2025 atmospheric-river storm, replayed over its 216 hourly frames: precipitation as falling drops and discharge as bars at the 7 USGS gauges and 3 virtual sensors inside the overview box.

Below the ground it shows Vs, Vp, Vp/Vs, density, units, alteration and the strain fields of Sect. 6.5, on a vertical section along the terrain cut and on a horizontal depth slice. For the strain fields, the depth slice also carries their orientation bars. A panel under the subsurface controls shows the relocated catalogue of Sect. 9.2, as located by ComCat, in the 1D model and in rainier3d, with optional lines from each 1D location to its 3D location. Its map data are built by scripts S8, S11, S18, S24 and S26 and published as a release asset named in `web/viewer/DATA_RELEASE`.

10 Limitations

- **Rock-physics parameters.** The unit parameters of Table 9 are chosen within published ranges and scaled by three global multipliers, of which V_0 and the rock Vs are resolved by the travel times and P* is not (Table 16). The alteration factors and the Q rule are likewise author choices. The planned replacement is a table of laboratory and field measurements on Cascade and analogue volcanic rocks (Watters et al., 2000; Heap and Violay, 2021), against which the calibrated V_0 can be tested. Per-lithology multipliers need more stations on plutonic and sedimentary units.
- **Unsourced parameters.** Values chosen by the authors carry the registry key `m1_placeholder`: all 16 unit rows of `configs/petrophysics.csv`, the alteration, magma-body and Q factors of `configs/perturbations.yaml`, three alteration thresholds in `configs/units.yaml`, and the quality-control, strain, region, load and volume choices of `configs/gnss.yaml`. Replacing them from the literature is tracked work.
- **Geometry by rules.** Contacts are vertical and unit bases are flat; the fusion smooths them below the critical zone with author-chosen cutoffs (Sect. 5.3.3), but does not place them. Structural modelling from the map contacts and orientation measurements would replace the column rules.
- **Hidden alteration and deep bodies.**
  - The alteration field covers only the top ~150 m. The buried alteration of the upper west flank (Finn et al., 2001) is not represented.
  - The magma body is a prescribed sphere whose size and anomaly are author choices; the tomography of Sect. 5.3.7 has almost no resolution at its depths.
  - The Southern Washington Cascades Conductor (Stanley et al., 1996) is not represented.
- **Regional correction.** It is one depth profile for the whole box. The iMUSH comparison shows that it holds beneath Rainier but makes Vs 5–6% too fast at 2–6 km in the south, so it should vary laterally.
- **Top 300 m.** The calibrated geology, critical zone included, is 18% faster in Vp and 23% faster in Vs than the shallowest layer of the Cascadia model, and the travel times constrain it only beneath the stations. Near-surface Vs from dense nodal and fibre arrays, or station terms estimated with a prior on their size, can settle which is right. Station terms are not estimated in the calibration.
- **Depth coverage of the calibration.** Only 30 of the 88 events are deeper than 11 km. Larger pick sets, such as the curated PNSN dataset (Ni et al., 2023), and Rainier-specific models (Obrebski et al., 2015; Flinders and Shen, 2017) would constrain the deep correction.

- **Reference 1D model.** The PNSN one-dimensional model used for comparison is read from a local table whose identity (P3 Puget Sound or C3 Cascades) is not documented [TODO: confirm with PNSN and cite].
- **Regional model below 9.9 km.** The regional model there is CRESCENT Vs with Brocher's Vp; the deep level of the Cascadia model (10.8–59.4 km) is not used.
- **Depth to rock.** The two estimates differ by a factor of 13.9 at the median (SOLUS100 1.41 m, capped at 2.01 m; SoilGrids 2017 19.6 m), and the water table lies between them in 92.8% of the cells (Sect. 4.3). No local data constrain the thickness of soil, regolith and weathered rock between 2 m and the top cells of L1, and the unconsolidated deposits of the geology model take map-unit thicknesses for eight symbols and fixed defaults (10–15 m) for the rest.
- **Resolution.** L1 is 250 m × 50 m, so thin deposits fall below the cell size.
- **Glaciers.** IceBoost exceeds the 1981 radar thicknesses on Emmons and Winthrop glaciers; its total should be compared with the lidar-based ice volume of Sisson et al. (2011).
- **Relocated catalogue.** Only the 2023–2025 events of magnitude 1 or larger are relocated. The 500 older ComCat events at or above the ground are not relocated until their picks are cached, and 10 events stop at the base of the search volume in rainier3d.
- **Geodesy and strain.** The GNSS network does not resolve strain on the edifice. The strain in the volume uses no Green's functions of the heterogeneous model and no rheology beyond linear elasticity (Sect. 6.5.1):
  - the geodetic strain rate at depth is assigned (depth-invariant horizontal rate, plane stress), not inverted from sources;
  - the edifice-load stress is that of a uniform half-space with a flat surface, converted to strain with the local stiffness, with a confined-overburden approximation inside the cone.
- **Hydrology.** The model has no hydrological state beyond the critical-zone columns (Sect. 7, Sect. 5.2), and no process of the model responds to precipitation.
- **Mass-movement catalogue.** Events from reports and papers are not yet digitised, the Exotic Seismic Events Catalog v3 has no scripted download, three landslide crowns sit on the 30 m DEM, and the Washington landslide inventory states no licence.
- **Vegetation layers.** Only the GEDI L3 canopy height is rebuilt by script (S28). The 10 m lidar grids were made in QGIS, and the GEDI L2B plant area index, L4B biomass and the soil map are read from delivered files.
- **Licences.** The lidar canopy layers (Washington DNR Lidar Portal terms not confirmed) and the soil-map image (source undocumented) are shown in the viewer but not published. The licences of CRESCENT Gen0, IceBoost v2, RGI 6.0, GlaThiDa, the Synoptic station catalogue and the Fan et al. (2017) water table are not yet recorded in the registry.

11 Conclusions

rainier3d assembles, in one reproducible structure, what is openly known about the subsurface and surface of Mount Rainier. Seismic monitoring at the volcano relies on one-dimensional models, and the regional three-dimensional models do not resolve the edifice or the near surface; rainier3d fills the space between the two.

- **Subsurface.** A geology-driven velocity model, fused with the regional tomography and calibrated on PNSN P and S−P travel times with every event relocated in 3D. On the held-out events the calibration lowers the RMS residual from 0.120 to 0.093 s for P and from 0.325 to 0.188 s for S. It raises the zero-pressure Vp of the rock units by a factor of 1.44 and makes the regional Vs beneath Rainier 5–8% faster at 2–7 km, a correction that the iMUSH tomography supports where it resolves Vs.
- **Critical zone.** Soil, Quaternary cover, weathered and fractured rock in fine columns down to 150 m, with porosity, permeability, saturation and effective stress, joined to the rock below with one water table and one pore pressure.
- **Tomography.** A local-earthquake tomography of 3770 PNSN earthquakes on a 250 m grid, which leaves the critical zone as built, lowers the held-out residuals further, to 0.0725 s for P and 0.1209 s for S, and resolves lateral structure of about 12 km between sea level and 6 km depth (Supplement S1).
- **Surface.** Layers for geology, ice, soil, water, vegetation, terrain and imagery on a common grid, and a catalogue of mass movements.
- **Geodesy and load.** GNSS strain rates, refreshed weekly, and the stress of the edifice load at depth, both carried into the model volume as strain.
- **Earthquake locations.** The 2023–2025 PNSN catalogue relocated with NonLinLoc in the 1D and 3D models with the same picks: the median RMS falls from 0.119 to 0.109 s, and no event lies above the ground in rainier3d, against four pinned at the topography in the 1D model.

Every cached input is checksummed and fetched by script from its archive, except four model inputs and four viewer files read from local or delivered copies (Sect. 3). Each parameter names its source or is marked as an author choice (`m1_placeholder`), and the products can be downloaded with one command in the formats that seismological codes read. The mass-movement catalogue carries time, and the critical-zone columns carry the state variables on which a groundwater model can run. The next steps are a laterally varying regional correction, an uncertainty for every gridded product, the events digitised from the literature, and a water budget on the critical-zone columns that would let precipitation change the seismic properties.

12 Code and data availability

The code, configuration, source registry and this paper are at https://github.com/Denolle-Lab/mt-rainier-digital-model (BSD-3-Clause). The 3D viewer in `web/viewer/` and the vendored canopy-storage code in `third_party/canopy-storage_seismic` are under the MIT licence. The derived products are published as assets of a versioned release of the same repository (CC-BY 4.0). The current release, `products-v1.0.0`, predates the critical-zone model; the release described in Sect. 9 will be built from the commit that accompanies this paper [TODO: products release and commit]. Its seven products are the model, the uniform grids, the strain in the volume, the edifice-load stress, the alteration field, the mass-movement catalogue and a GNSS snapshot of 1 September 2026. They are listed with SHA-256 checksums in `SHA256SUMS` and `src/rainier3d/products.json`, downloaded with `rainier3d fetch`, and described product by product, with

command-line and Python examples, in `docs/products.md`. The GNSS product is refreshed weekly in the release `gnss-latest`, with dated copies. The relocated 2023–2025 catalogue and the summaries of the December 2025 event are the assets of the releases `relocated-catalog-2023-2025-v1` and `storms-dec2025-v1`, also under CC-BY 4.0. The viewer reads its map data from the release named in `web/viewer/DATA_RELEASE`. The software and the derived products will be deposited on Zenodo as two versioned records (`docs/doi.md`) once the subsurface stress model is in the release [TODO: concept and version DOIs, and the software tag].

The supplement (Supplement S1, `docs/paper/rainier3d_supplement.md`) describes the local-earthquake tomography; its figures and an interactive view of the rays are published at https://denolle-lab.github.io/mt-rainier-digital-model/tomography/, and the construction of the model, stage by stage, is recorded in `docs/model_construction.md`.

Appendix A lists the input data sets with their DOIs or service addresses. `docs/data_policy.md` gives the licence tier of each. `docs/data_manifest.csv` gives the checksum of each cached input.

Three inputs require attribution notices:

- GPS time series are provided by the Pacific Northwest Geodetic Array, Central Washington University.
- The imagery layers contain modified Copernicus Sentinel data (2023, 2025).
- MRMS precipitation is NOAA open data, whose terms request attribution.

The code and this paper were written with an AI coding assistant (Claude, Anthropic) under the direction of the authors (Sect. 3.4). All numbers are produced by the scripts named in the text. DOIs were resolved at doi.org or against the Crossref and DataCite registries (`docs/citations.csv`).

Appendix A: Data sets

Table A1: Input data sets.

Data set	Role in the model	Reference
Cascadia velocity model v1.7	regional Vp and Vs to 9.9 km	Wirth et al. (2025); Wirth Moriarty et al. (2025)
CRESCENT Gen0 community velocity model	regional Vs below 9.9 km	He et al. (2026); He (2026)
Washington 1:100,000 surface geology (GeMS)	surface units	Washington Geological Survey (2026a)
Geology of Mount Rainier National Park	stratigraphy; map overlay	Fiske et al. (1963)
USGS 3D Elevation Program	elevation	U.S. Geological Survey (2026c)
IceBoost v2 per-glacier thickness; RGI 6.0	ice thickness; glacier identifiers	Maffezzoli (2025); RGI Consortium (2017)

Data set	Role in the model	Reference
GlaThiDa; 1981 radar surveys	glacier thickness check	World Glacier Monitoring Service (WGMS) (2020); Driedger and Kennard (1986)
1996 helicopter EM and magnetic survey	alteration	Rystrom et al. (2000)
iMUSH local-earthquake tomography	independent check	Incorporated Research Institutions for Seismology (2020); Ulberg et al. (2020)
PNSN origins and phase picks (ComCat)	calibration and validation	U.S. Geological Survey (2026a)
PNSN one-dimensional velocity model, western Washington	reference model for validation and relocation	[TODO: citable reference; registry key pnsn_1d_wa has no BibTeX entry]
FDSN station metadata	station coordinates, sensor inventory	EarthScope Consortium (2026)
EarthScope GNSS site metadata	GNSS site positions	ear
2025 Rainier NP node array (FDSN network Z5)	geophone node positions and dates	Schmandt (2025)
Paradise–Nisqually Entrance DAS channel table	fibre route	[TODO: citable reference; registry key das_paradise_nisqually is a local file with no BibTeX entry]
Synoptic weather, SNOTEL and streamflow stations (gaia-hazlab catalogue)	sensor inventory in the viewer	syn, TODO: licence
PANGA and UNR GNSS daily positions	velocities, strain	Pacific Northwest Geodetic Array (2026); Blewitt et al. (2018)
SOLUS100	soil thickness; texture, bulk density, rock fragments and organic carbon at 0–1.5 m; depth to a restrictive layer	Nauman (2024)
POLARIS v1.0	soil hydraulic conductivity and van Genuchten parameters to 2 m	Chaney et al. (2019)
SoilGrids250m 2017 depth to bedrock	regolith thickness	Shangguan et al. (2017, TODO: licence)
USGS global hybrid Vs30	Vs30	Heath et al. (2020)

Data set	Role in the model	Reference
US water-table depth	water table	Ma et al. (2026a)
Global water-table depth	water table, comparison	Fan et al. (2017)
NHDPlus High Resolution	streams	U.S. Geological Survey (2026b)
MRMS multi-sensor QPE, 1 h, Pass 2	event precipitation	Zhang et al. (2016)
USGS instantaneous discharge	event river gauges	U.S. Geological Survey (2026e)
seis-hydro-2-sed virtual discharge and storm windows	event virtual sensors	sei
ETH global canopy height 2020	canopy height	Lang, Nico et al. (2022)
NLCD 2021	land cover	U.S. Geological Survey (2023)
Copernicus Sentinel-2 L2A	imagery, NDVI, NDSI, leaf area index	European Space Agency (2025); European Space Agency (2023)
GEDI L2B, L3, L4B	plant area index, canopy height, biomass	Dubayah et al. (2021b); Dubayah et al. (2021a); Dubayah et al. (2025)
Washington DNR Lidar Portal, 2022–2023 "Wali" acquisition	canopy height and vegetation cover (viewer only)	can
Soil-map image of the park	display only	[TODO: source and legend; registry key soil_map_image]
Washington State Landslide Inventory Database	landslide and debris-flow catalogue	Washington Geological Survey (2026c)
Seismogenic mass movements, western United States	dated, seismically located events	Allstadt et al. (2017)
Mount Rainier volcano hazards, digital data	lahar inundation zones	Hoblitt et al. (1998); Schilling et al. (2008)
Quaternary active faults of Washington	faults	Washington Geological Survey (2026b)
NSHM 2023 fault sections database	fault check (no section in the box)	Hatem et al. (2021)
USGS The National Map imagery (USGSImageryOnly)	viewer aerial imagery	[TODO: add a registry key]

Author contributions. MD conceived the project, designed the model, its pipeline (scripts S0–S29), the calibration and the validation, directed the development of the code and wrote the paper. DY designed and built the three-dimensional viewer (`web/viewer/`). MKö

produced the vegetation products of the canopy-storage project (lidar canopy height and cover, Sentinel-2 leaf area index and the gridded GEDI products) and wrote their download and gridding code, vendored in `third_party/canopy-storage_seismic`. MH, SH and MKi contributed to the design of the model and the interpretation of its results through discussions in person and on Slack; SH also specified the terrain-geometry layers (surface and bedrock slope, local relief and valley depth) and their comparison with the mass-movement catalogue. [TODO: confirm that all authors reviewed and edited the manuscript.]

Competing interests. The authors declare that they have no conflict of interest.

Acknowledgements. This work was supported by the Jerome and Linda Paros Geohazard Center at the University of Washington.

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Canopy height model and vegetation cover fraction, 10 m, from airborne lidar DSM/DTM tiles of the WA DNR Lidar Portal (2022-2023 "Wali" acquisitions, e.g. Rainier Wali 2022; NAD83(HARN) Washington South ftUS), gridded in QGIS for the canopy-storage project (M. Koepfli), https://lidarportal.dnr.wa.gov/(tiles);https://github.com/koepflma/canopy-storage_seismic(private;non-seismic_code/CHM/,commit292ebe4).

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